

Lifetime Tester

User Manual

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Captronics Systems Pvt Ltd

	Name	Signature	Date
Prepared By	Ananth A		
Reviewed By	Prabhakaran T J		
Approval By	Midhunseelan O S		

Customer

	Name	Signature	Date
Reviewed By	Abhay Kumar		14.02.2025
Approved By			

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Terms, Definitions and Abbreviations

Table 1: Terms, Definitions and Abbreviations

Sl. No	Term	Explanation
1	ATE	Automated Test Equipment
2	UM	User Manual
3	BOM	Bill of Materials
4	UUT	Unit Under Test
5	DUT	Device Under Test
6	ITA	Interchangeable Technology Adapter
7	HOST	The Server computer used as a controller for the system
8	Platform	Phoenix Based Equipment Rack
9	CSPL	Captronic Systems Pvt. Ltd.
10	DAS	Data Acquisition System
11	GUI	Graphical User Interface
12	H/W	Hardware
13	LabVIEW	Laboratory Virtual Instrumentation Engineering Workbench
14	NI	National Instruments
15	PXI	PCI Extension for Instrumentation
16	PXIe	PXI Express
17	NI Matrix	National Instruments Matrix
18	PK Matrix	Pickering Matrix
19	DI	Digital Input
20	AI	Analog Input
21	DO	Digital Output
22	AO	Analog Output
23	CAN	Controller Area Network
24	FGEN	Function Generator
25	Scope	Oscilloscope
26	DC	Direct Current
27	DMM	Digital Multimeter
28	DIO	Digital Input and Output
29	MIO	Multifunction Input and Output
30	Calib	Calibration
31	BOB	Brake Out Box
32	OVP/OCF	Over Voltage Protection/ Over Current Protection

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1 Safety Precautions



(Always read these instructions before using this equipment)

For personnel and product safety please read the contents of these operating instructions carefully before using. Please save this manual to make it accessible when required and always forward it to the end user.

⚠ CAUTION Indicates that the incorrect handling may cause hazardous conditions. Always follow the instructions because they are important to personal safety. Otherwise, it could result in electric shock, fire, erroneous operation, and damage of the instrument.

2 Normal service conditions

⚠ CAUTION

Use the instrument in an environment that meets the Normal service conditions as following points:

- Ambient temperature: -5 to 50°C, average day temperature: 35°C or lower
- Humidity: 30~85%RH, non-condensing.
- Altitude: 1000m or less
- Pollution Degree: 2
- Atmosphere without corrosive gas, dust, salt, oil mist.
- A place without excessive shocks or vibration.
- Do not expose to rain and water drips.
- An area where no pieces of metal and no inductive substances disperse.
- Do not expose to strong electromagnetic field and ambient noises.

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3 Installation Instructions

⚠ CAUTION
This instrument should be installed and used by a qualified electrician. ● Verify the following points. ✓ Ratings Voltage 220-240 V AC phase-neutral / 415-480V AC phase-phase ✓ Frequency 50/60Hz

4 Introduction

The purpose of the document is to describe the user instructions for the product tooling Test Base Equipment rack referred to as “Lifetime Tester”. It is developed for testing content.

The heart of the system is the PXI based data acquisition unit mounted inside the console panel. The console panel features control push buttons, digital indications and monitors for monitoring and controlling the test systems.

5 Scope

The scope of this document is restricted to describing the tool functionality of Lifetime Tester.

The system is designed to provide the following features to the User.

- 1. Developing Custom Test Sequence in NI TestStand**
- 2. Executing the developed TestStand Sequence Files in LabVIEW Environment**
- 3. Manual access to individual hardware present in the system**
- 4. Report storage for Used to SQL Data Management.**

The application software is used for TestStand sequence execution, data logging, offline data analysis and generating reports.

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6 Tools Overview

Test Platform provides a standard set of measurements and control equipment to allow the testing for Lifetime Tester.

7 Tools Specifications

7.1 Functional Specifications

The table below will give details about the functional specifications of the Lifetime Tester.

Table 2: Functional Specifications

F1	The platform shall utilize a 19" equipment rack for housing equipment with a maximum height of 42U (73.5").
F2	The platform shall include a PXIe 1085 Chassis for connecting modular instruments to the system.
F3	The PXI chassis shall include below hardware functions at a minimum:
F4	• PXIe 8861 – Controls PXI and PXI express systems.
F5	• PXI 6514 DIO - Provides Digital input and output capabilities to the PXIe systems.
F6	• PXIe 6363 MIO - Provides Analog, Digital input and Analog, Digital output capabilities to the PXIe systems.
F7	• PXIe 6345 AI - Provides Analog input, PWM Input and output capabilities to the PXIe systems.
F8	• PXI 4065 – DMM to perform voltage and current measurements.
F9	• PXIe 5413 FGEN – Waveform generator capable of generating user-defined, arbitrary waveforms and standard functions including sine, square, triangle, and ramp.
F10	• PXIe 5110 Oscilloscope – Acquires and analyses time and frequency-domain analog signals as part of a PXI system.
F11	• PXIe 6739 AO - Provides Analog output, PWM Input and output capabilities to the PXIe systems.

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F12	• PXIe 6738 AO - Provides Analog output, PWM Input and output capabilities to the PXIe systems.
F13	• PXIe 6612 CI/CO - Provides PWM Input and output capabilities to the PXIe systems.
F14	The Platform shall include Hitech High Voltage Programmable Power Supply- 400W Auto ranging Dual Output Power Supply, 60V, 10A.
F15	The Platform shall include Hitech – DC Programmable Power Supply, 80 V, 170 A, 5000 W, 400 VAC.
F16	The Platform shall include one fixed 24V (2 no's), + - 15 V SMPS Power supply.
F17	The Platform shall include a Windows 10 compatible PC.
F18	The Platform shall include a wall mount sit stand workstation, monitor holder gas strut arm with keyboard tray rotate LCD bracket.
F19	The keyboard and mouse will be mounted in a monitor holder gas strut arm tray rotate.
F20	The Platform shall include an LCD monitor.
F21	The Platform shall provide castors for easy relocation.
F22	The Platform shall provide a power distribution unit with emergency Power OFF button.
F23	The Platform shall provide proper grounding connections for all internal instrumentation using a copper rod from rack top to rack back side.
F24	The Platform shall include an Uninterruptible Power Supply system.

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7.2 Non-Functional Specifications

Items and features outlined in this section are considered during the design of the test system and were incorporated in the final design document.

Table 3: Non-Functional Specifications

Term	Explanation
D1	The platform could operate in -20 to 70deg Celsius and should operate from -20 to 60 deg Celsius.
D2	The platform shall operate in 5% to 50% humidity, non-condensing.
D3	The platform is intended for level hard surfaces
D4	Equipment shall be placed in the rack to keep center of gravity as low as possible to prevent a tipping hazard
D5	The keyboard and mouse can be concealed for user convenience
D6	Interchangeable Technology Adapter
D7	The platform shall operate over the AC voltage range of 390 - 440 Vac 50/60Hz, <65 Amps

7.3 Compliance Specifications

The table below will give details about the compliance specifications of the Lifetime Tester.

Table 4: Compliance Specifications

Term	Explanation
E1	The Platform shall provide warning signs to expose pitch hazards.
E2	All calibrated equipment shall be calibrated per manufacturers specifications.
E3	All equipment shall be maintained per manufacturers specifications

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8 Pictorial View of 6 Head Lifetime Tester

8.1 6 Head LTT Instrument Rack.



Figure 1: Lifetime Tester.

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8.2 Lifetime Tester Power supply Rack

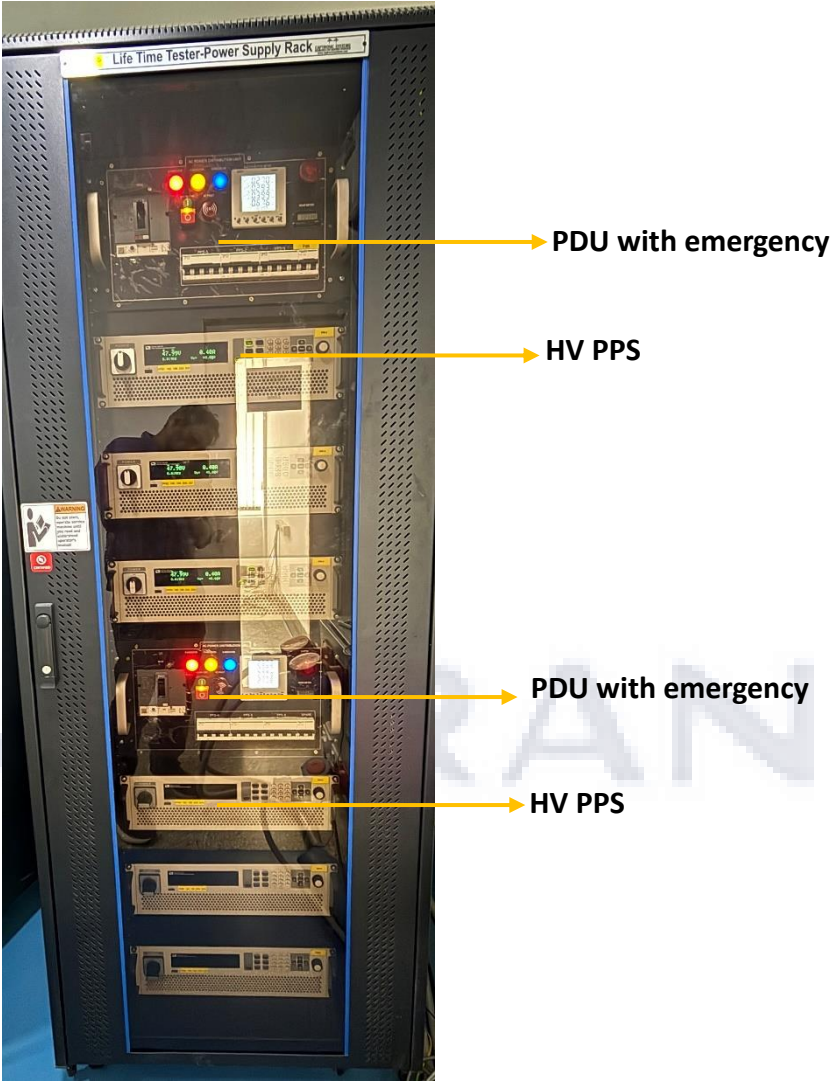


Figure 2: Lifetime Tester Power supply Rack.

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9 Pictorial View of 3 Head Lifetime Tester.

9.1 3 Head LTT Instrument Rack.

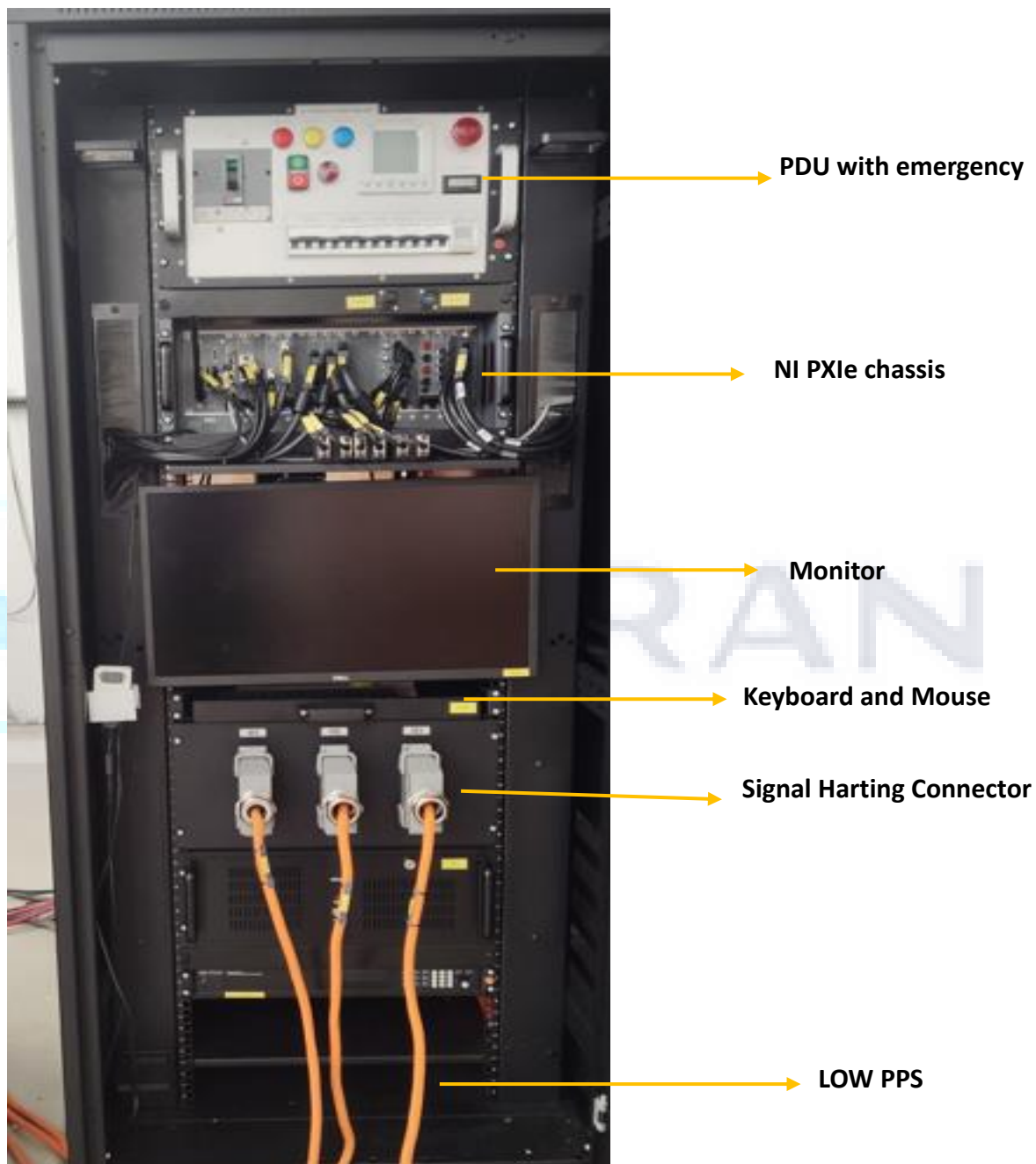


Figure 3: 3 Head Lifetime Tester.

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9.2 3 Head LTT Power supply Rack and RL Load.



Figure 4: 3 Head Lifetime Tester Power supply Rack and RL Load.

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10 Procedure to Power ON Console

10.1 Prior Inspection

- Prior to applying power to the system. Users should make several final inspections of the hardware components and interconnections.
- Visually inspect the system to ensure that all hardware components are present. Verify correct model numbers for each component.
- Inspect all components and NI modules to ensure that they are installed in the correct slot locations and placed securely in position and Refer the Figure 5.

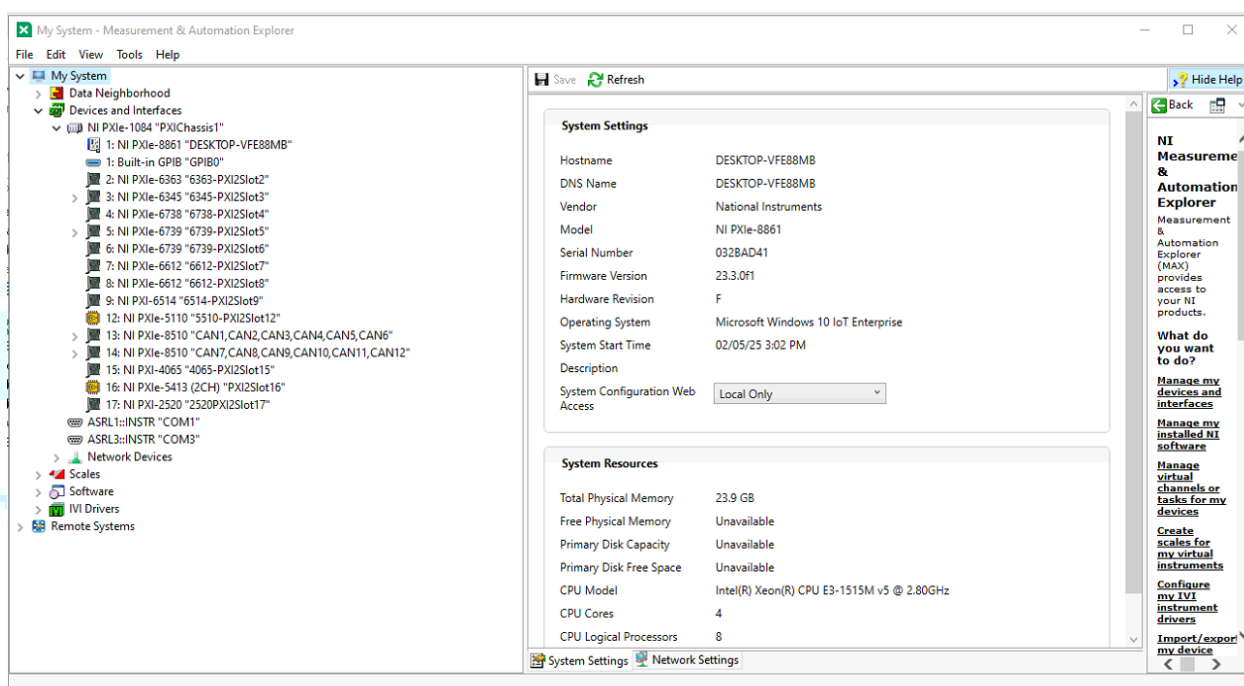


Figure 5: NI MAX Card Information

List the NI Hardware Slot details:

- 1) Slot 1- PXIe-8861- Desktop PC.
- 2) Slot 2- PXIe-6363- Analog Input.
- 3) Slot 3- PXIe-6345- Analog Input.
- 4) Slot 4- PXIe-6738- Analog Output.
- 5) Slot 5- PXIe-6739- Analog Output.
- 6) Slot 6- PXIe-6739- Analog Output.
- 7) Slot 7- PXIe-6612- PWM Input and Output.
- 8) Slot 8- PXIe-6612- PWM Input and Output.

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9) Slot 9- PXI-6514- Digital Input and Output.

10) Slot 12- PXIe-5110- Oscilloscope.

11) Slot 13-PXIe-8510- CAN.

12) Slot 14-PXIe-8510- CAN.

13) Slot 15-PXIe-4065- Digital Multimeter.

14) Slot 16-PXIe-5413- Waveform Generator.

15) Slot 17-PXI-2520- Relay Output

- Check that the incoming power is correctly wired to the power supply and that the system power is properly routed and connected to each instrument.
- Check that the output wiring connections are in place and properly terminated at the DUT end.
- Connect 415 (390-440) VAC supply to main inlet (MCCB) and turn ON main disconnection switch.



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10.2 Turn ON the System

10.2.1 Turn ON the Main Toggle Switch

- Turn ON the Main Switch in the Instrument Rack PDU panel by toggle switch to up position. Refer the main switch control fig.6.



Figure 6: Main Power Switch

10.2.2 Verifying the ON the Three Phase Indicator

- Check the status of the RYB indication present in the AC PDU box. Refer Fig. 7.



Figure 7: RYB Power ON Indicator

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10.2.3 Turn ON the Rig Push Button Switch

- If the Mains on indicator does not turn on check the input power supply connection

Press the RIG ON push button

- Press the rig ON push button on the AC PDU box to turn ON the console. Refer the push button control fig.8.



Figure 8: Rig ON/OFF Push Buttons

10.2.4 Ensure the Turn ON the MCB.

- Ensure the PPS 1, PPS2, PPS3, PXI, Rack, Circuit MCBs and ELR are ON. Refer fig 9

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Figure 9: MCB ON Condition

- Turn ON the main MCB and individual MCB's which is controlling the instrument.
- Check instruments are getting powered ON.

10.2.5 Turn ON the UPS Switch

- Turn ON the UPS Switch in the Instrument Rack PDU panel. Refer to the main switch control fig.11.



Figure 10: UPS On Indication

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10.2.6 Turn ON the NI PXI.

- Ensure the PXI are ON. Refer fig 10.



Figure 11: PXI ON Indication

- Turn ON the PXI power and wait for a few seconds.
 - Ensure that the computer and monitor are ON. Ensure that the display cables are connected.
 - Login to PC using the default password “VTLTT”.

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10.2.7 Turn ON the Low Voltage PPS Switch

- Turn ON the LV PPS Switch in the Instrument Rack PDU panel. Refer to the main switch control fig.6.
- Press the Switch once the power is on, and the LED display will light up Fig 12



Figure 12: Low Voltage PPS On Indication

10.2.8 Turn ON the Power Supply Rack

- Turn ON the PDU Switch and MCB in the Power Supply Rack PDU panel. Refer to the main switch control Turn ON the. Follow steps mentioned in section 10.2
- Turn ON the High Voltage Programmable Power supplies.

10.2.9 Turn ON all the High Voltage PPS Switch

- Turn ON the High Voltage Programmable Power supplies.
- Turn the knob clockwise once the power is on, and the LED display will light up Fig 13.



Figure 13: High Voltage PPS On Indication

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11 Software Details

11.1 How to Launch LTT Application

- Run the LTT.exe application as an administrator. It's located as a shortcut on the PC desktop.
- Application also located here - "D:\eDCU LTT\VT LTT Project\Exe\LTT.exe"
- Splash screen figure 14 appears in the desktop for few seconds.

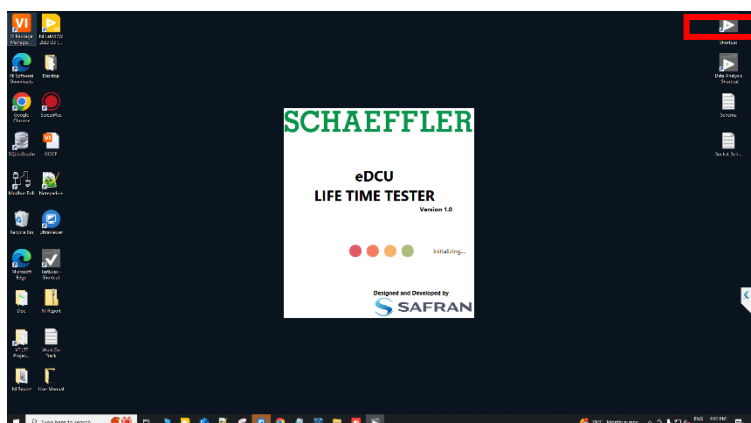


Figure 14: Lifetime Tester Splash Screen

11.2 User Interface Flow

The Main Screen of Lifetime Tester provides the user to enter different software features of the system such as Admin Login, User Account Manager, Setup, Auto Test and Diagnostic Test. The software user interface flow is given as schematic.

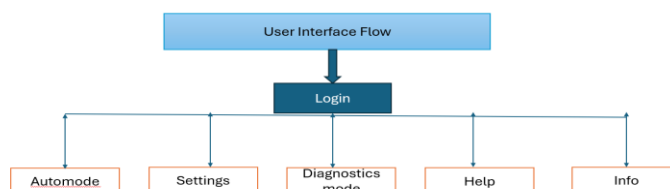


Figure 15: User Interface Flow

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11.3 Main Screen

The Main Screen for ATE Tester has the following buttons as shown below. Only Login and Exit options are enabled till logged in.

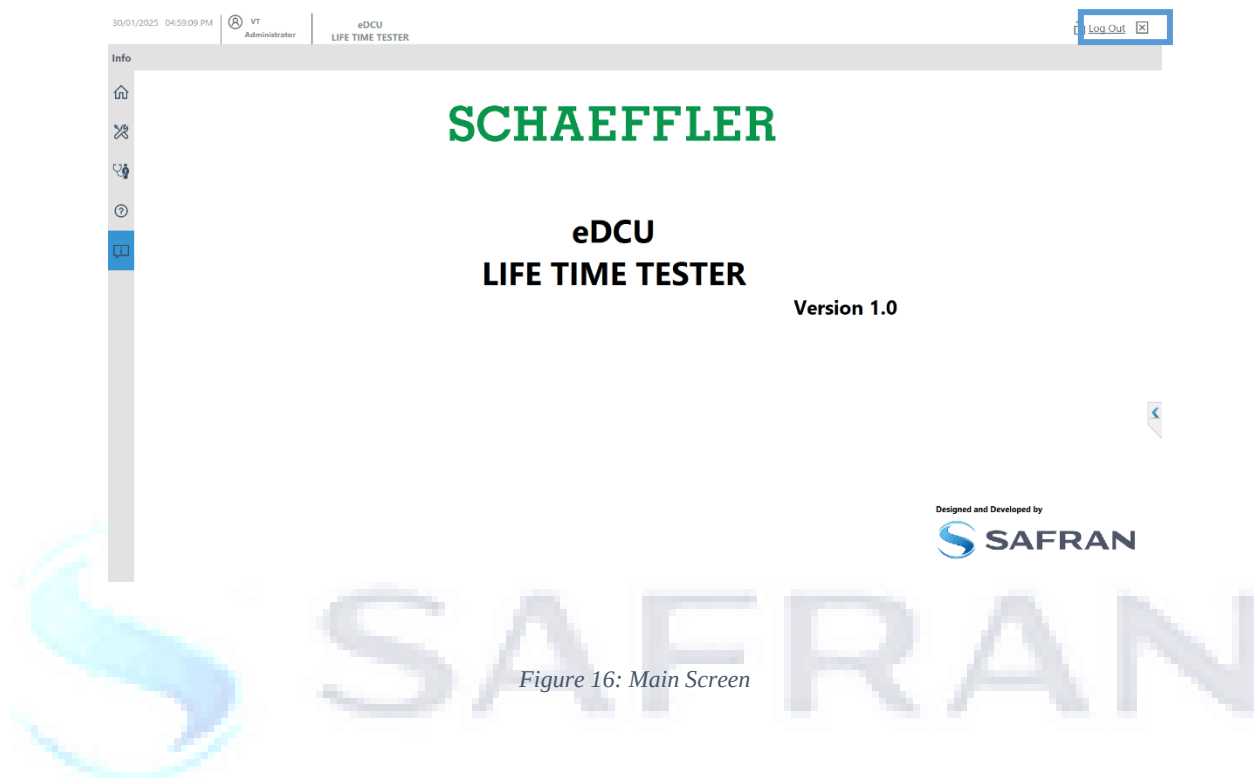


Figure 16: Main Screen

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11.4 How to Login and Manage Users

11.4.1 Login Procedure

Once you launch the application the automatically login screen will appear. In case it didn't appear you can login by clicking the "Login" button in main screen (Fig: 17), below shown User Login window will appear.

- 1) Enter the Username - "VT" (default)
- 2) Enter the Password - "VT" (default)
- 3) Click the Sign In button.
- 4) Finally click the Exit Button.

Figure 17: Login Panel

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11.4.2 Creating a New User

By clicking the “Add user” Button from the main screen (Fig: 18), below shown Add user Screen (Fig: 18) will appear. It will be accessed only by admin to add, delete or edit the user accounts.

- 1) Click the Add User Button.
- 2) Enter the Username
- 3) Enter the Password.
- 4) Confirm the Password Again.
- 5) Select the appropriate profile (Operator, Specialist, Administrator) based on the required access level. Difference between all the profile is mentioned below.
 - If your profile is set to "**Operator**," you can access the following pages: Home, Help, and Info.
 - If your profile is set to "**Administrator**," you can access the following pages: Home, Help, Info, and Settings.
 - If your profile is set to "**Specialist**," you can access the following pages: Home, Help, Info, Settings, and Diagnostics.
- 6) Click the “OK” button.
- 7) Finally click the Exit Button.

The screenshot displays the 'Access Privilege' interface. At the top, a yellow header bar contains the title 'Access Privilege'. Below this is a navigation bar with icons for Logout, Add User, Find User, Delete User, Change User, and Exit. The 'Add User' button is highlighted with a green callout '1'. The main form area contains the following fields: 'User Name' (with '123' entered), 'Password' (with '***' entered), 'Confirm Password' (with '***' entered), and 'Profile' (a dropdown menu set to 'Operator'). The 'Password' and 'Confirm Password' fields are grouped with a green callout '3&4'. The 'Profile' dropdown is highlighted with a green callout '5'. Below the form is an 'Ok' button highlighted with a green callout '6'. At the bottom of the screen, a status bar shows 'Login Successful!'. A green callout '7' points to the 'Exit' button in the navigation bar.

Figure 18: Add User Screen

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11.4.3 Change User / Admin Password

The admin password can be changed by using the existing password. To change the password, select the user and click “Change Password” Button in User Account Manager Window (Fig:10), below shown Change Password Screen (Fig: 19) will appear.

- 1) Click the Change User Button.
- 2) Enter the Username.
- 3) Enter the Current Password.
- 4) Enter the New Password.
- 5) Enter the Confirm the Password.
- 6) Click the Change Button.
- 7) Finally click the Exit Button.

Figure 19: Change User / Admin password

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11.4.4 Delete Username from List

The admin is a default user ID which cannot be deleted from the user list. To delete the user details, select the user and click “Delete User” button below shown Account Manager Screen (Fig: 20). The Confirmation screen will appear, Click Ok to confirm for delete the selected user details.

- 1) Click the Delete User Button.
- 2) Select the User details.
- 3) Click the Delete User Button.
- 4) Finally Click the Exit.

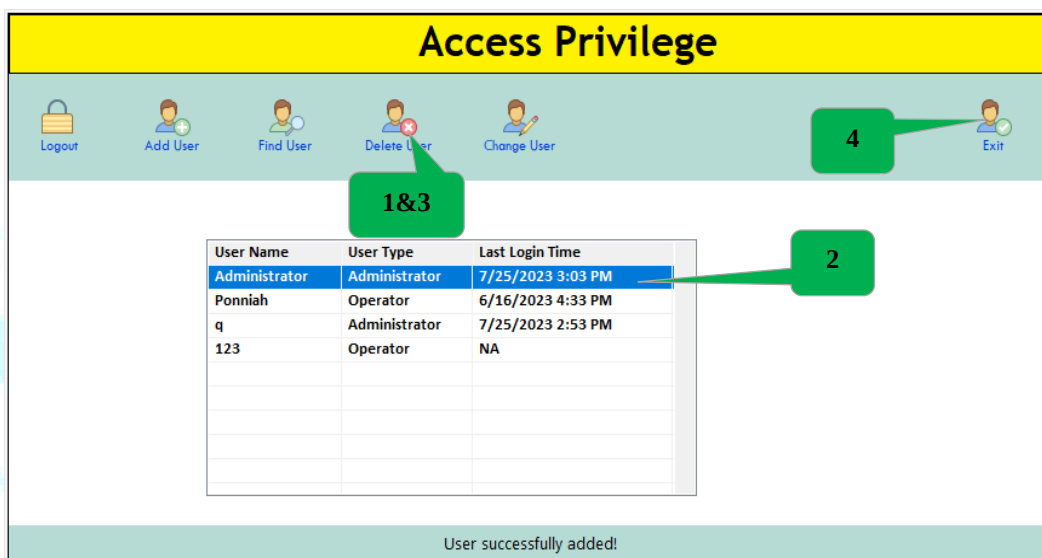


Figure 20: Delete username from the list.

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11.4.5 Find Username from List

The Find the from the user list. To Find the user details for Username, User Type and Last Login time. The Confirmation screen will appear Fig 21, Click Ok to confirm for delete the selected user details.

- 1) Click the Find User Button.
- 2) Select the User details.
- 3) Finally Click the Exit.

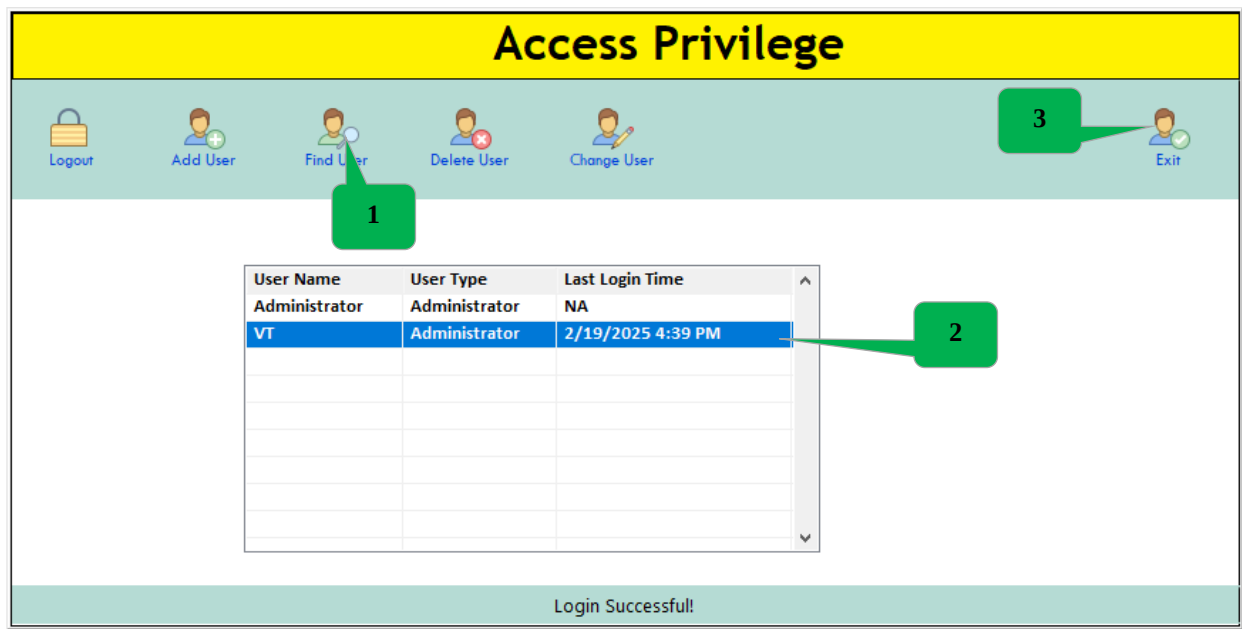


Figure 21: Find username from the list.

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11.5 Procedure to Change System Configuration

Clicking the 'Settings' button from the main window (Fig. 16) will open the window shown below (Fig. 22). Select the schema and table names in the respective fields, then choose 'Import' or 'Export' to apply the changes. This module imports and exports data based on the selected schema and table files.

- 1) Click the Setting Screen.
- 2) Click the Database Configuration Tab Button.
- 3) Select the List the Schema Name.
- 4) Select the List of the Table Name.
- 5) Click Export and Import the File(.CSV).
- 6) Click the Refresh button and show the data in Window.

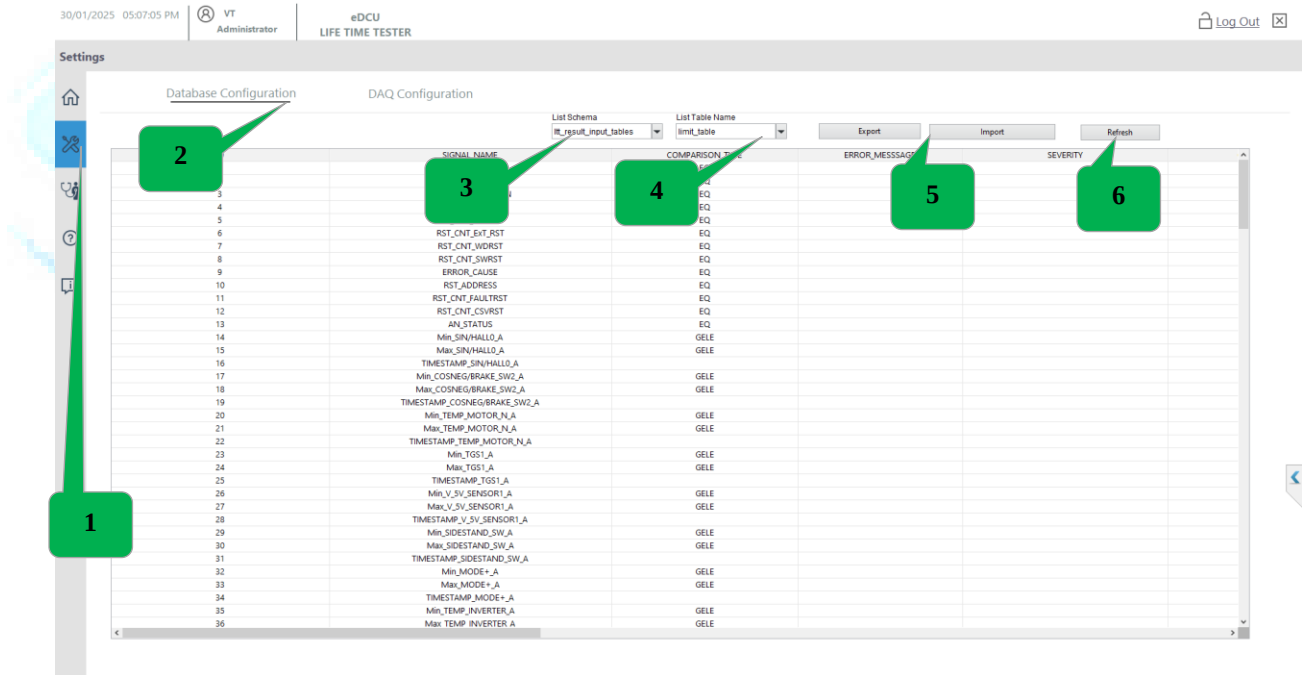


Figure 22: Settings Screen

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11.5.1 Procedure for Import data to change existing configuration

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Click the Database configuration Tab Button.
- 3) Select the List of the schema.
- 4) Select the List of the Table Names.
- 5) Click the Import Button.
- 6) Choose the respective path and select the file(.CSV).
- 7) Selected file shows the file name.
- 8) Click the OK button

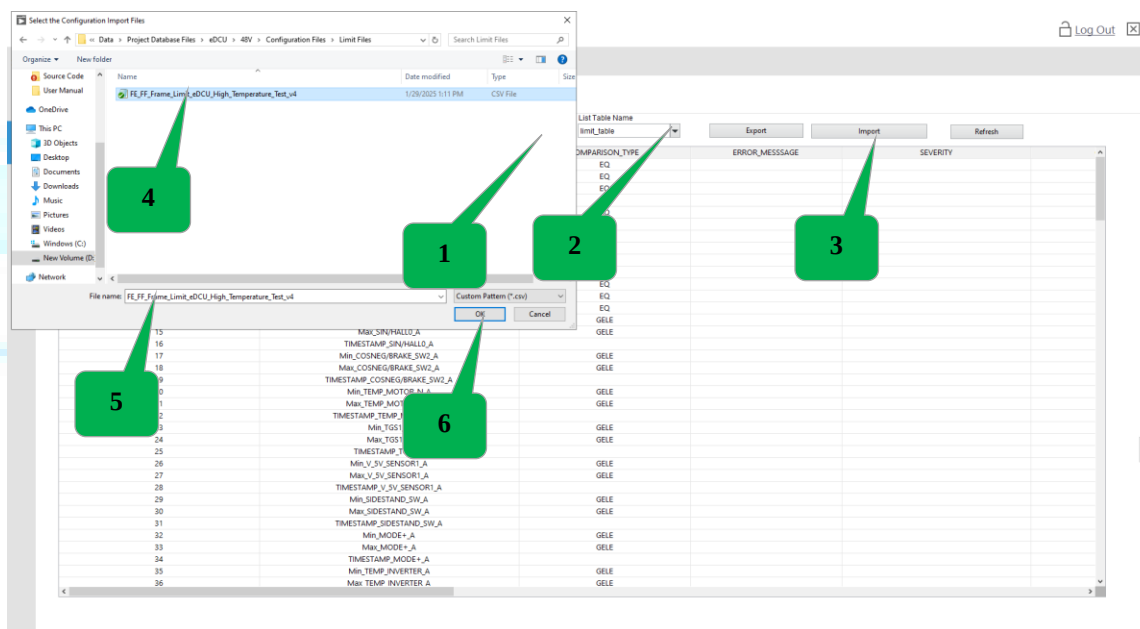


Figure 23: Setting Import Data

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11.5.2 Procedure Export data in .csv file

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Click the Database configuration Tab Button.
- 3) Select the List of Schema List.
- 4) Select the List of Table List.
- 5) Click the Export button.
- 6) Select the Folder and Enter file name.
- 7) Show the File name(.CSV).
- 8) Click the OK button.

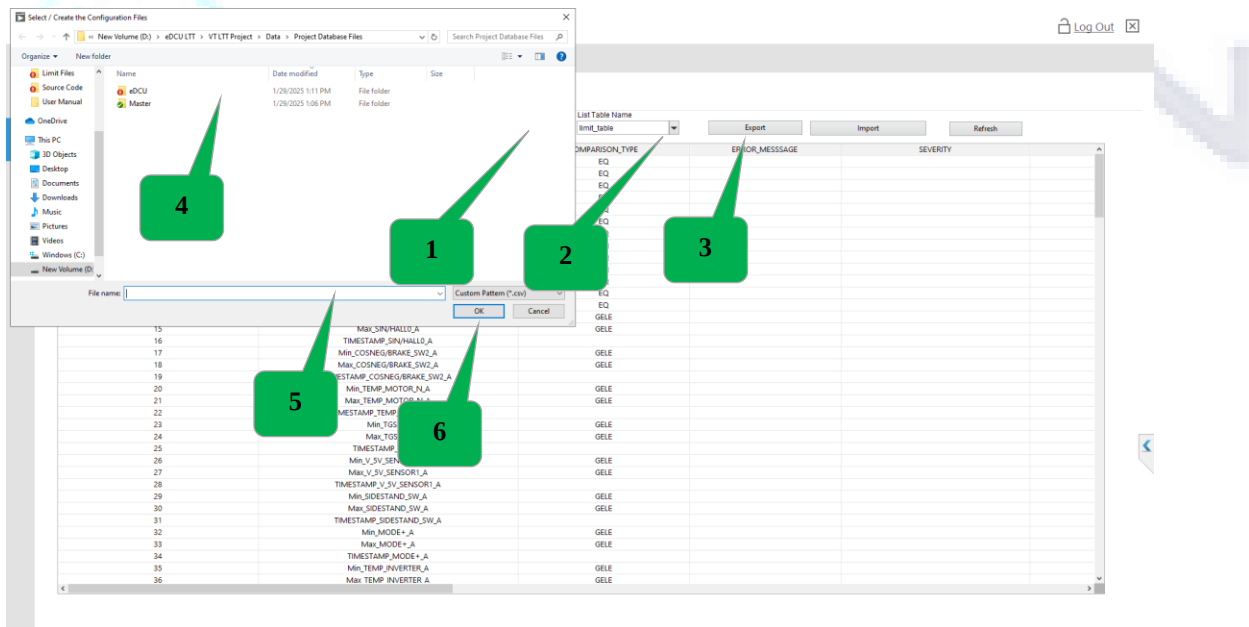


Figure 24: Settings Export Data

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11.5.3 Procedure the Setting Screen Update.

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Click the Database configuration Tab Button.
- 3) Select the List of Schema Names.
- 4) Select the List of Table Names.
- 5) Click the Refresh button.
- 6) Show the Data for Selected the Schema and Tables Name

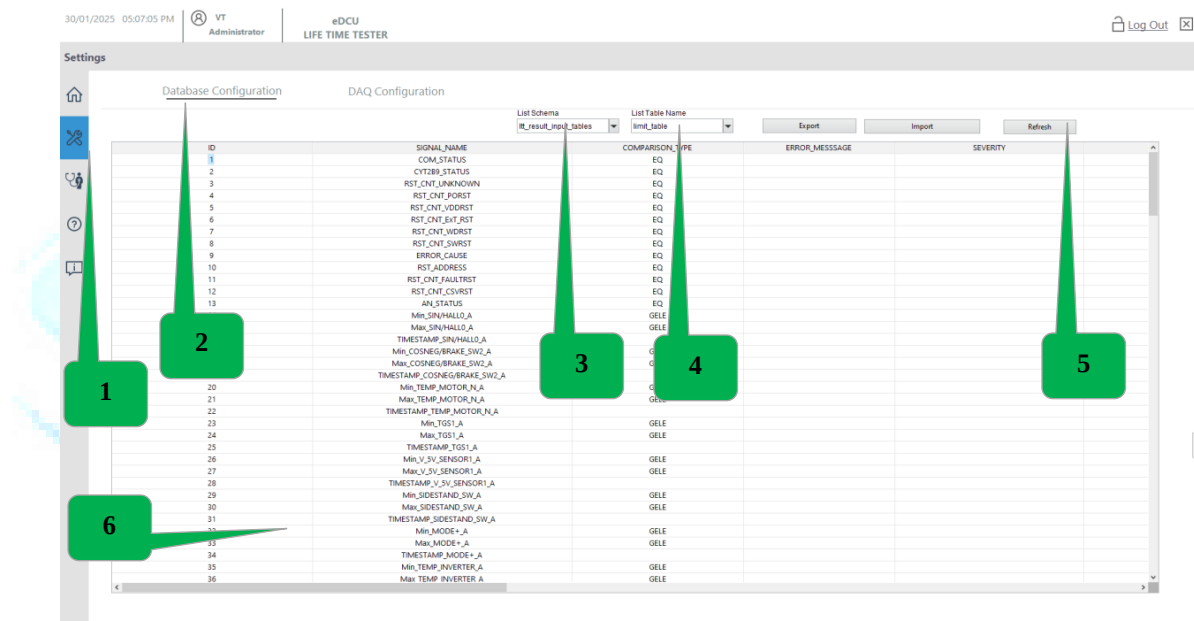


Figure 25: Settings Screen Update View

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11.5.4 Procedure the Setting DAQ Configuration Screen Update.

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Click the DAQ configuration Tab Button.
- 3) Select the List of DUT ID.
- 4) Select the List of Channel Type.
- 5) **Rename the Signal Name** as per the required specifications.
- 6) Provide options to **Enable or Disable** signals.
- 7) Click the **Refresh button** to update the status.
- 8) Ensure that the **Renamed Signal Name**, along with the **Enable and Disable options**, are properly updated.
- 9) Display the **data for the selected DUT ID and Channel Types** for reference.

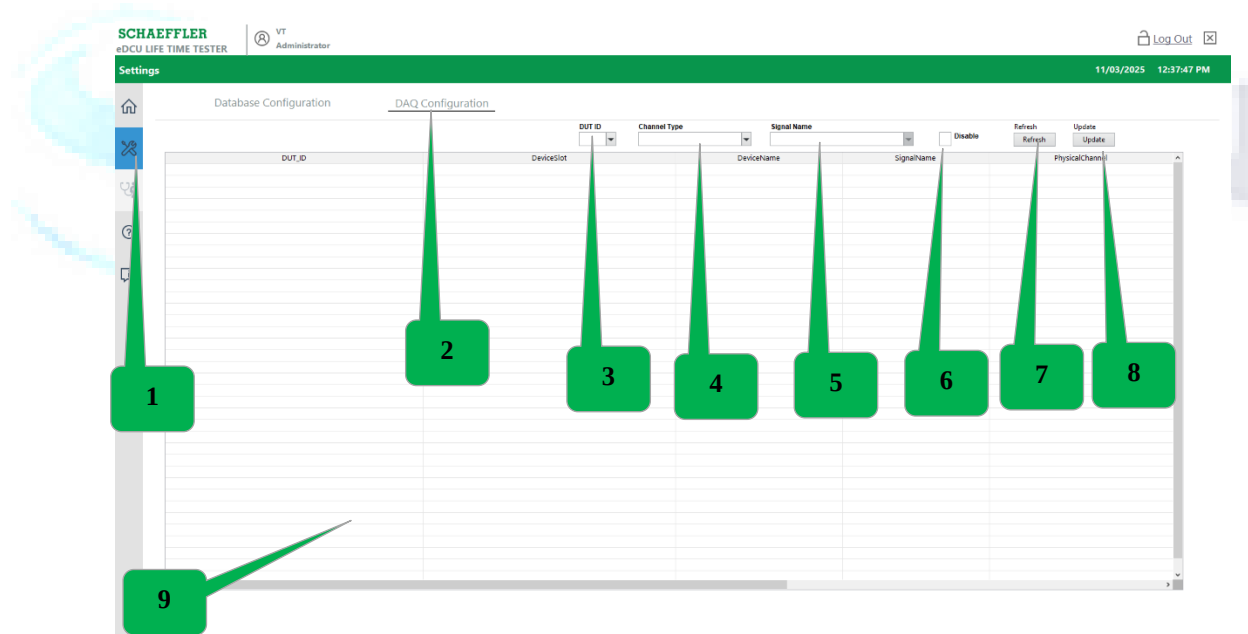


Figure 26: DAQ Configuration Settings Screen Update View

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11.5.5 Procedure the Select the DUT ID and Channel Type Screen Update.

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Next, click the **DAQ Configuration** tab button to access the configuration settings.
- 3) From the available options, **select the list of DUT IDs**.
- 4) Then, choose the **list of Channel Types** that you wish to configure.
- 5) Finally, **display the data** for the selected DUT ID and Channel Types to review the current settings.

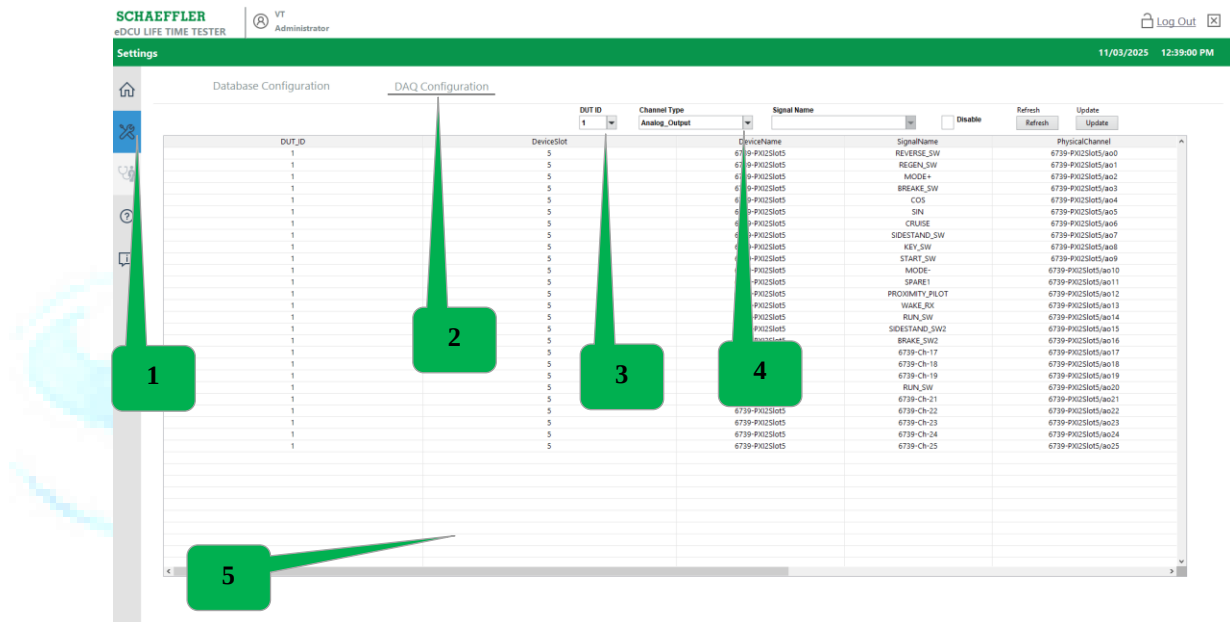


Figure 27: Select DUT ID and Channel Type Update View

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11.5.6 Procedure the Double Click the Select the Specific Row Signals.

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Next, click the **DAQ Configuration** tab button to access the configuration settings.
- 3) Select and **double-click** the **Signal Row** to view its details.
- 4) Display the **Signal Name** for the selected signal.
- 5) Show the **Enable or Disable** status for the signal.

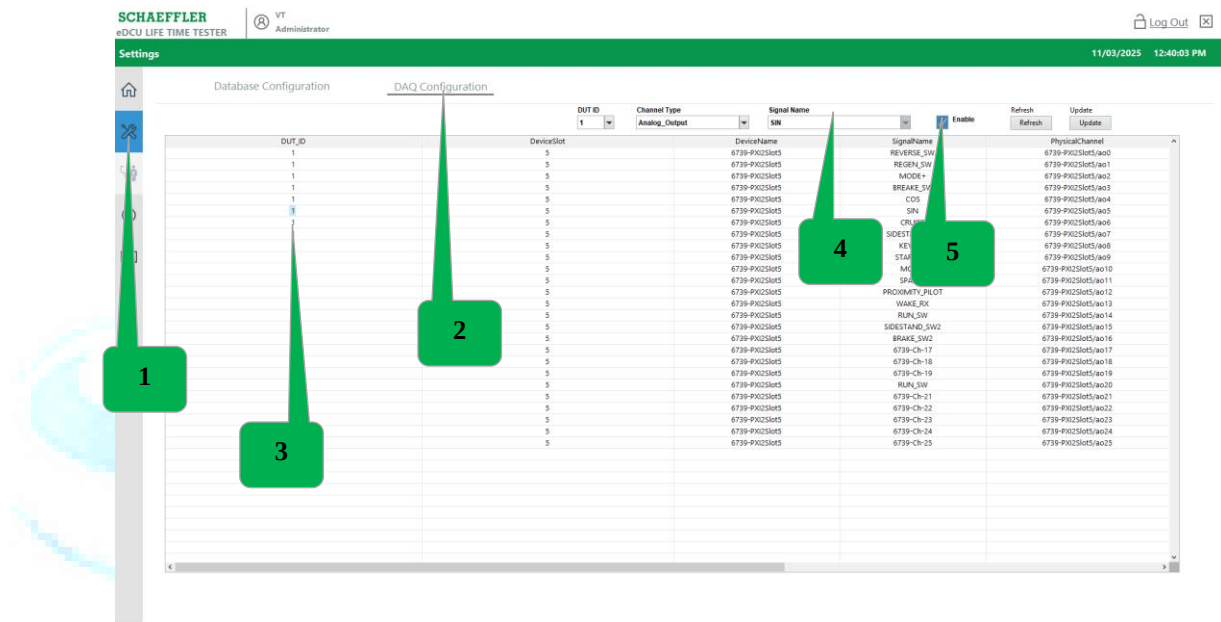


Figure 28: Select the Row Signals Update View

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11.5.7 Procedure the Update Signal Name and Enable and Disable Option.

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Click the **DAQ Configuration** tab button to access the configuration settings.
- 3) **Rename the Signal Names** as needed.
- 4) Provide the option to **Enable or Disable** the signal using the respective buttons.
- 5) After making the changes, **click the Update** button to save the modifications.

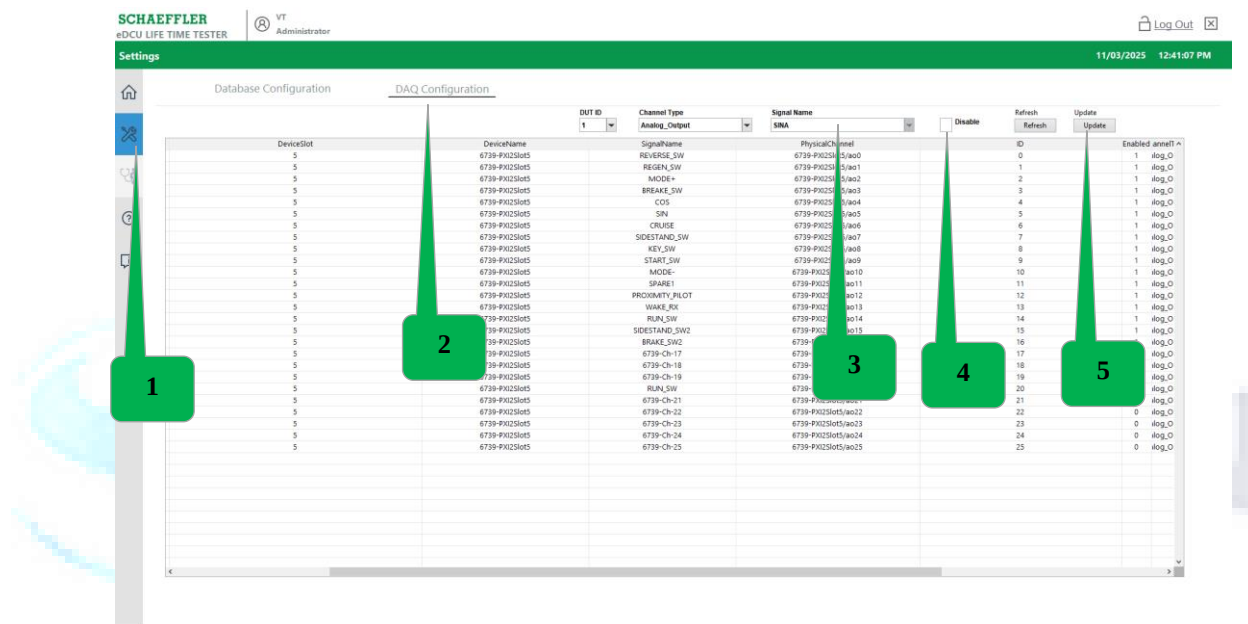


Figure 29: Settings Screen Update View

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11.5.8 Procedure the Setting Screen - Refresh the DAQ Configuration.

- 1) Click the Settings Screen button Refer the Fig 22.
- 2) Click the **Database Configuration** tab button to access the configuration settings.
- 3) **Select** the list of **DUT IDs** from the available options.
- 4) Choose the **List of Channel Types** that you need to configure.
- 5) Click the **Refresh** button to update the data.
- 6) **Display the data** for the selected **Schema and Table Names** based on the DUT ID and Channel Type.

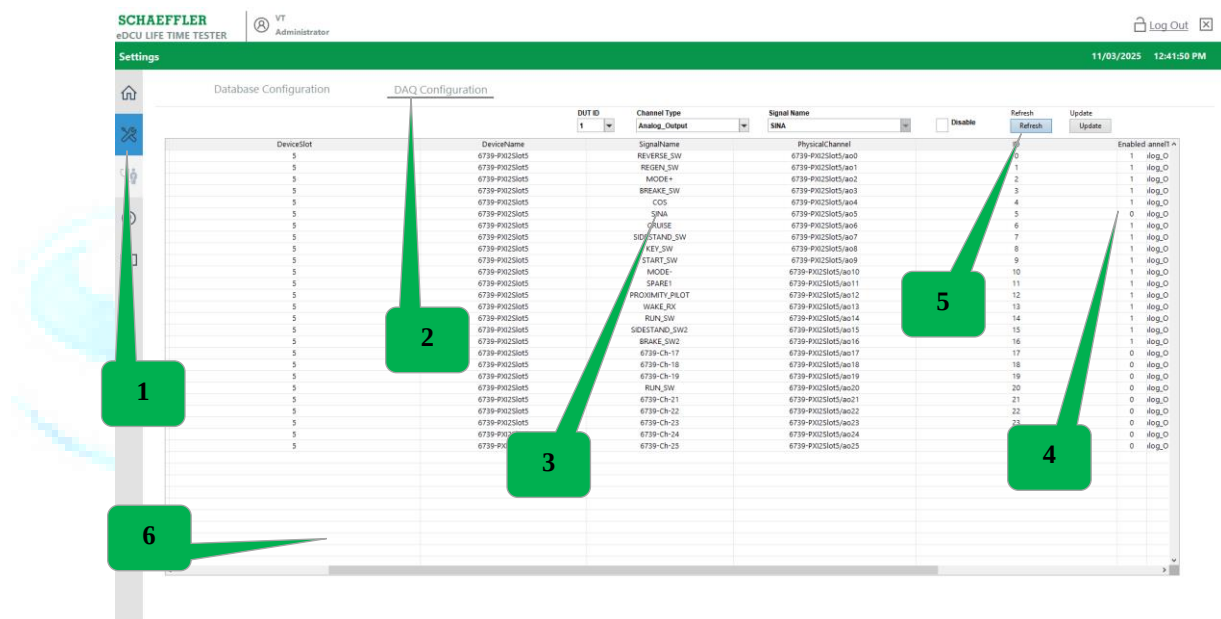


Figure 30: Refresh the DAQ Configuration

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11.6 Procedure for Auto Mode (To Start New/Existing Test)

Auto mode is used to test the test case by selecting and applying various sequences to run Refer the Figure 31.

- 1) Click the Home symbol button to enter Auto mode.
- 2) After clicking the Home button, the Test Stand Login pop-up will appear.
- 3) Click the 'OK' button (with the password field empty).

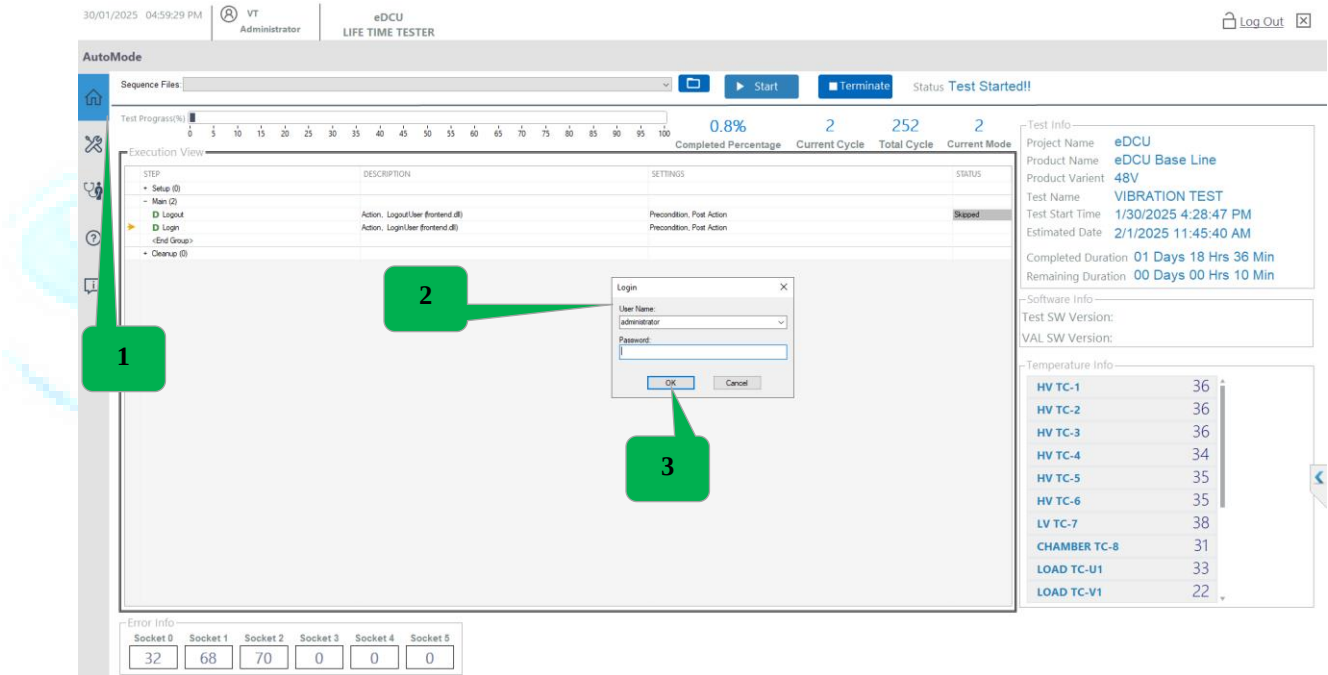


Figure 31: Auto Mode and Login Test Screen

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11.6.1 Auto Mode Details Screen

- 1) Click the Open Sequence button for Purpose and Select the Sequence file Location is D:\eDCU LTT\VT LTT Project\Client Sequence\eDCU\eDCU LifeTimeTest_VAL12.seq
- 2) Show the Selected Sequence file was loaded.
- 3) Click the Start button purpose for Test Start.
- 4) Click the Terminate button for purpose stop the test.
- 5) Status for Test running status update.
- 6) Show for Test progress update and Complete percentage, Current cycles, Total Cycles and Current Mode.
- 7) Show for Test Information in Project Name, Product Name, Product Info, Test Name, Test Start Time, Test Estimated Time, Test Completed Time and Remaining Duration.
- 8) Show for Software Information in Test SW Version and VAL SW Version.
- 9) Show for Temperature Information in Power Supply Temperature, Chamber Temperature and Load bench temperature.
- 10) Show for Execution View Windows for Sequence current step running execution View.
- 11) Show for the Error Information for socket-based failure count.
- 12) Click the Exit button to close the application entirely.

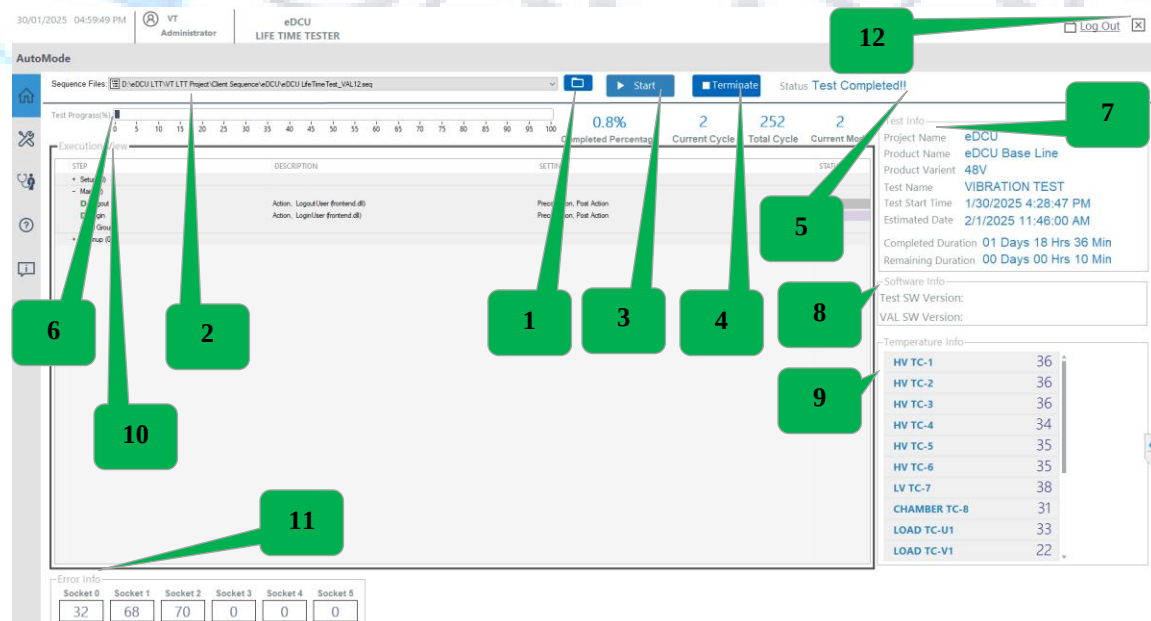


Figure 32: Auto Mode Screen User Interface Information

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11.6.2 Sequence File Loading and Start the Test.

- 1) Click the Open Sequence button and Select the Sequence file Location is

D:\eDCU LTT\VT LTT Project\Client Sequence\eDCU\eDCU LifeTimeTest_VAL12.seq

- 2) Show the Selected Sequence file was loaded.
- 3) Click the Start the Button

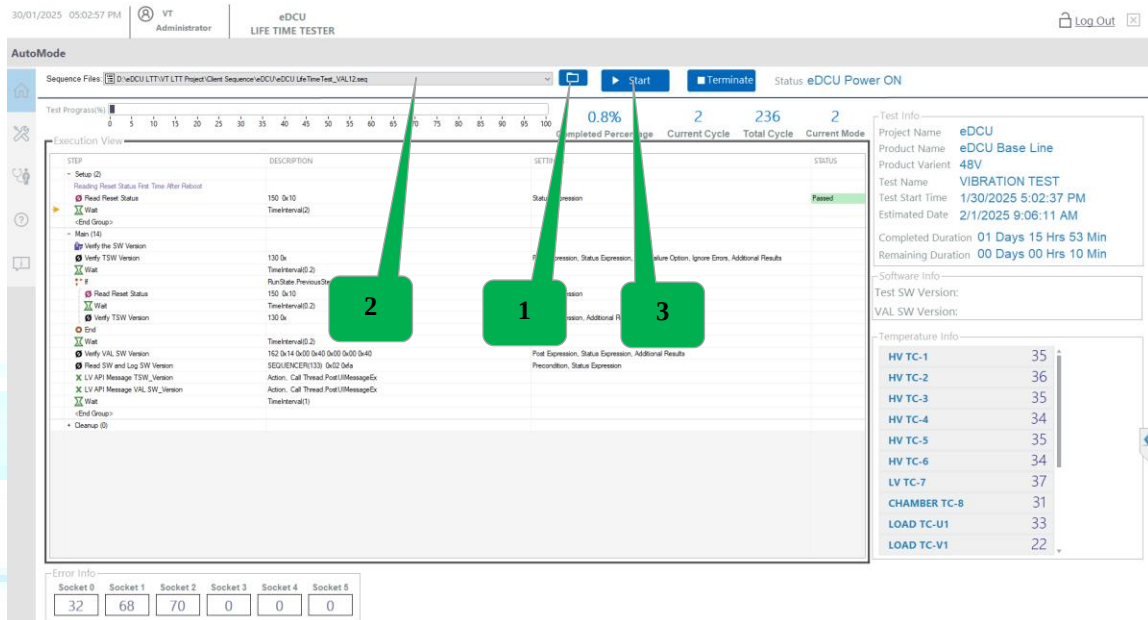


Figure 33: Sequence File Loading and Start Test

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11.6.3 Number of Socket Entry

- 1) Enter the number of sockets (eDCUs connected).
- 2) Click the OK Button.

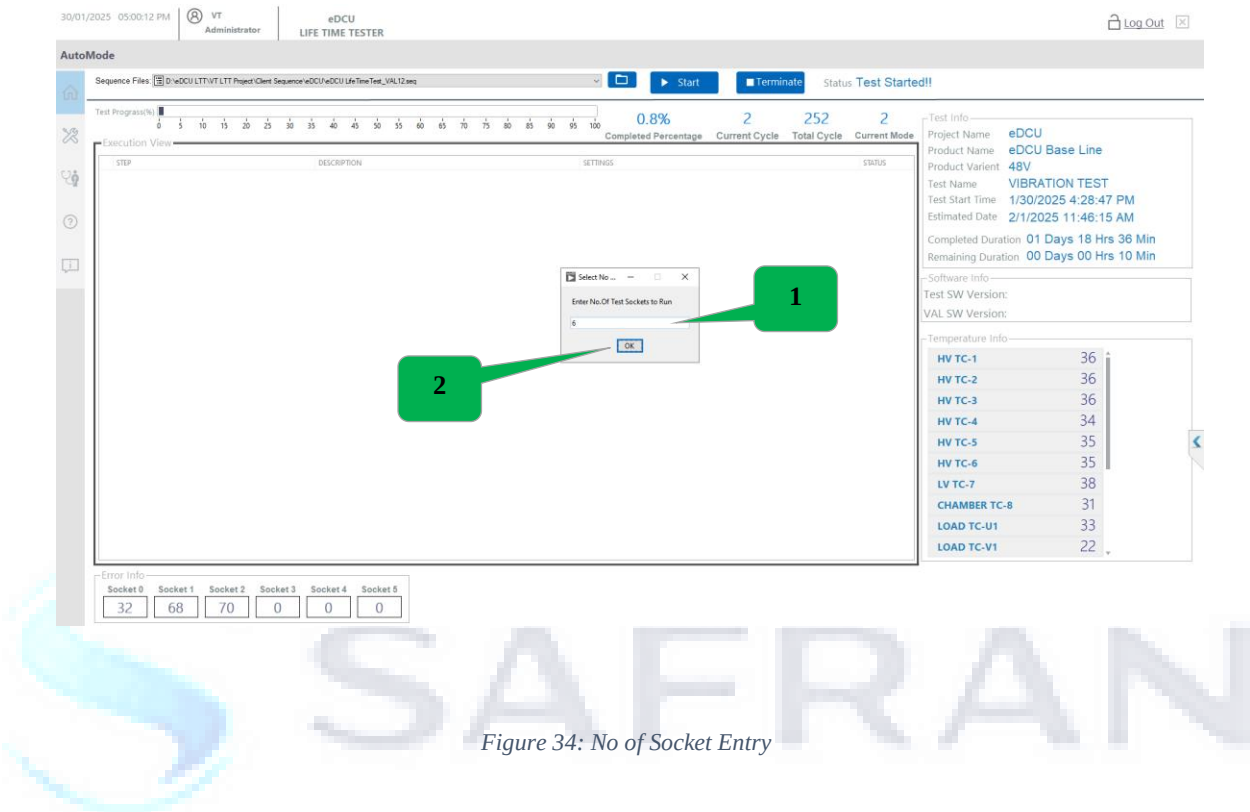


Figure 34: No of Socket Entry

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11.6.4 Serial Number Entry Screen

- 1) Enter the eDCU serial number for each socket.
- 2) Tick 'Disable Test Socket' option in case you don't want to run that socket.
- 3) Click 'GO' to proceed with the test.
- 4) Clicking 'STOP' will abort the test.

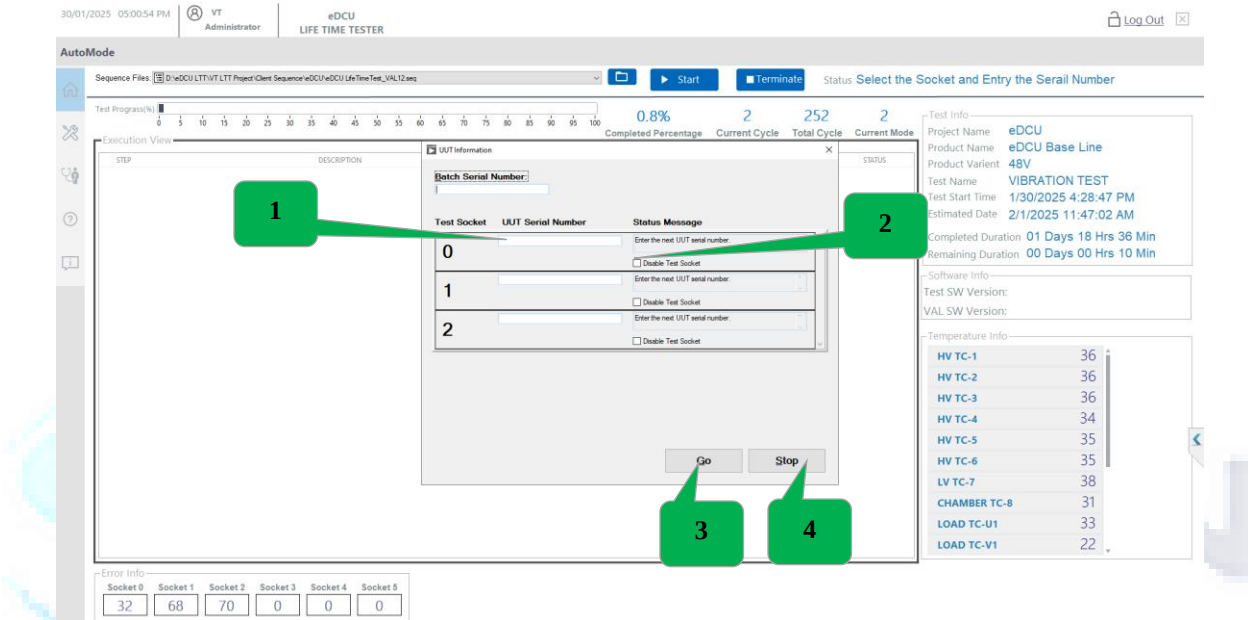


Figure 35: Serial number entry

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11.6.5 Test Selection on Screen(New Test/ Resume test)

- 1) The Test Selection window offers options to select a test, start a new test, continue the last test, or cancel.
- 2) Clicking the 'New Test' will initiate a fresh test.
- 3) Clicking 'Continuous Last Test' will resume the previous test from the last cycle.
- 4) Clicking 'Cancel' will abort the current test.

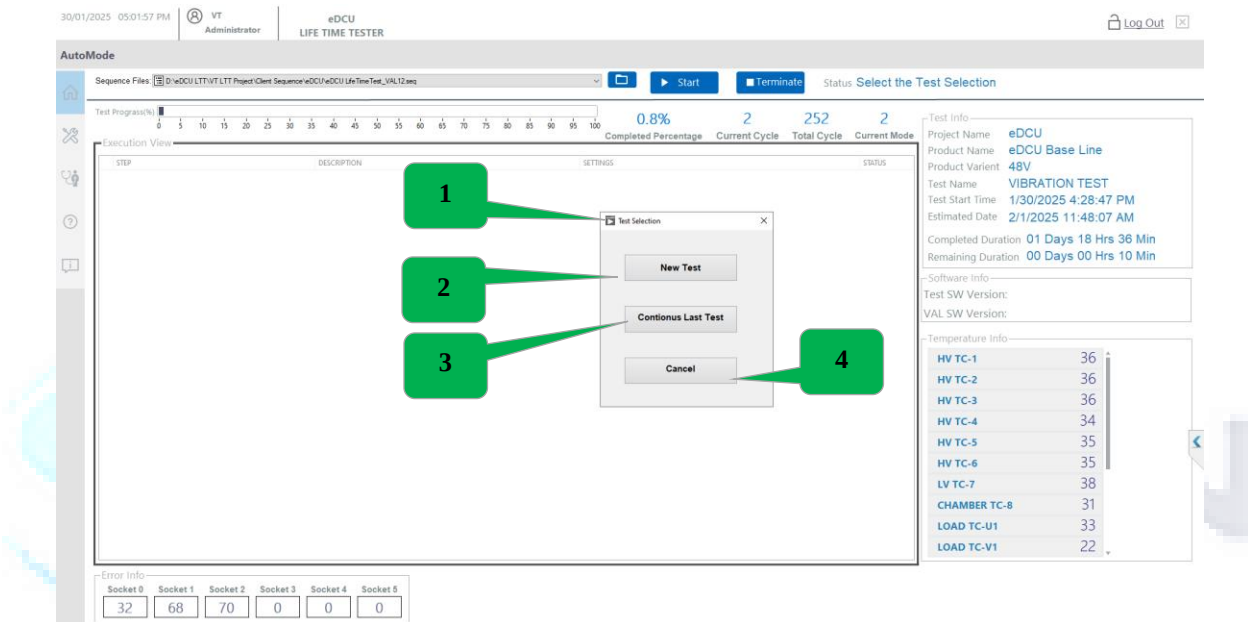


Figure 36: Test selection

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11.6.6 New Test

Follow the procedure outlined in sections 11.6.1 to 11.6.5, refer to the images provided for guidance

- 1) Enter the project name.
- 2) Select/Enter the product name.
- 3) Select/Enter the product variant.
- 4) Select the test name.
- 5) Select/Enter the test phase.
- 6) Provide the test description.
- 7) Enter the temperature chamber detail if applicable.
- 8) Verify the total number of cycles.
- 9) Click the "OK" button.

The screenshot shows the 'Test UUT Info Window' dialog box. The fields and their corresponding callout numbers are:

- 1: Project Name (eDCU Base Line)
- 2: Product Name (eDCU)
- 3: Product Variant (48V)
- 4: Test Name (HIGH TEMPERATURE LIFE TEST)
- 5: Test Phase (DV)
- 6: Test Description (Sample Trial is Runtime)
- 7: Temperature Chamber Name (Without Chamber)
- 8: # Total Cycles (4800)
- 9: OK button

The background interface shows the 'Sequence Files' list with the selected file 'D:\eDCU\LTT\VT Project\Client Sequence\eDCU LifeTimeTest_VAL12.seq'. The status bar at the bottom shows 'Completed Percentage' (0.8%), 'Current Cycle' (2), 'Total Cycle' (236), and 'Current Mode' (3).

Figure 37: New Test UUT Test Info

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11.6.7 Continuous/Resume Test

Follow the procedure outlined in sections 10.6.1 to 10.6.5, refer to the images provided for guidance

- 1) The last schema is automatically updated with test UUT information. Select from dropdown incase you wish to start another test.
- 2) Verify the Project Name.
- 3) Verify the product name.
- 4) Verify the product variant.
- 5) verify the test name.
- 6) Verify the test phase.
- 7) Provide the test description.
- 8) Enter the temperature chamber detail if applicable.
- 9) Verify the current and total cycles.
- 10) Click the "OK" button.

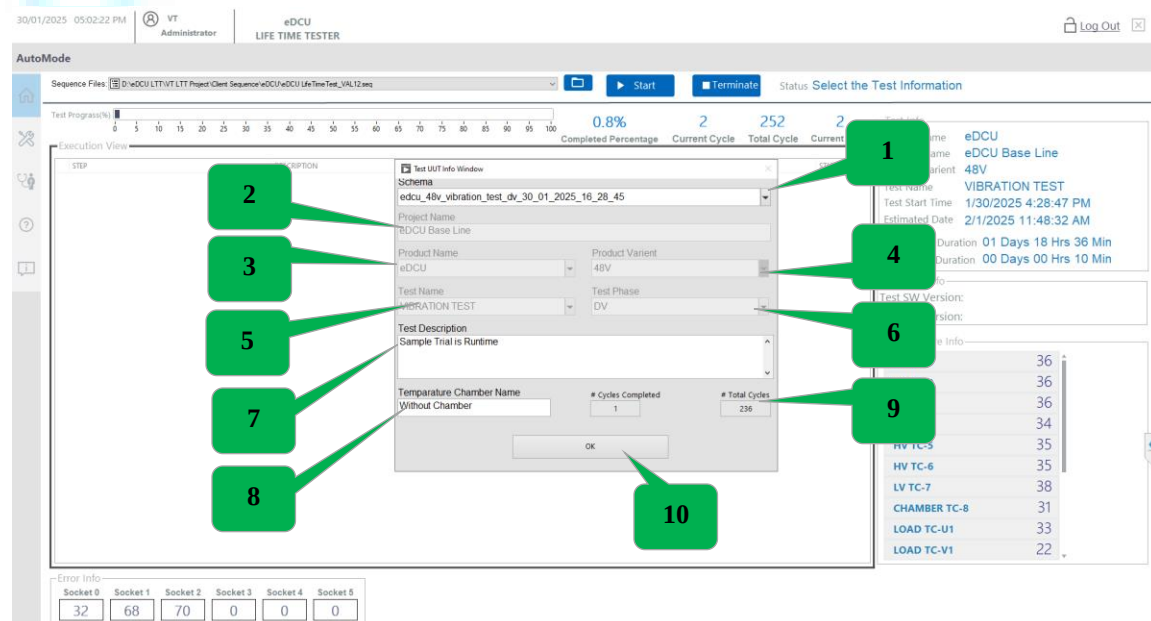


Figure 38: Continuous Test Selection

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11.6.8 Verify Sequence Running

- The test execution window is updated to display the current step being processed.
- The main rack displays the tower lamp status: Green indicates the test is running, while Red and the buzzer signal a test failure.

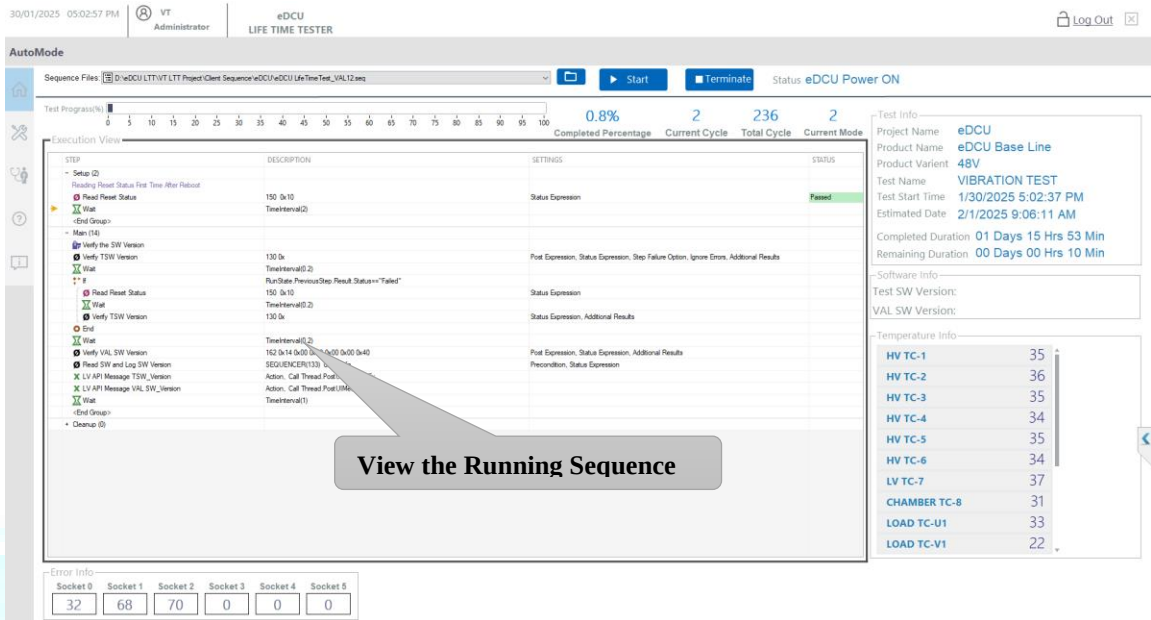


Figure 39: Sequence Running



Figure 40: Tower Lamp Status

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11.6.9 Verify Wait Monitor step while a mode is running.

- 1) Display the current state.
- 2) Display the running test mode.
- 3) Display the running cycle.
- 4) Display the current and target temperatures.
- 5) Display the wait time and remaining time.
- 6) Display the eDCU RMS current values.
- 7) Display the high voltage PPS current values.
- 8) Display the status update.
- 9) Display the raw power status.
- 10) Display the monitor state for RMS current and chamber.

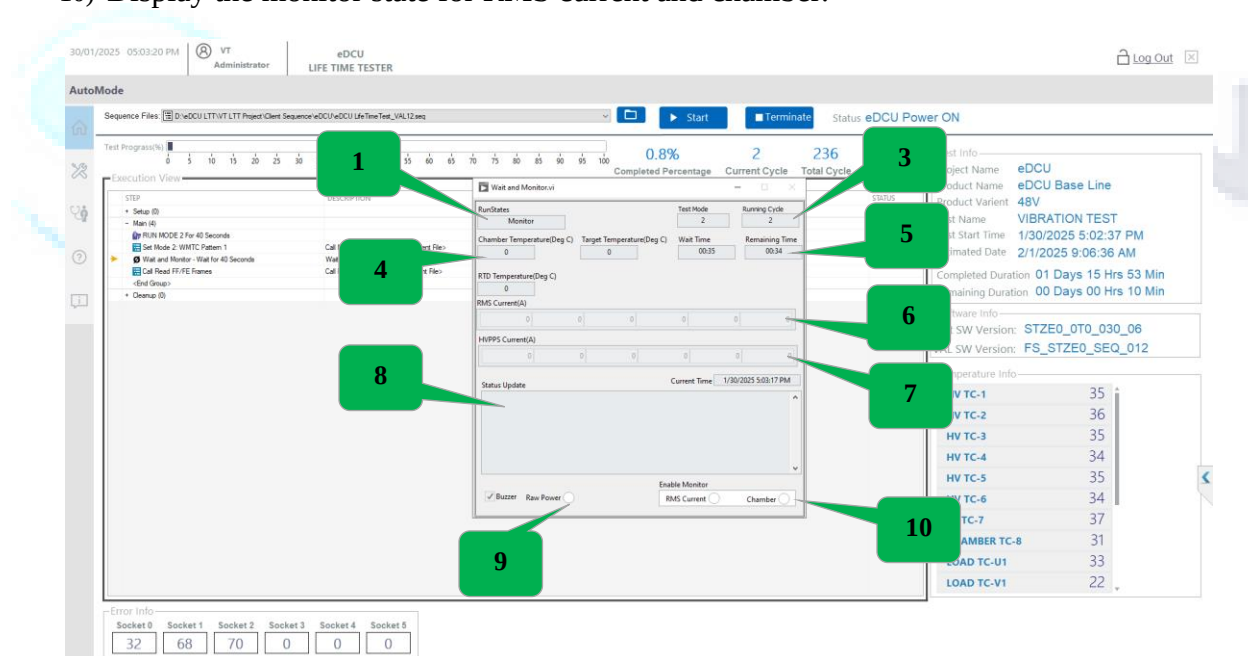


Figure 41: Wait Monitor

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11.6.10 Procedure for Terminating a Test

- 1) Click the **"Terminate" Button**: Press the **"Terminate"** button to end the test.
- 2) Confirm **Termination**: If prompted, click **"OK"** or **"Yes"** to confirm the test termination.
- 3) Verify **Test Status**: Ensure the test status reflects that it has been successfully terminated.

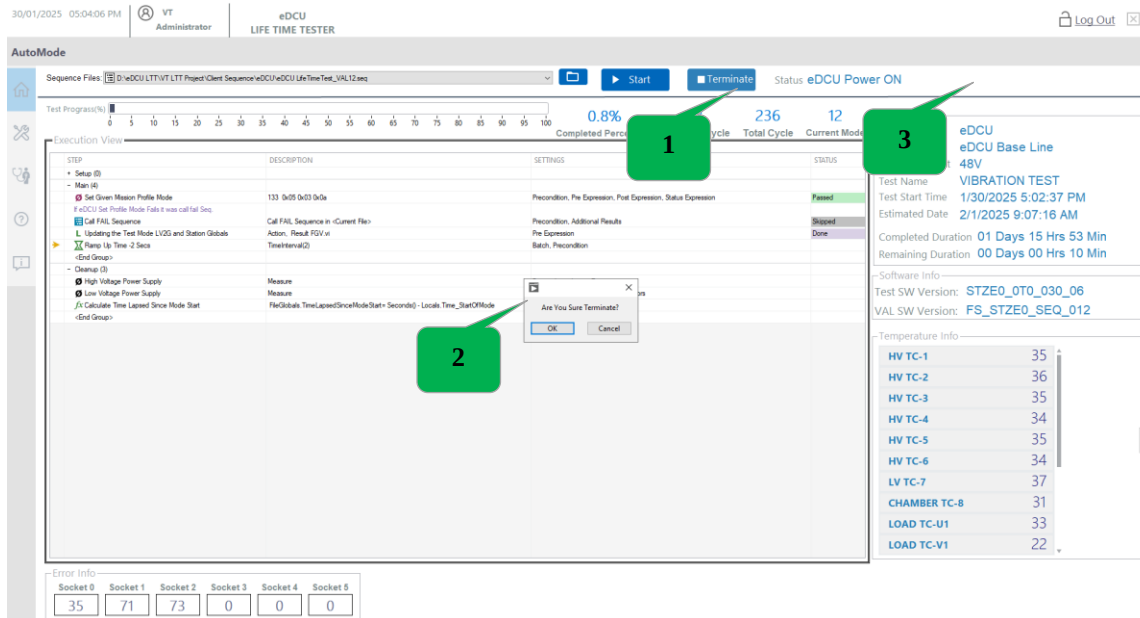


Figure 42: Test Terminate

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11.6.10.1 Verify Test Termination Completion.

- Verify **Test Status** Ensure the test status reflects that it has been successfully terminated.

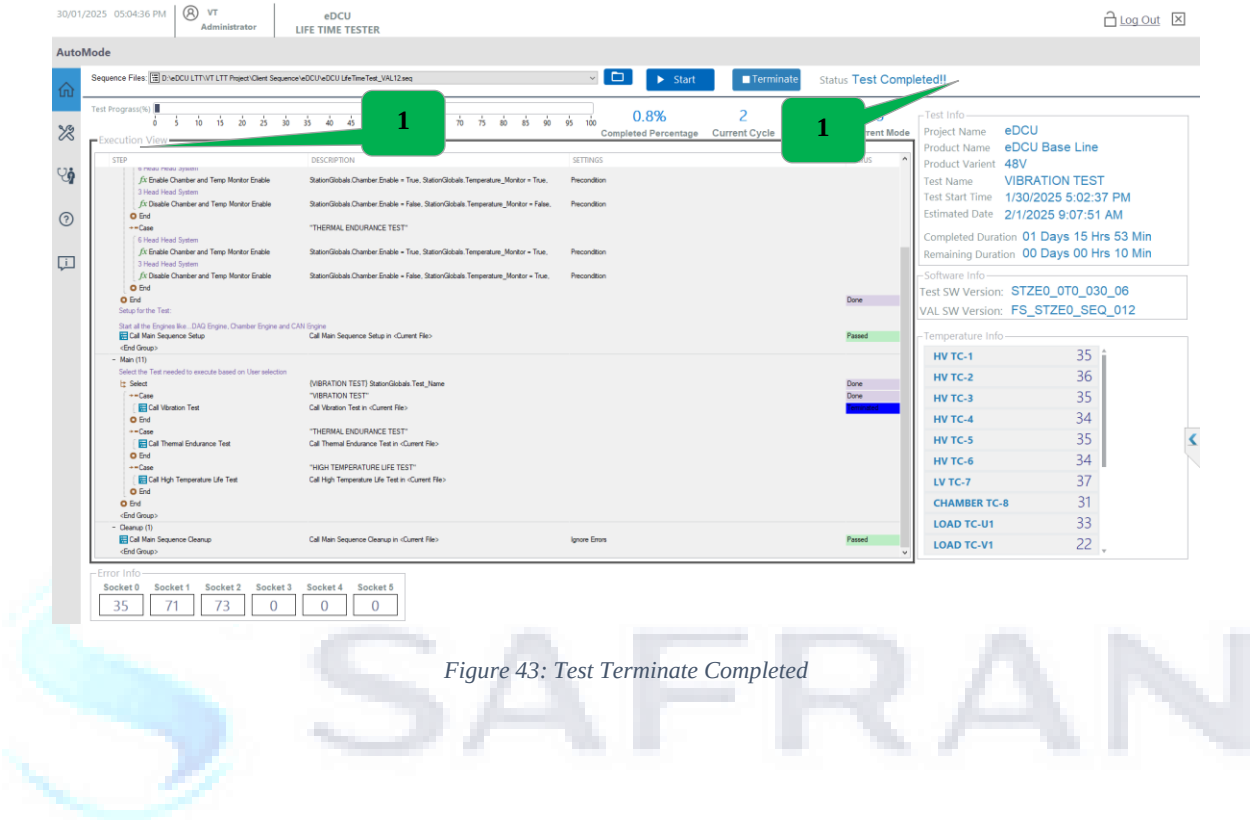


Figure 43: Test Terminate Completed

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11.6.12 Procedure for Exiting the Application.

- 1) Terminate the test incase test is running.
- 2) Click the 'Exit' button.
- 3) Click the OK to confirm the exit or Cancel button if it was by mistake

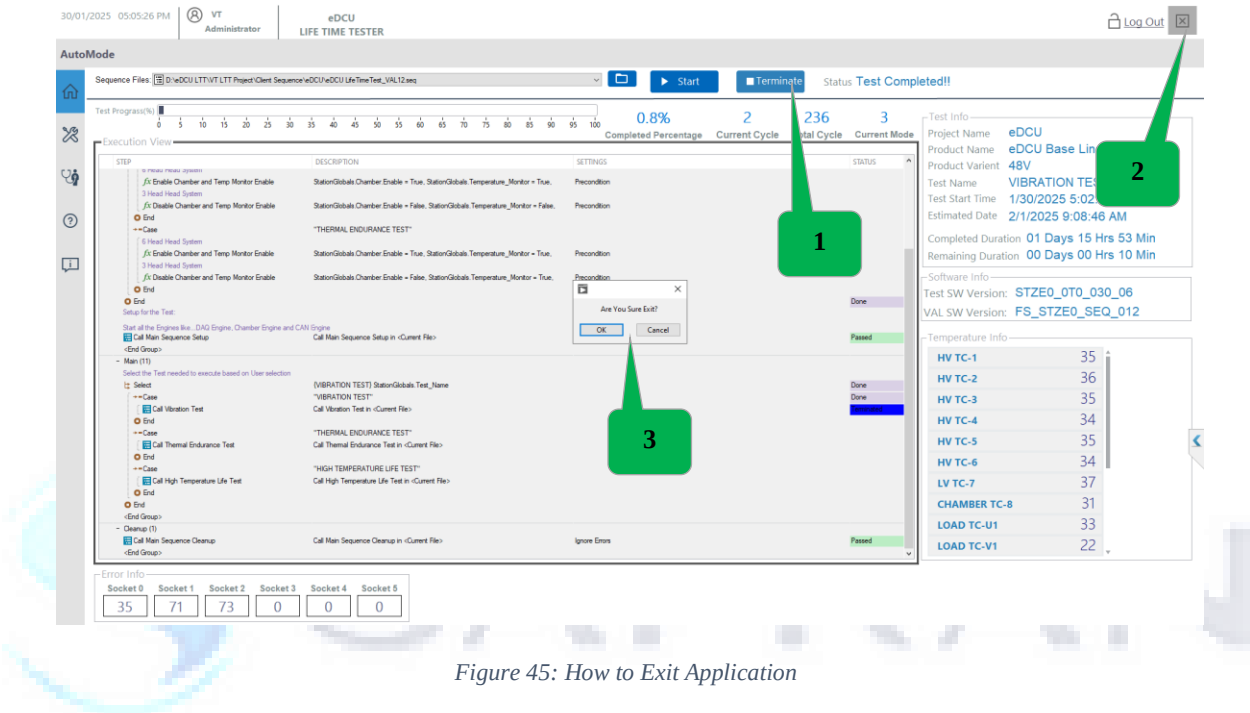


Figure 45: How to Exit Application

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11.7 Loopback test screen

Loopback signals are used to identify and verify the signals for Analog Input and Output, PWM Input and Output, as well as Digital Input and Output.

- 1) Use the Loopbox to connect to the Harting connection as shown in the figure 46.
- 2) To start the test procedure for the Loopbox, refer to Section 11.6.
- 3) Click the Open Sequence button for Purpose and Select the 6 Head Sequence file
Location is D:\eDCU LTT\VT LTT Project\Client Sequence\Selftest\6 Head/ LTT Self Test -Box 6 Head.seq
- 4) Click the Open Sequence button for Purpose and Select the 3 Head Sequence file
Location is D:\eDCU LTT\VT LTT Project\Client Sequence\Selftest\3 Head/ LTT Self Test -Box 3 Head.seq



Figure 46: Loopback Box

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11.8 Diagnostic test screen

Diagnostic screens are used for identifying and troubleshooting issues. Refer the Figure 47.

- 1) Click the Diagnostic symbol Button.
- 2) Show the Diagnostic Screen window and Diagnostic Instruments.
- 3) Digital Input Button.
- 4) Digital Output Button.
- 5) Relay Output Button.
- 6) Analog Input Button.
- 7) Analog Output Button.
- 8) PWM Input Button.
- 9) PWM Output Button.
- 10) CAN Diagnostic Button.
- 11) PPS Diagnostic Button.
- 12) Temperature Controller Diagnostic Button.

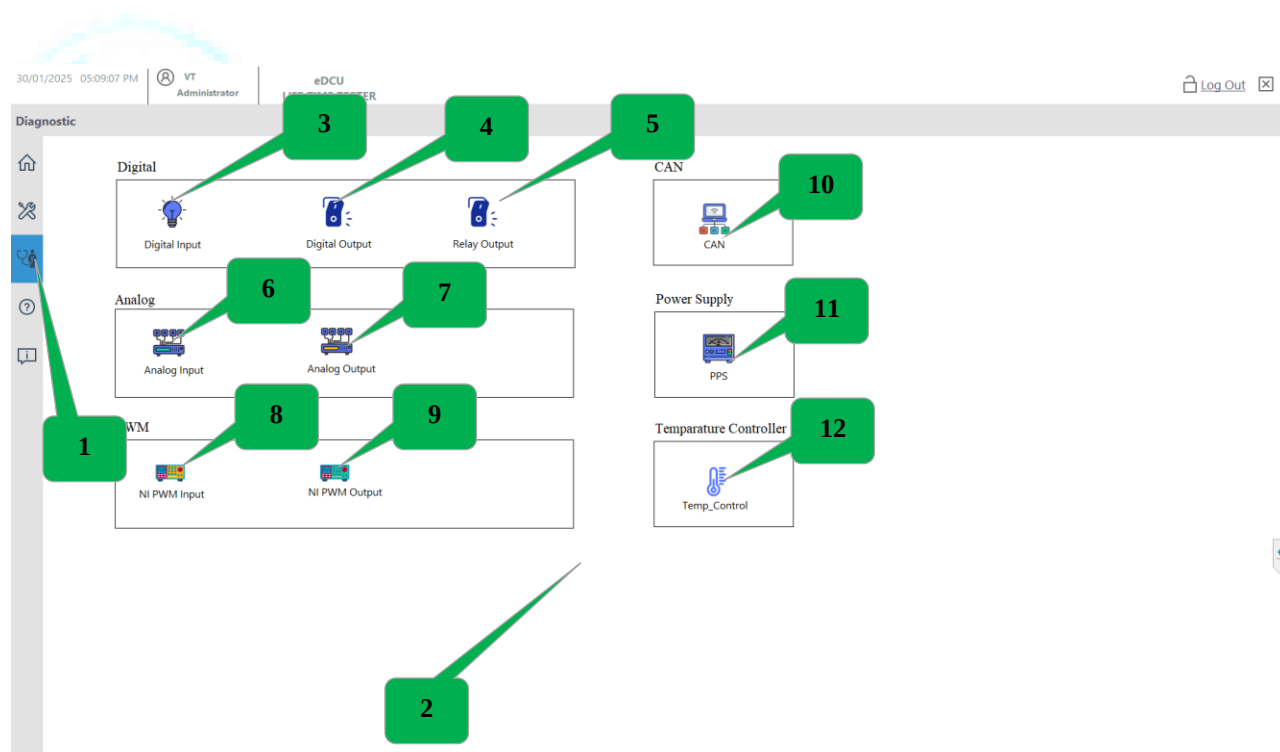


Figure 47: Diagnostic Test Screen

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11.8.1 Digital Input Diagnostic Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the Digital Input Diagnostic Button and Refer Fig 48.
- 3) Select the Operation for this NI Digital Input Card Selection.
- 4) Click the Start Button use Signal Acquire the Digital Input Signals.
- 5) Show the Digital Input status (ON or OFF).
- 6) If need the stop the acquire the Digital Signal Click Stop Button.
- 7) Finally Click the Close Button.

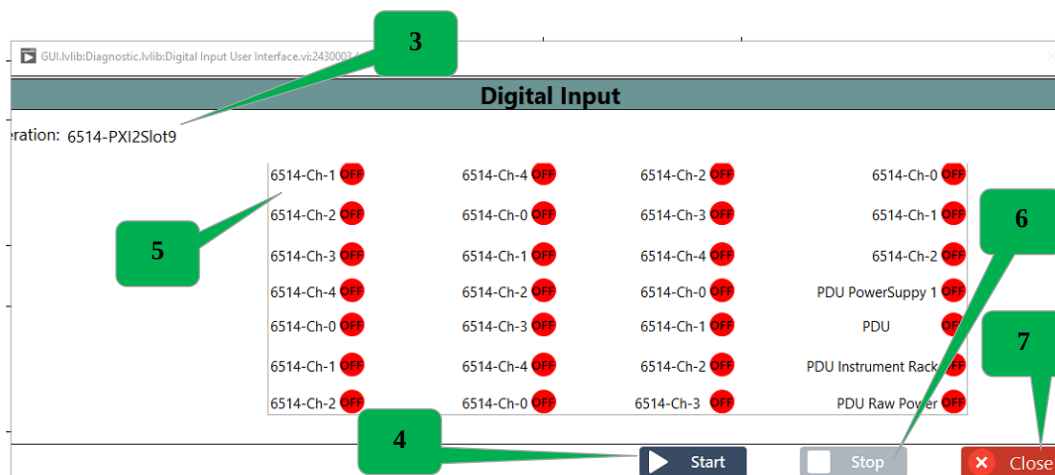


Figure 48: Digital Input

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11.8.2 Digital Output Diagnostic Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the Digital Output Diagnostic Button and Refer Fig 49.
- 3) Select the Operation for this NI Digital Output Card Selection.
- 4) Click the Start Button use Signal Write the Digital Output Signals.
- 5) Enable Digital Output Control (ON or OFF) as needed.
- 6) Click the Write Button
- 7) If need the stop the Control the Digital Signal Click Stop Button.
- 8) Finally Click the Close Button.

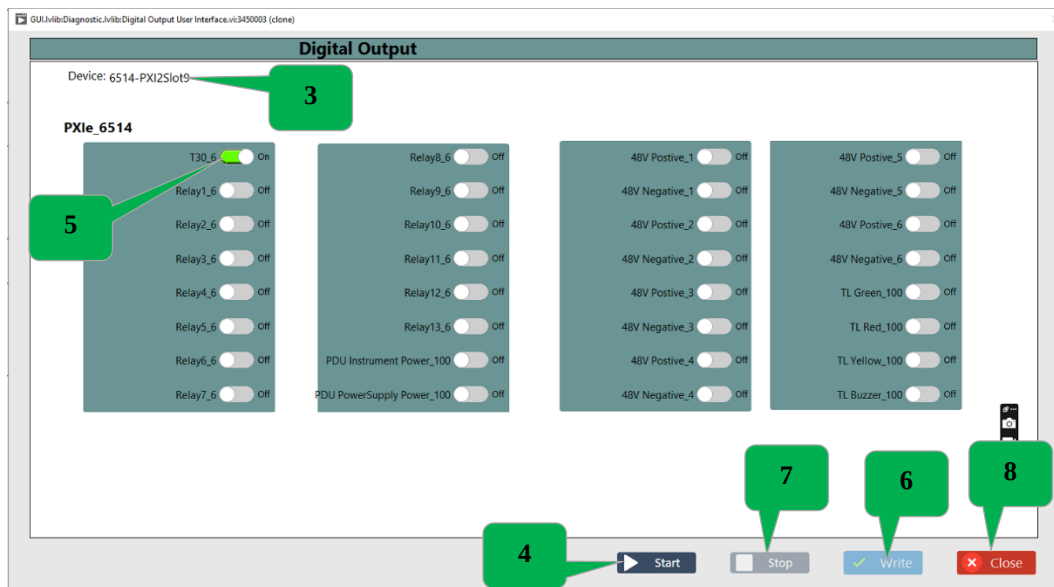


Figure 49: Digital Output

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11.8.3 Relay Output Diagnostic Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the Relay Output Diagnostic Button and Refer Fig 50.
- 3) Select the Operation for this NI Relay Output Card Selection.
- 4) Click the Start Button use Signal Write the Digital Output Signals.
- 5) Enable Digital Output Control (ON or OFF).
- 6) Click the Write Button
- 7) If need the stop the Control the Digital Signal Click Stop Button.
- 8) Finally Click the Close Button.

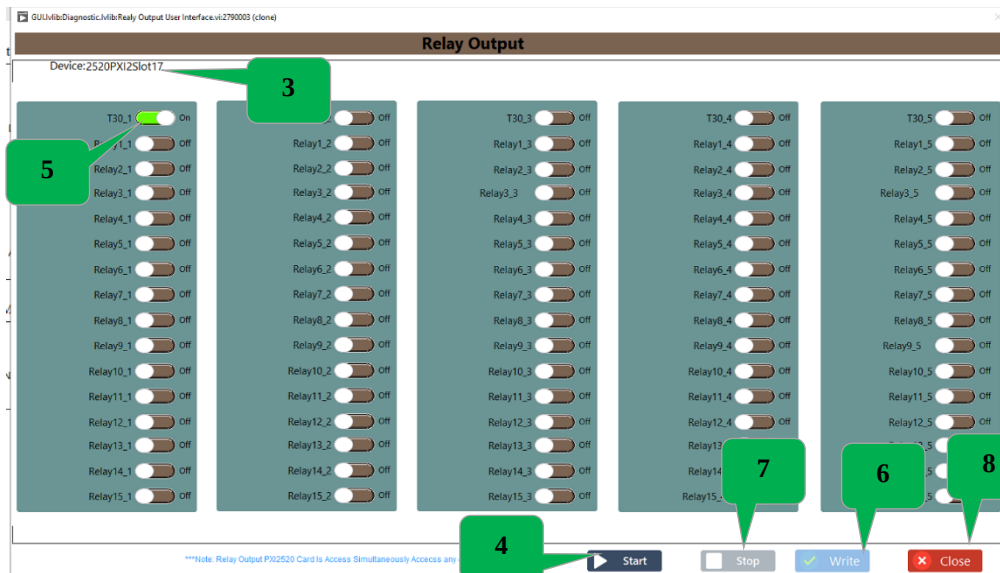


Figure 50: Relay Output

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11.8.4 Analog Input Diagnostic Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen
- 2) Click the Analog Input Diagnostic Button and Refer Fig 51.
- 3) Select the Operation for this NI Analog Input Card Selection.
- 4) Click the Start Button to use Signal Acquire the Analog Input Signals.
- 5) Show the Analog Input Read Data.
- 6) If need the stop the acquire the Analog Signal Click Stop Button.
- 7) Finally Click the Close Button.

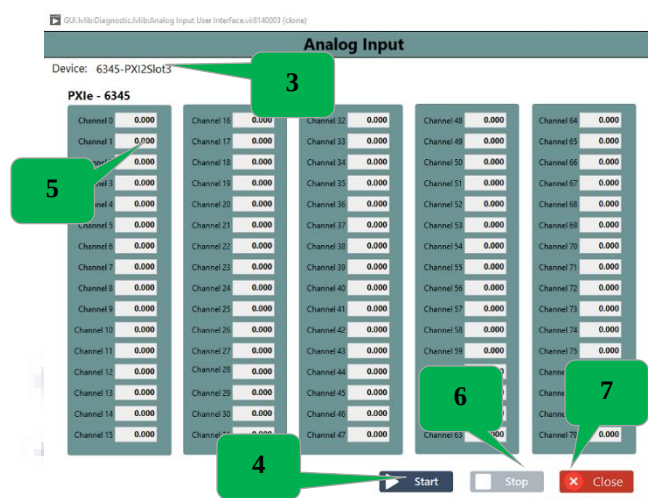


Figure 51: Analog Input

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11.8.5 Analog Output Diagnostics Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the Analog Output Diagnostic Button and Refer Fig 52.
- 3) Select the Operation for this NI Analog Output Card Selection.
- 4) Click the Start Button to use Signal Write the Analog Output Signals.
- 5) Entry Analog Output Write Data.
- 6) Click the Write Button
- 7) If need the stop the Control the Analog Signal Click Stop Button.
- 8) Finally Click the Close Button.

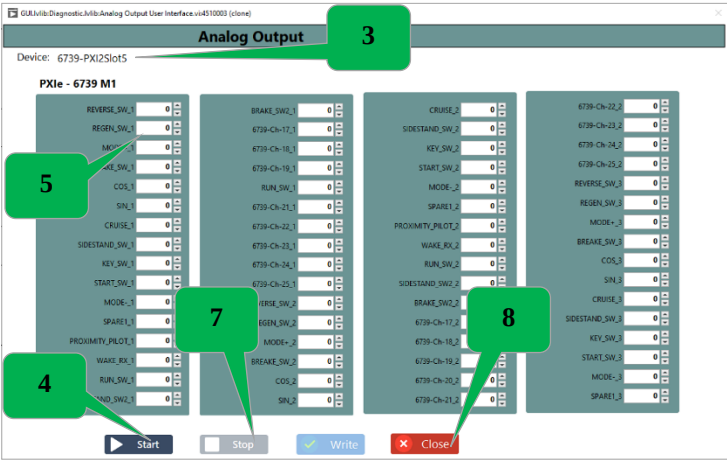


Figure 52: Analog Output

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11.8.6 PWM Input Diagnostics Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PWM Input Diagnostic Button and Refer Fig 53.
- 3) Select the Operation for this NI PWM Input Card Selection.
- 4) Select the Terminal Configuration.
- 5) Tick the Enable or Disable the Channel.
- 6) Select the Physical Channel.
- 7) Click the Start Button to use Signal Acquire the PWM Input Signals.
- 8) Show the PWM Input Read Data Show the Frequency and Duty Cycles in Max 0.99.
- 9) If need the stop the acquire the PWM Signal Click Stop Button.
- 10) Finally Click the Close Button.

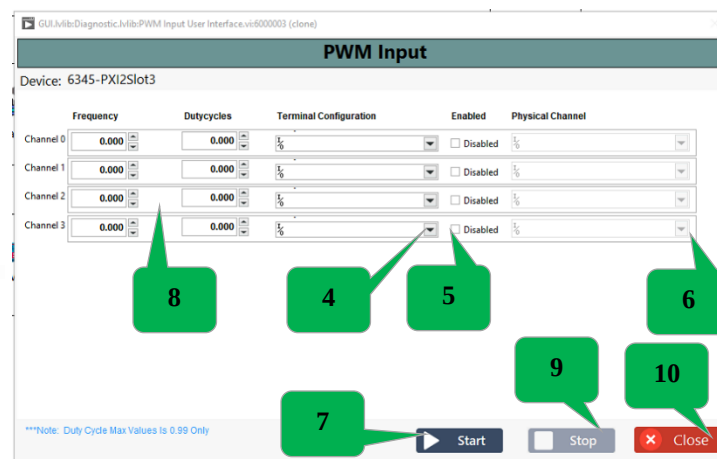


Figure 53: PWM Input

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11.8.7 PWM Output Diagnostics Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PWM Output Diagnostic Button and Refer Fig 54.
- 3) Select the Operation for this NI PWM Output Card Selection.
- 4) Select the Terminal Configuration.
- 5) Tick the Enable or Disable the Channel.
- 6) Select the Physical Channel.
- 7) Enter the Channel Frequency and Duty Cycles (0.99 Max).
- 8) Click the Start Button for Signal Write the PWM Output Signals
- 9) If need the stop the Write the PWM Signal Click Stop Button.
- 10) Finally Click the Close Button.

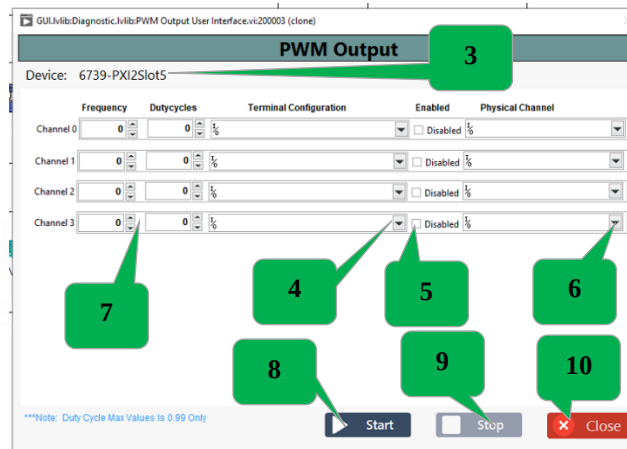


Figure 54: PWM Output

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11.8.8 CAN Diagnostics Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the CAN Diagnostic Button and Refer Fig 55.
- 3) Click the Configuration Tab button.
- 4) Select the XNET Interface CAN port for Ex CAN2 (Socket 0) and CAN4 (Socket 1).

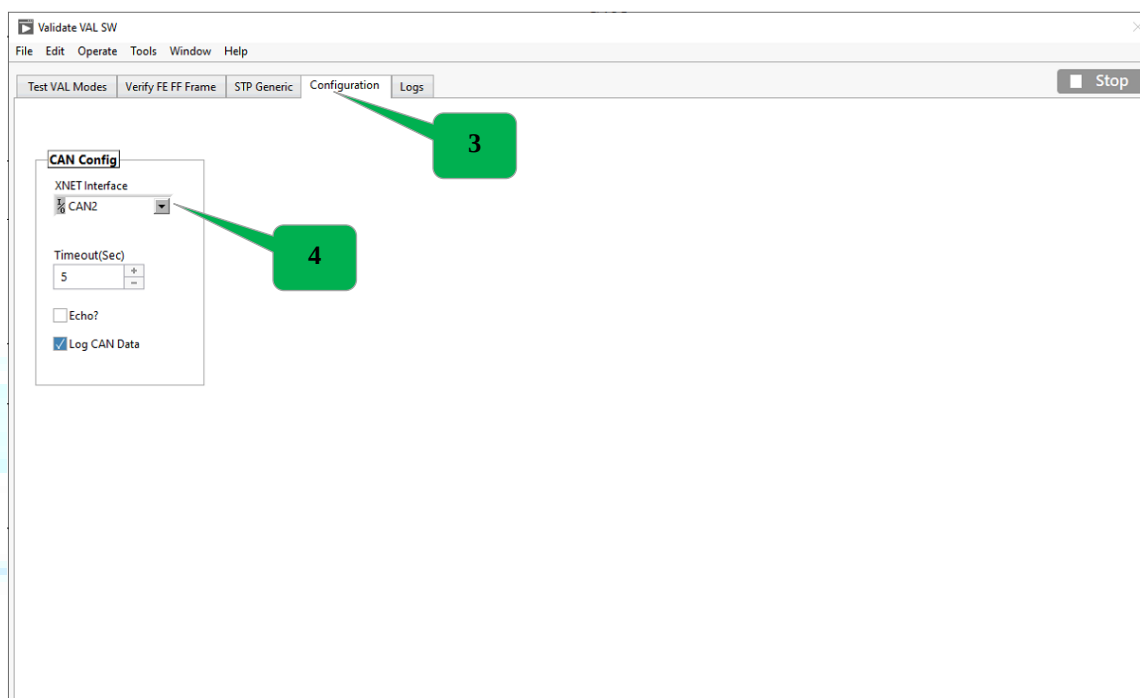


Figure 55: CAN Configuration

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11.8.8.1 Check VAL Software and Mode Testing

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the CAN Diagnostic Button and Refer Fig 55.
- 3) Configure the CAN Channel and Refer Fig 56.
- 4) Click the Test VAL Modes Tab Control button.
- 5) Click the TSW version button.
- 6) Show the TSW version.
- 7) Click the VAL SW Version button.
- 8) Show the VAL SW Version.
- 9) Select the VAL Mode.
- 10) Click the Start button and Mode is running.
- 11) Click the Start button and Mode is Stop.
- 12) Click the Latest RMS Current button.
- 13) Show the Latest RMS Current Data.
- 14) Show the CAN Read Raw Data.
- 15) Show the Response Status.
- 16) Finally Click the Stop button.



Figure 56: Test Software and Mode Write and eDCU Response Read

11.8.8.2 Read FE FF Frames Data

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the CAN Diagnostic Button and Refer Fig 55.
- 3) Configure the CAN Channel and Refer Fig 57.
- 4) Click the Verify FE FF Frame Tab Control button.
- 5) Select the Frame Type and FE Frame or FF Frame.
- 6) Select the Data Source.
- 7) Enable or Disable the Raw Data.
- 8) Click the Refresh button.
- 9) Show Data Size.
- 10) Show the Response Status.
- 11) Show the Frame Data.
- 12) Finally Click the Stop button.

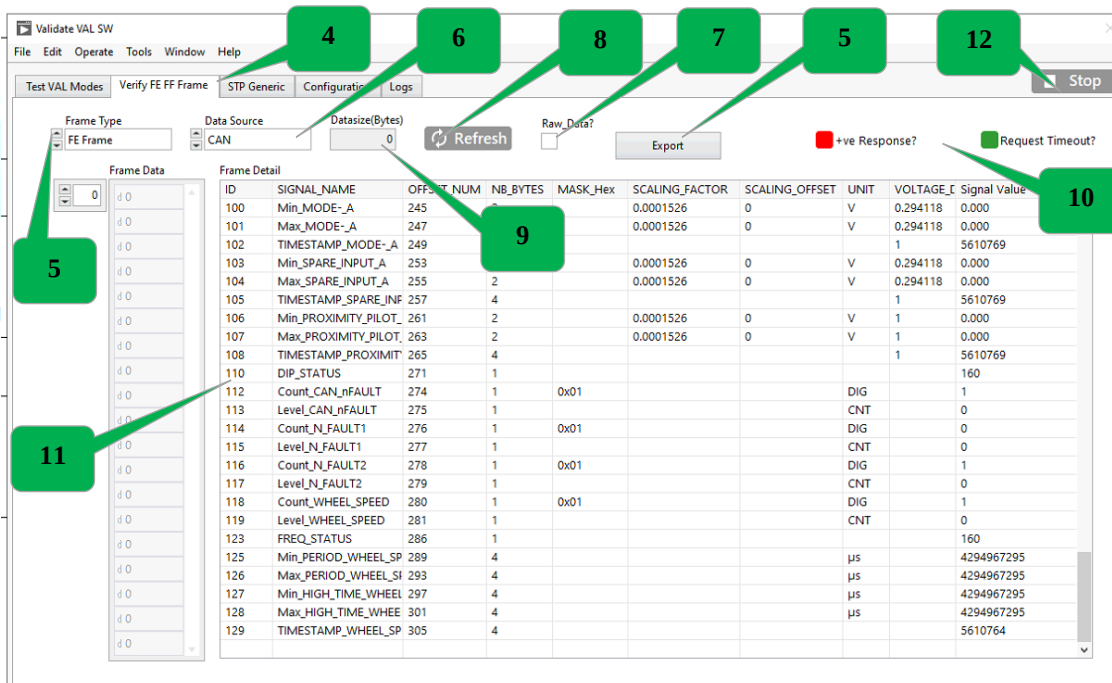


Figure 57: CAN Verify FE FF Frame Read

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11.8.8.3 CAN STP Generic Request/Response

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the CAN Diagnostic Button and Refer Fig 55.
- 3) Configure the CAN Channel and Refer Fig 58.
- 4) Select the Command.
- 5) Enter Valid Write Data.
- 6) Click the Sent button.
- 7) Show the Response Data (Hex).
- 8) Show the Response Data (ASCII).
- 9) Show the Response Status.
- 10) Finally Click the Stop button.



Figure 58: CAN STP Generic Read

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11.8.8.4 Monitor CAN Log

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the CAN Diagnostic Button and Refer Fig 59.
- 3) Click the Log Tab Control button.
- 4) Show the CAN View Table for Verify the CAN Read and Write Data.
- 5) Finally Click the Stop button.

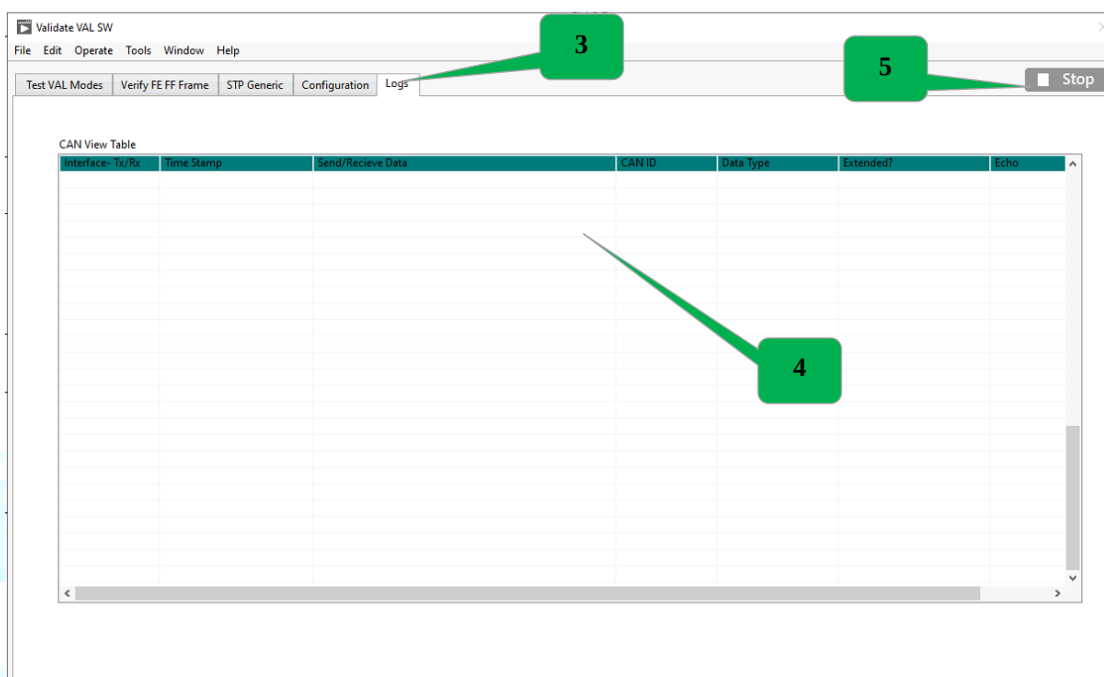


Figure 59: CAN Log Read

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11.8.9 High Voltage PPS Diagnostic Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.

11.8.9.1 HV PPS Initialization

- 3) Select the Operation “Init”.
- 4) Select the HVPPS Based socket Number
- 5) Click the Apply.

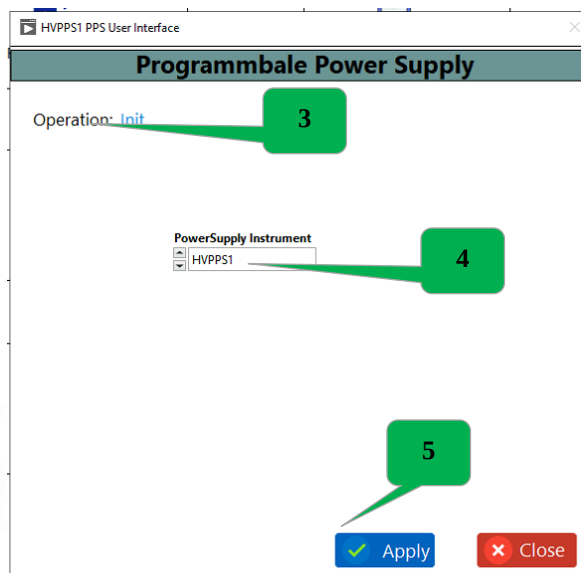


Figure 60: HV PPS Initialize

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11.8.9.2 HV PPS Self-test

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 58.
- 4) Select the Operation “Self-Test”.
- 5) Click the Apply.
- 6) Show the Power supply Status

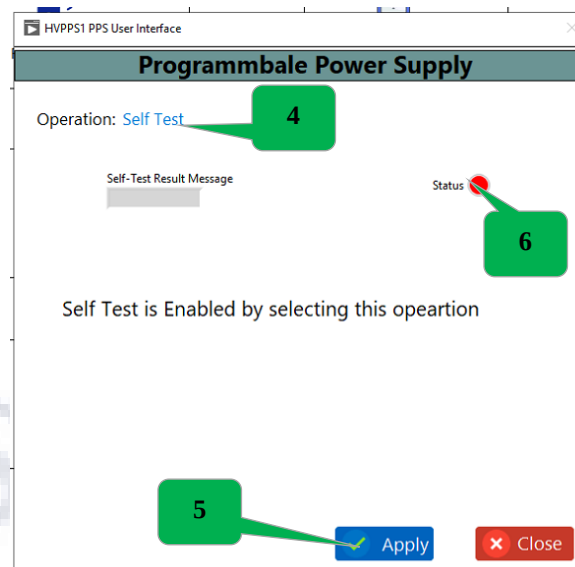


Figure 61: HV PPS Self-Test

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11.8.9.3 HV PPS Configure

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 58.
- 4) Select the Operation “Configure”.
- 5) Pre-Configure the Data for based on Power supply Selection.
- 6) Enable or Disable the Output.
- 7) Click the Apply.

Operation: [Configure](#)

Output Priority Mode: Voltage

Voltage Level (V): 48

Current Limit (A): 150.00

Current Level (A): 155.00

Voltage Limit (V): 55.00

Voltage Limit OVP (V): 60.00 ☐ OVP

Current Limit OCP (A): 160.00 ☐ OCP

OVP Delays (S): 1.00

OCP Delays (S): 1.00

Figure 62: HV PPS Configure

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11.8.9.4 HV PPS Measure Current

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 60.
- 4) Select the Operation “Measure Current”.
- 5) Show the Measure Current Data.
- 6) Click the Apply.

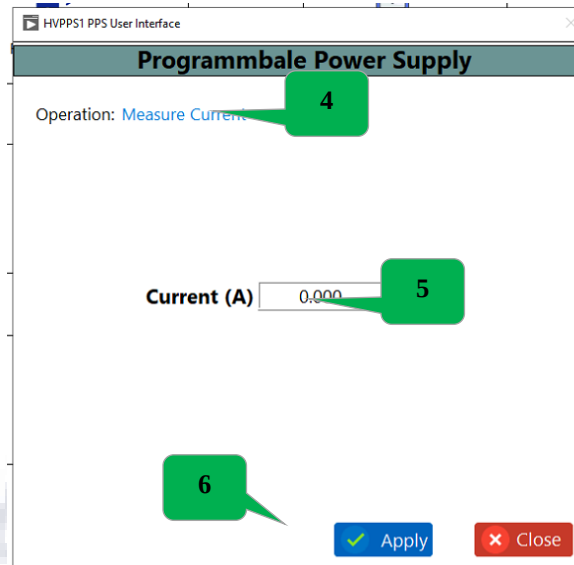


Figure 63: HV PPS Measure Current

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11.8.9.5 HV PPS Close

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 60.
- 4) Select the Operation “Close”.
- 5) Click the Apply
- 6) Finally Click the Close Button.

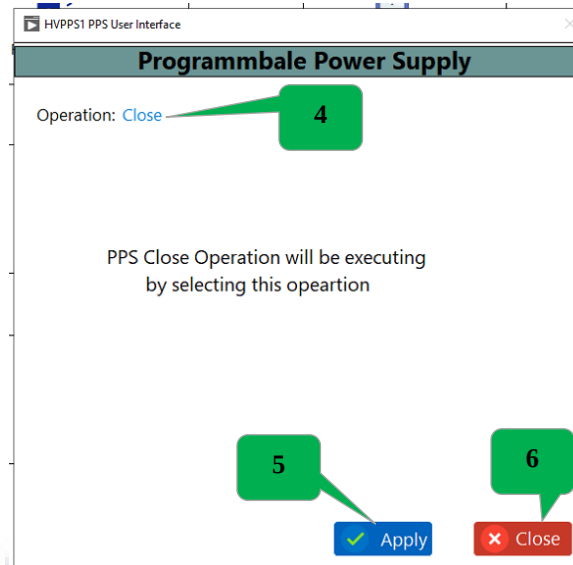


Figure 64: HV PPS Close

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11.8.10 Procedure LV PPS Diagnostic Screen

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.

11.8.10.1 LV PPS Initialization

- 3) Select the Operation “Init”.
- 4) Select the LVPPS
- 5) Click the Apply.

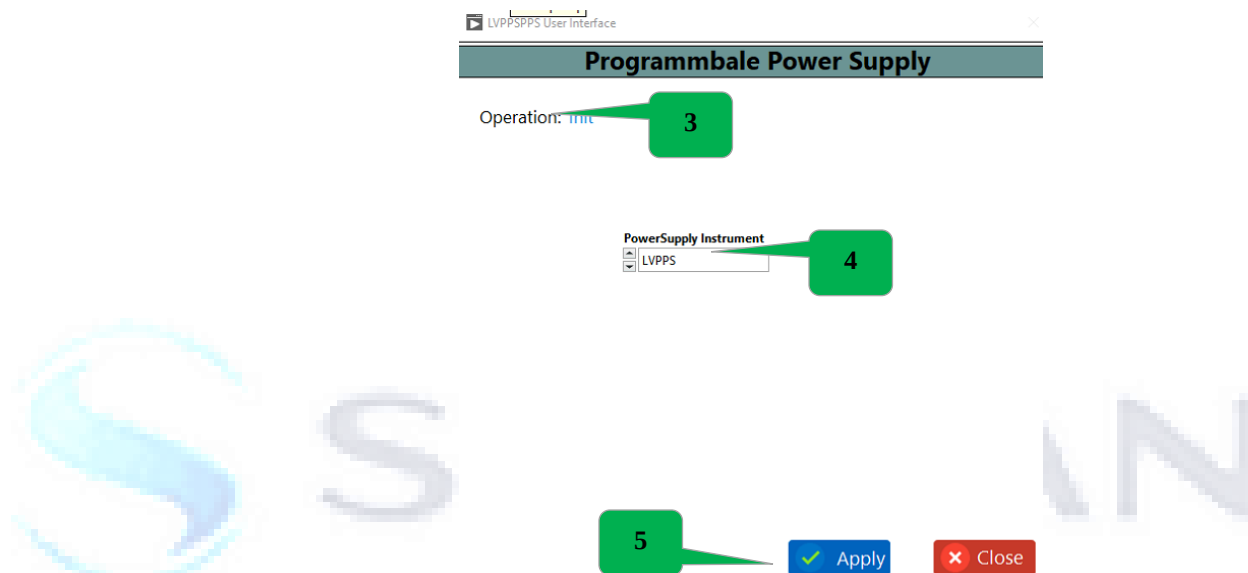


Figure 65: LV PPS Initialize

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11.8.10.2 Procedure LV PPS Self-test

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 65.
- 4) Select the Operation “Self-Test”.
- 5) Click the Apply.
- 6) Show the Power supply Status

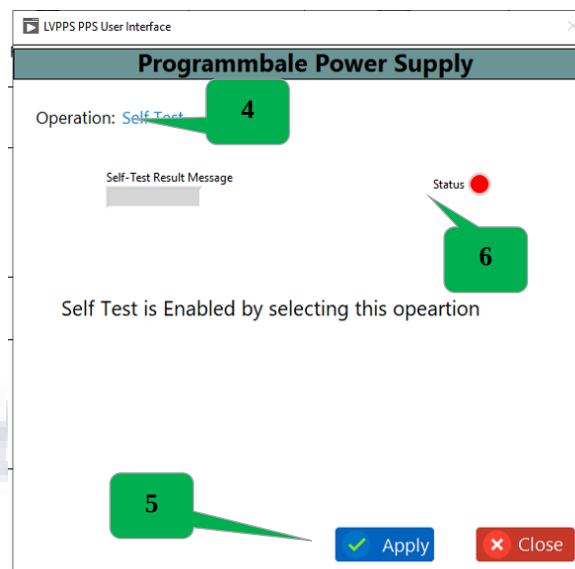


Figure 66: LV PPS Self-Test

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11.8.10.3 Procedure LV PPS Configure

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 65.
- 4) Select the Operation “Configure”.
- 5) Pre-Configure the Data for based on Power supply Selection.
- 6) Enable or Disable the Output.
- 7) Click the Apply.

LVPPS PPS User Interface

Programmable Power Supply

Operation: **Configure** (4)

Output Priority Mode: Voltage (dropdown) ☒ Output (6)

Voltage Level (V): 12

Current Limit (A): 6.00

Current Level (A): 8.00

Voltage Limit (V): 15.00

Voltage Limit OVP (V): 15.00 ☐ OVP

Current Limit OCP (A): 8.00 ☐ OCP

OVP Delays (S): 1.00

OCP Delays (S): 1.00

☒ Apply (7)

Figure 67: LV PPS Configure

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11.8.10.4 Procedure Configure LV PPS Measure Current

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Figure 47: Diagnostic Test Screen.
- 3) Click the PPS Init and Refer Fig 65.
- 4) Select the Operation “Measure Current”.
- 5) Show the Measure Current Data.
- 6) Click the Apply.

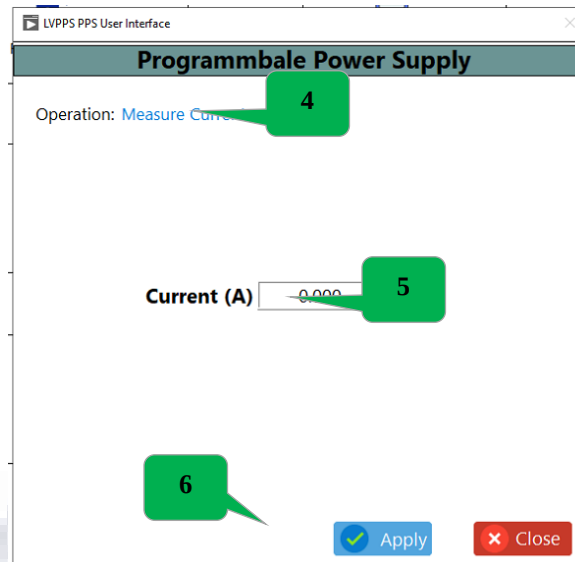


Figure 68: LV PPS Measure Current

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11.8.10.5 Procedure LV PPS Close

- 1) Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
- 2) Click the PPS Diagnostic Button and Refer Fig 60.
- 3) Click the PPS Init and Refer Fig 65.
- 4) Select the Operation “Close”.
- 5) Click the Apply
- 6) Finally Click the Close Button.

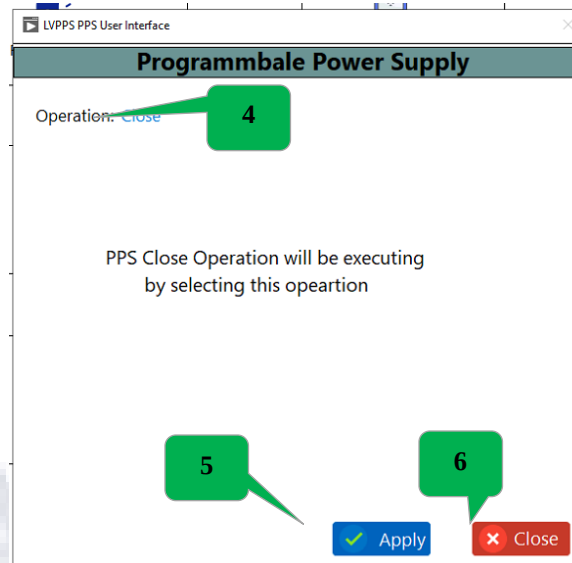


Figure 69: LV PPS Close

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11.9 Temperature Diagnostic Screen

In this Screen we can Diagnostic power supply Temperature, Chamber Temperature, Load rack Temperature. We can read or write the Temperature Controller.

1. Click the Diagnostic Screen Button and Figure 47: Diagnostic Test Screen.
2. Click the Temperature controller and Figure 47: Diagnostic Test Screen.
3. Select the Operation in Read or Write.
4. Select the Slave Unit ID.
5. If Operation is “Written” and Enter the Write Starting Address.
6. If Operation is “Writing” Enter the Set Limit Temperature.
7. Click the Apply button.
8. If Operation is “Read” and Enter the Read Starting Address, Number of Holding.
9. Click the Apply button.
10. If Operation is “Read” Show the Holding Registers Value.
11. Show the Current Temperature and Set Limit Temperature.
12. Finally Click the Close button.

The screenshot shows the 'Temperature Control' diagnostic screen. It features a yellow header bar with the title 'Temperature Control'. Below the header, the interface is divided into two main sections. The left section contains controls for 'Operation' (a dropdown menu set to 'Read'), 'Slave unit ID' (a numeric input field set to '1'), 'Write Starting Address' (a numeric input field set to '8451'), and 'Set Limit Temperature' (a numeric input field set to '27'). The right section contains controls for 'Read Starting Address' (a numeric input field set to '8192'), 'Number of Holding' (a numeric input field set to '5'), 'Holding Registers Value' (two numeric input fields, both set to '0'), 'Current Temperature' (a numeric input field set to '0'), and 'Set Limit Temperature' (a numeric input field set to '0'). At the bottom of the screen, there are two buttons: a blue 'Apply' button with a green checkmark icon and a red 'Close' button with a white 'X' icon. Numbered green callouts point to various elements: 1 points to the 'Operation' dropdown, 2 points to the 'Slave unit ID' field, 3 points to the 'Write Starting Address' field, 4 points to the 'Set Limit Temperature' field, 5 and 7 point to the 'Apply' button, 6 points to the 'Read Starting Address' field, 8 points to the 'Number of Holding' field, 9 points to the 'Current Temperature' field, and 10 points to the 'Close' button.

Figure 70: Temperature control

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12 Procedure for Flashing of VAL Software to DUT

This section outlines the process for flashing the TSW+VAL software to the DUT. There are two methods available: the Manual procedure or the Automated procedure.

12.1 Setup

- 1) Launch the LTT Application (refer: How to Launch LTT Application)
- 2) Disconnect the Harness form LTT and insert breakout box in between as shown in below image.
- 3) Connect Coredo Tool harness in the wire in breakout box, connection is as below:
 - a. Power to T30 of BoB box (refer BoB doc for pin assignment)
 - b. GND to GND of BoB box
 - c. CAN_H to CAN2_H BoB box
 - d. CAN_L to CAN2_L BoB box



Figure 71: Break out Box

- 4) Please refer to the procedure for the manual mode steps in the for Procedure for Manually Flashing of VAL Software to DUT.
- 5) Please refer to the procedure for the auto mode steps in the for Procedure for Automated Flashing of VAL Software to DUT.

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12.2 Procedure for Manually Flashing of VAL Software to DUT

Click the Diagnosis button to display the screen. To use flashing, access the driver programmable power supply and digital output. The user screen, as shown below, will appear. Please refer to Fig. 72.

- 1) Follow the section 12.1 instructions for HW setup
- 2) Click the Diagnosis symbol button.
- 3) Click the PPS button for HVPPS.
- 4) Click the Digital Output Purpose for HV Power Output Enable and Disable.
- 5) Click the PPS button for LVPPS.
- 6) Click the Relay Output Purpose for LV Power Output Enable and Disable.
- 7) Click the CAN button Purpose for Check eDCU SW version

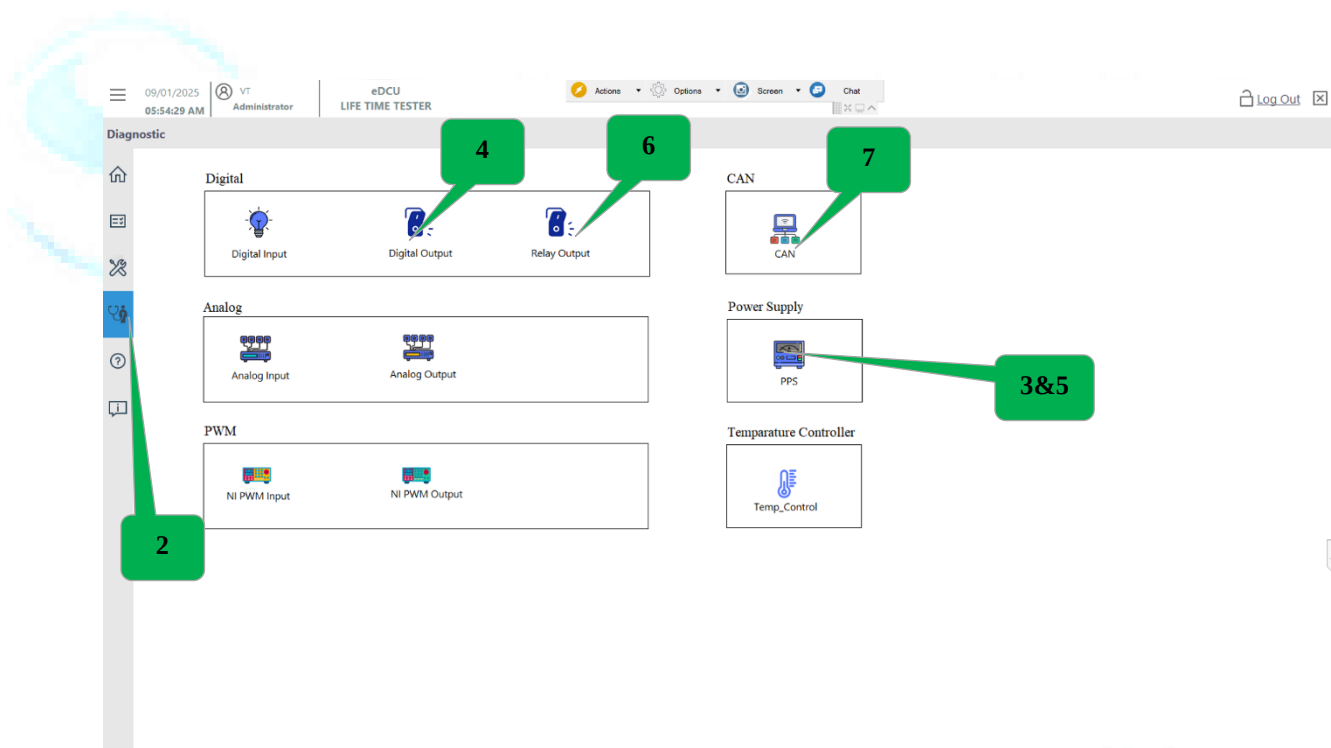


Figure 72: Diagnosis mode Flash

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12.2.1 Procedure of DUT Power ON.

1. Pre-Power Check.

- Before powering on the system, inspect all connections for correct placement and secure engagement.

2. Switch and Power Supply Status Turn Off Condition.

- Confirm that all DO switches and are OFF before applying power.
- Ensure the power supply is set to the correct voltage and current settings for the DUT.
- Inspect for any potential short circuits or incorrect wiring that may cause issues when power is applied.

3. DUT Connection.

- Ensure the DUT is Positive and Negative Connection and Load side U, V, W Connection then 40 Pin Connection Connected.
- Confirm there are no loose components or connections on the DUT.

4. Fixture Engagement.

- Verify that the DUT is properly engaged with the Harting Connection.
- Verify the Harting Connector connected to the LTT Side.
- Check the Connector for any mechanical or alignment issues that may cause improper contact.

5. Connections for Manual Test.

40-pin Connector:

- Verify the 40-pin connector is properly seated and locked in place.

UVW Cables.

- Check that the UVW cables (U, V, W phases) are correctly connected, ensuring proper phase alignment.

DC Link Cables.

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- Ensure that the DC link cables are connected securely without loose terminals or exposed wires.

12.2.2 Turn ON Power Supply (T30 and DC Link)

Click the PPS button and Selection list mention LVPPS, HVPPS (Based the socket selection HVPPS) the Run Cleanup and tick the Don't show again batch execution then click the OK. screen below the user screen will appear Refer the Fig 69.

- 1) Click the Diagnostic Screen Button and Refer Fig 69.
- 2) Select the Operation "Init".
- 3) Select the power supply for LVPPS or HVPPS based on Socket and Ensure the Hardware Connection.
- 4) Click the apply button.

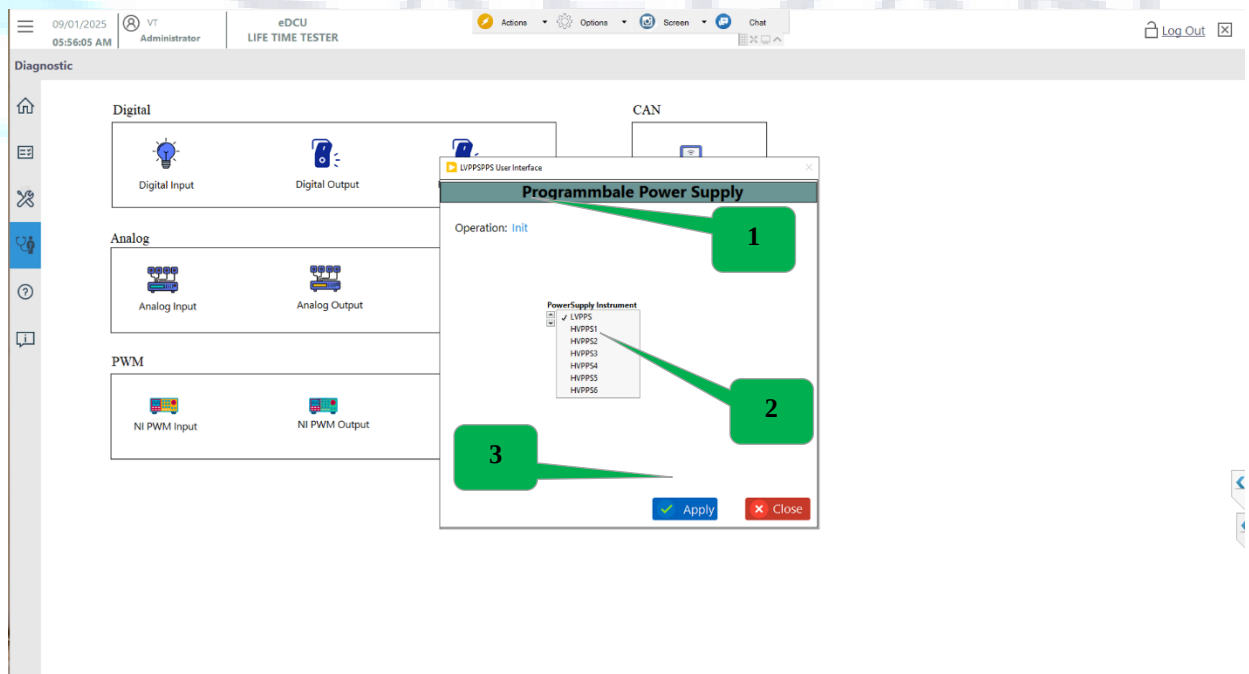


Figure 73: Power Supply Selection

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12.2.2.1 High Voltage PPS Initialize

HVPPS initialize the instrument then click the Apply. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72.
- 2) Select the Operation “Init”.
- 3) Select the PPS Selection HVPPS (Based Socket 0 Ex: - HVPPS1).
- 4) Final Click the Apply Button.
- 5) Title bar reflected on selected PPS to Verify

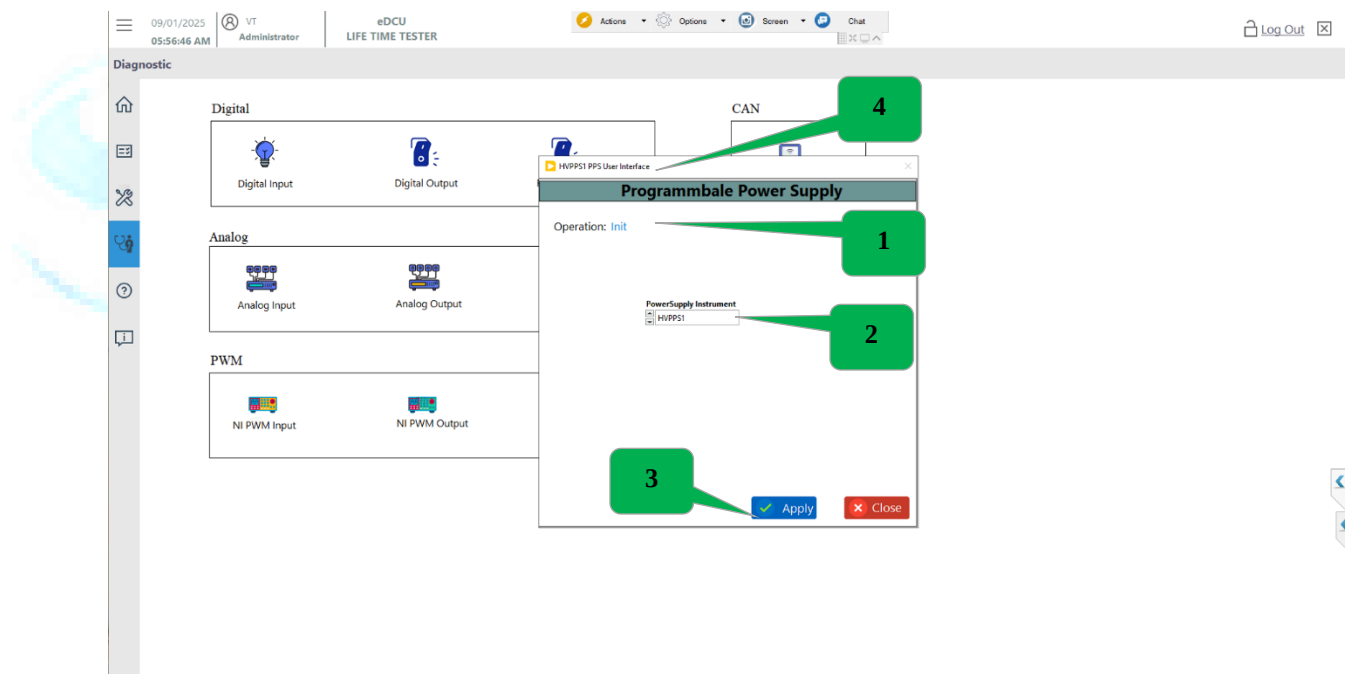


Figure 74: HV PPS Initialize

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12.2.2.2 High Voltage PPS Configure

HVPPS Pre configuration and Enable the Output Only to click the OK. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Select the Operation “Configure”
- 3) Based on power supply selection Data is Preconfigured
- 4) Configure the Enable Output Button.
- 5) Final Click the Apply Button.

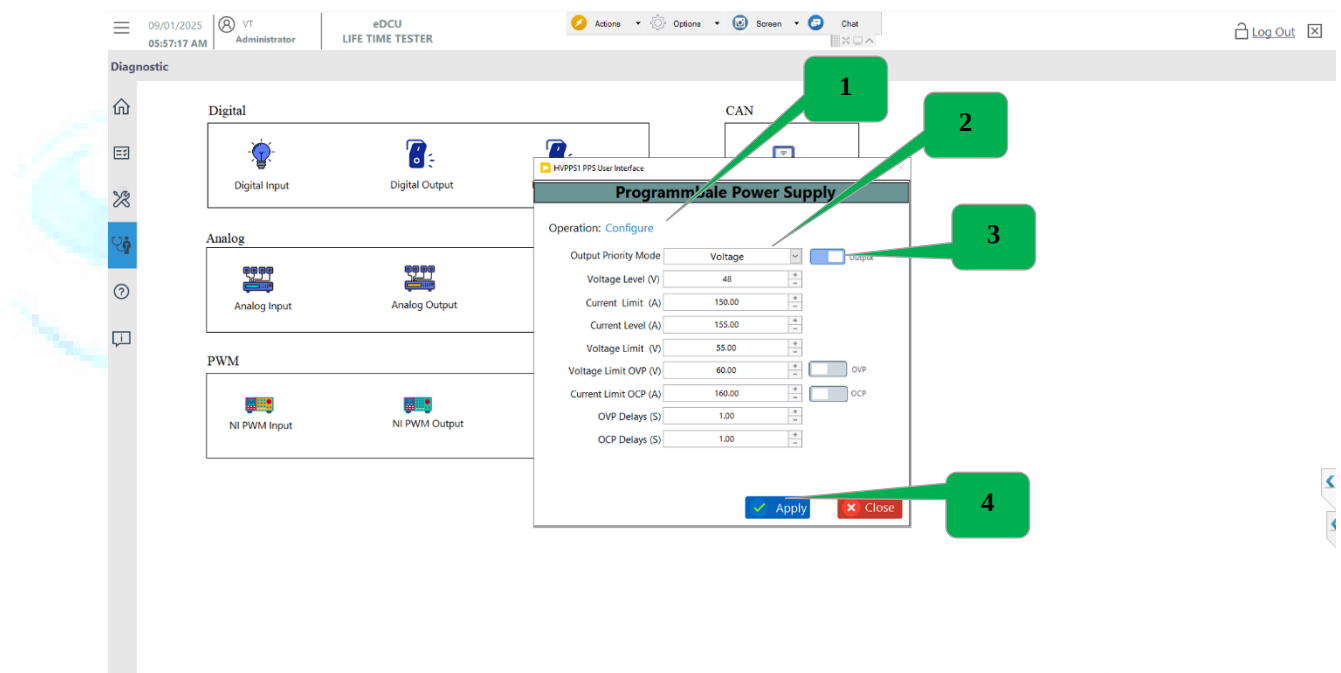


Figure 75: HV PPS Configure

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12.2.2.3 Turn ON High Voltage PPS Digital Output(Relay)

eDCU 48V Positive and Negative Digital Output Enabled and Write. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Enable the 48V Positive and Negative Output Based on Socket.
- 3) Click the Start Button.
- 4) Finally Click the Write Button.



Figure 76: HV PPS Relay Output

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12.2.2.4 Low Voltage PPS Initialize

LVPPS initialize the instrument then click the Apply. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Select the Operation “Init”
- 3) Select the PPS Selection LVPPS.
- 4) Final Click the Apply Button.

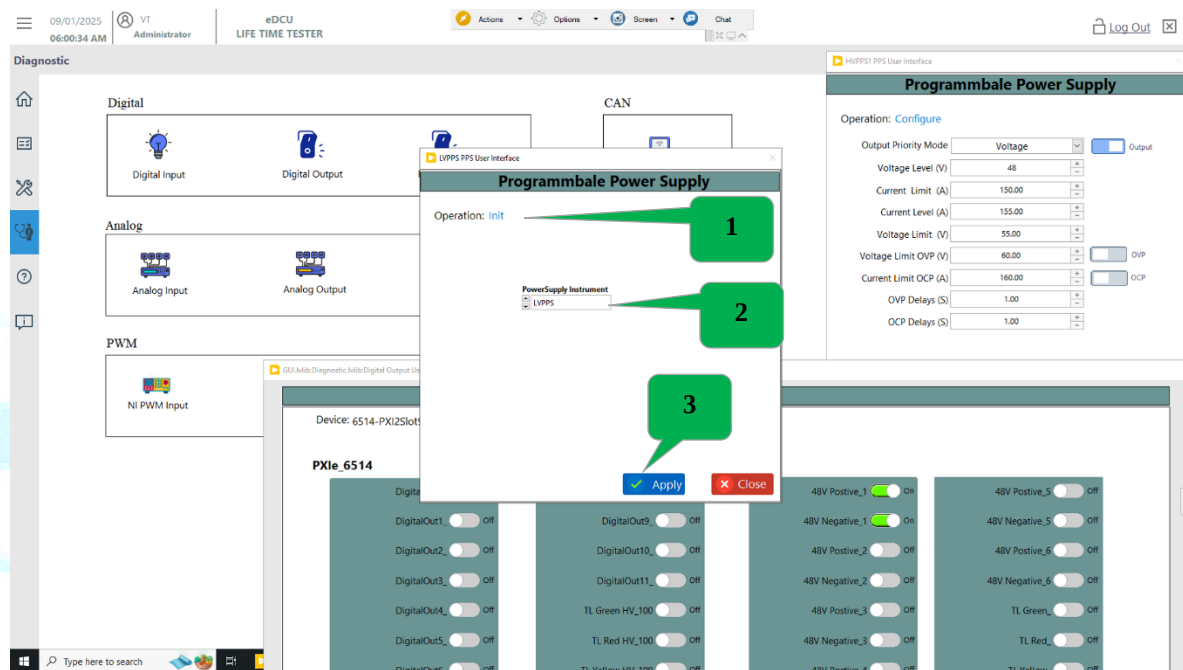


Figure 77: LV PPS Initialize

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12.2.2.5 Low Voltage PPS Configure

LVPPS Pre configuration and Enable the Output Only to click the OK. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Select the Operation “Configure”.
- 3) Based PPS Selection Configure the Data.
- 4) Configure the Enable Output Button.
- 5) Final Click the Apply Button.

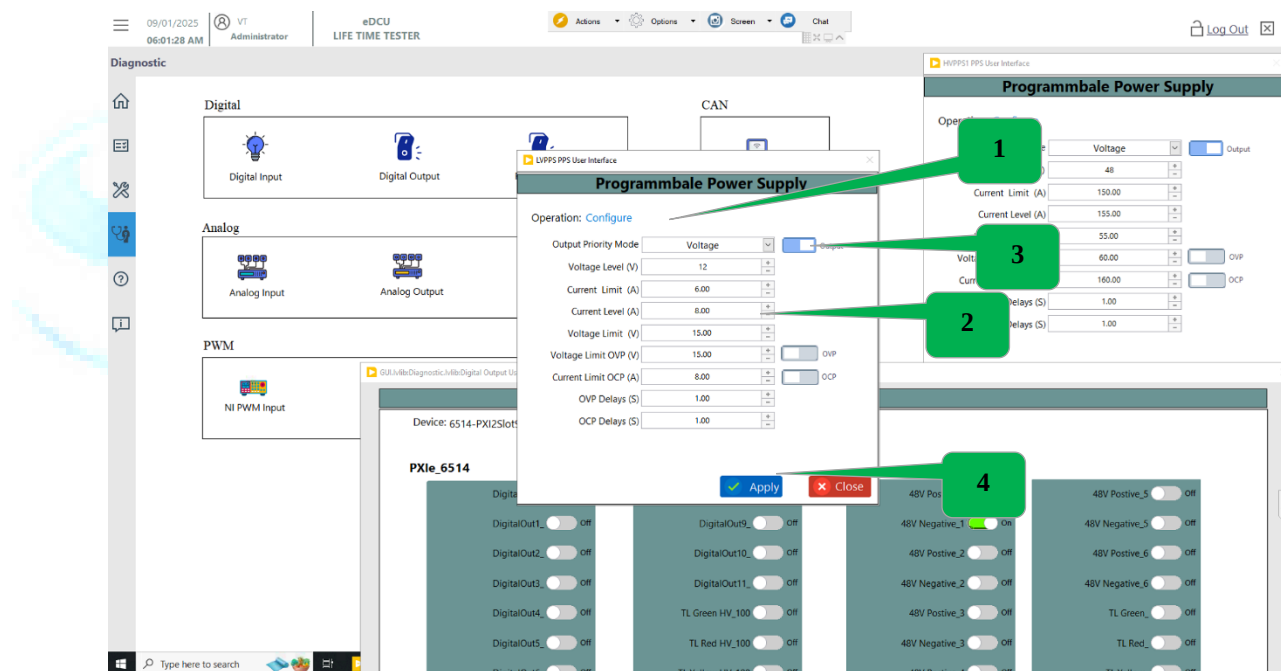


Figure 78: LV PPS Configure

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12.2.2.6 Turn ON Low Voltage PPS Relay Output

eDCU 12V Positive and Negative Relay Output Enabled and Write. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Enable the 12V T30 and T30 GND Output.
- 3) Click the Start Button.
- 4) Finally Click the Write Button.

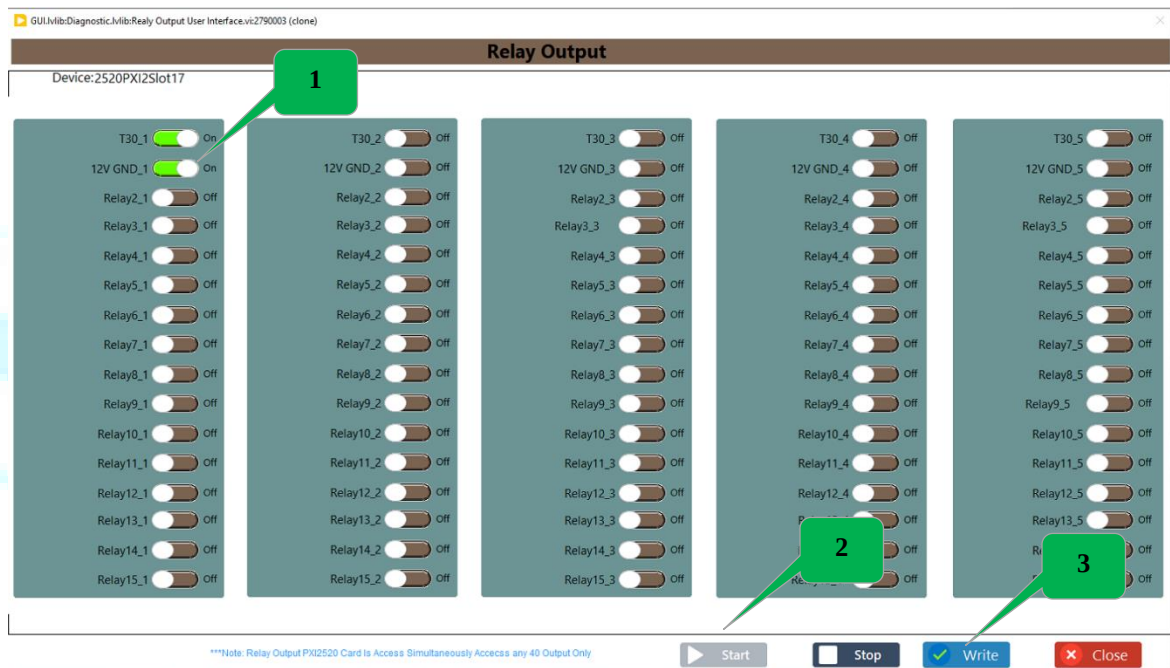


Figure 79: LV PPS Relay Output

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12.2.3 CAN Configure

CAN configure the port based on socket. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Click the Tab Configure
- 3) Based Socket CAN port selection (Ex: Socket0-CAN2, Socket1-CAN4)

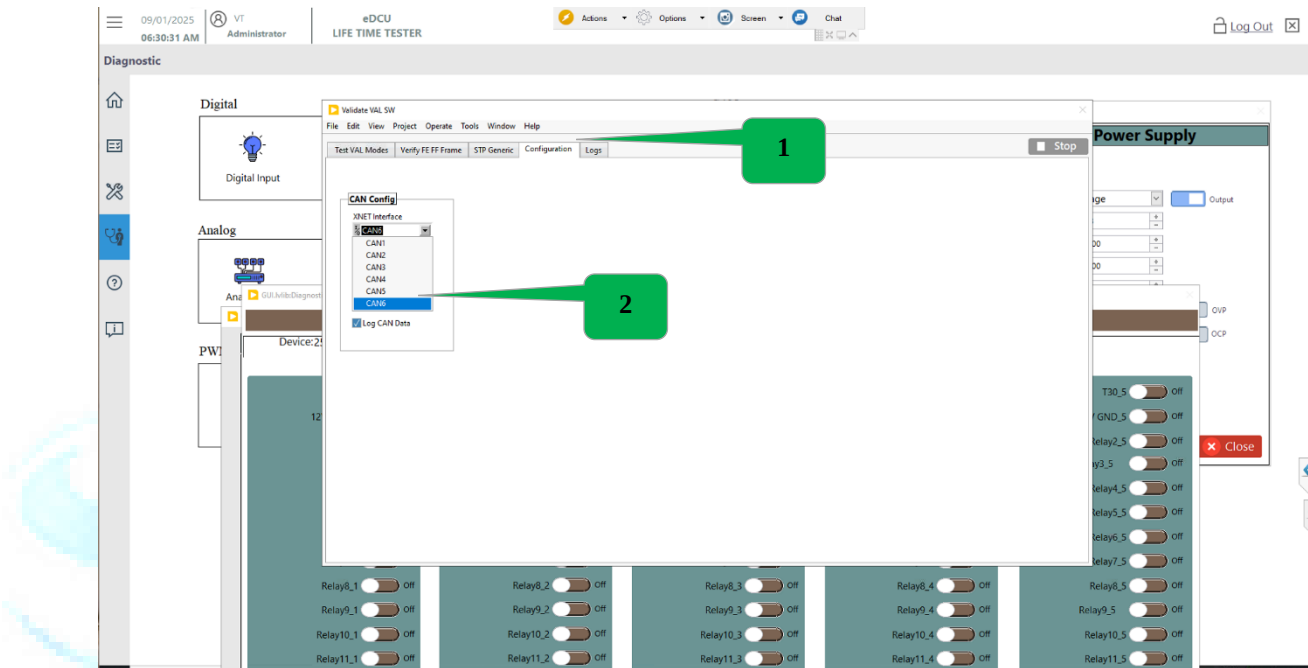


Figure 80: CAN Configuration

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12.2.3.1 Read SW Version

Click the CAN Read the VAL SW the read. screen below the user screen will appear.

- 1) Click the Diagnostic Screen Button and Refer Fig 72
- 2) Click the Tab Test VAL Modes Button
- 3) Click the Button TSW version read
- 4) Show the TSW version
- 5) Click the Button VAL version read
- 6) Show the VAL version

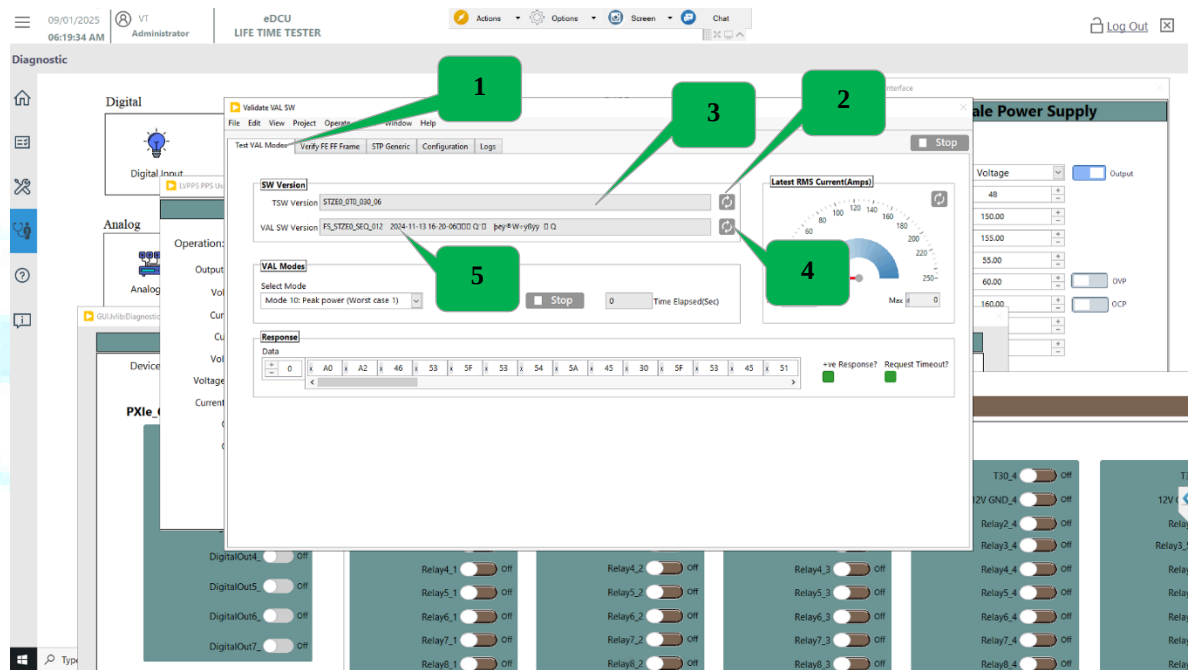


Figure 81: CAN Test VAL Modes

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12.2.4 Flashing New Software

Now, proceed with flashing of the new TSW/VAL SW. After completing the flashing process, perform a Power Cycle on DUT

12.2.5 Verify Updated SW after Flash

Once the eDCU is powered on, the TSW version and VAL SW version need to be read and compared to the flashing version for verification.

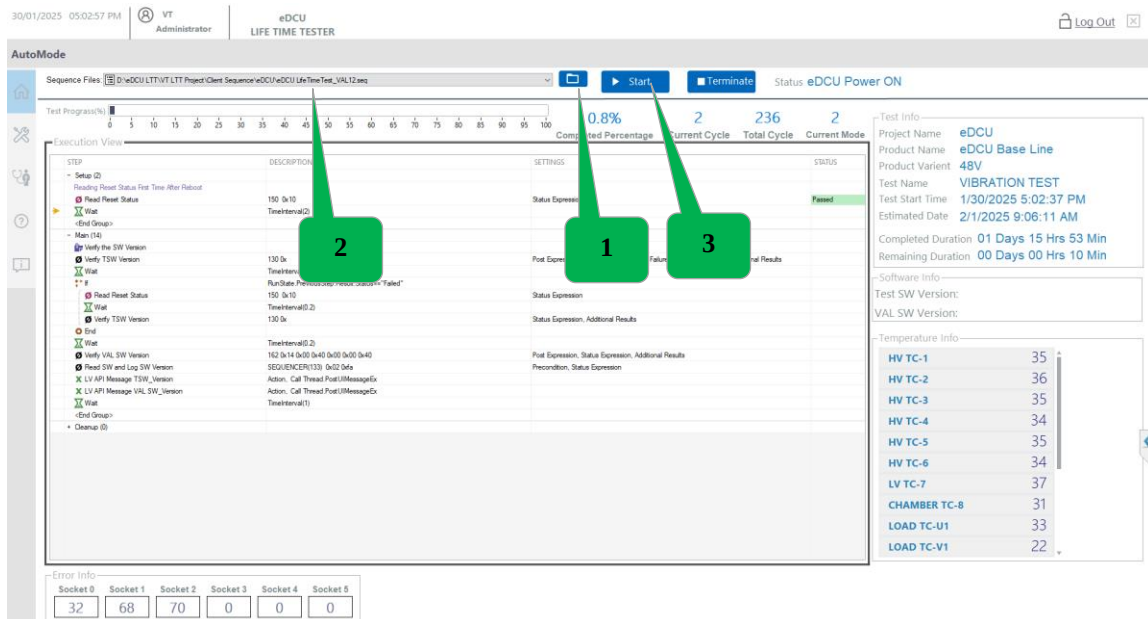
12.3 Procedure for Automated Flashing of VAL Software to DUT

Please refer to the procedure in the page/section for further details. Procedure for Auto Mode (To Start New/Existing Test)

12.3.1 Run the Software Flash Sequence

Select open the sequence File Path and the Software Flashing Sequence display the windows then Start the Test. “Running Display” screen below the user screen will appear and Refer the Figure 24.

- 1) Click the Open Sequence button and Select the Sequence file Location is
D:\eDCU LTT\VT LTT Project\Client Sequence\eDCU\Software Flash.seq
- 2) Show the Selected Sequence file was loaded.
- 3) Click the Start the Button.
- 4) Flow the Figure Next Steps.



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Figure 82: How to Flash Sequence

12.3.2 Read existing software

- 1) Existing Software read TSW and VAL SW Version showing Popup
- 2) Click the button “OK”.

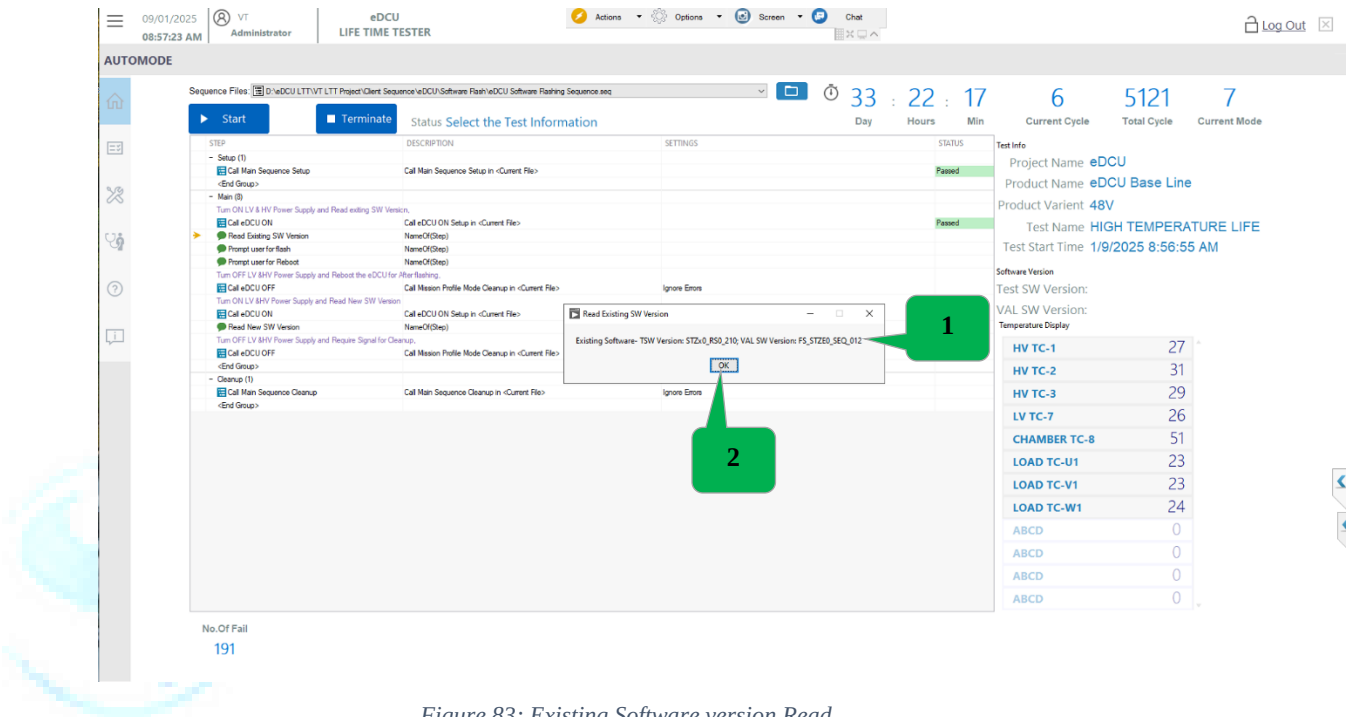


Figure 83: Existing Software version Read

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12.3.3 Flashing New Software

- 1) Proceed the flash the eDCU and After flashing showing Popup.
- 2) click the “OK” Button.

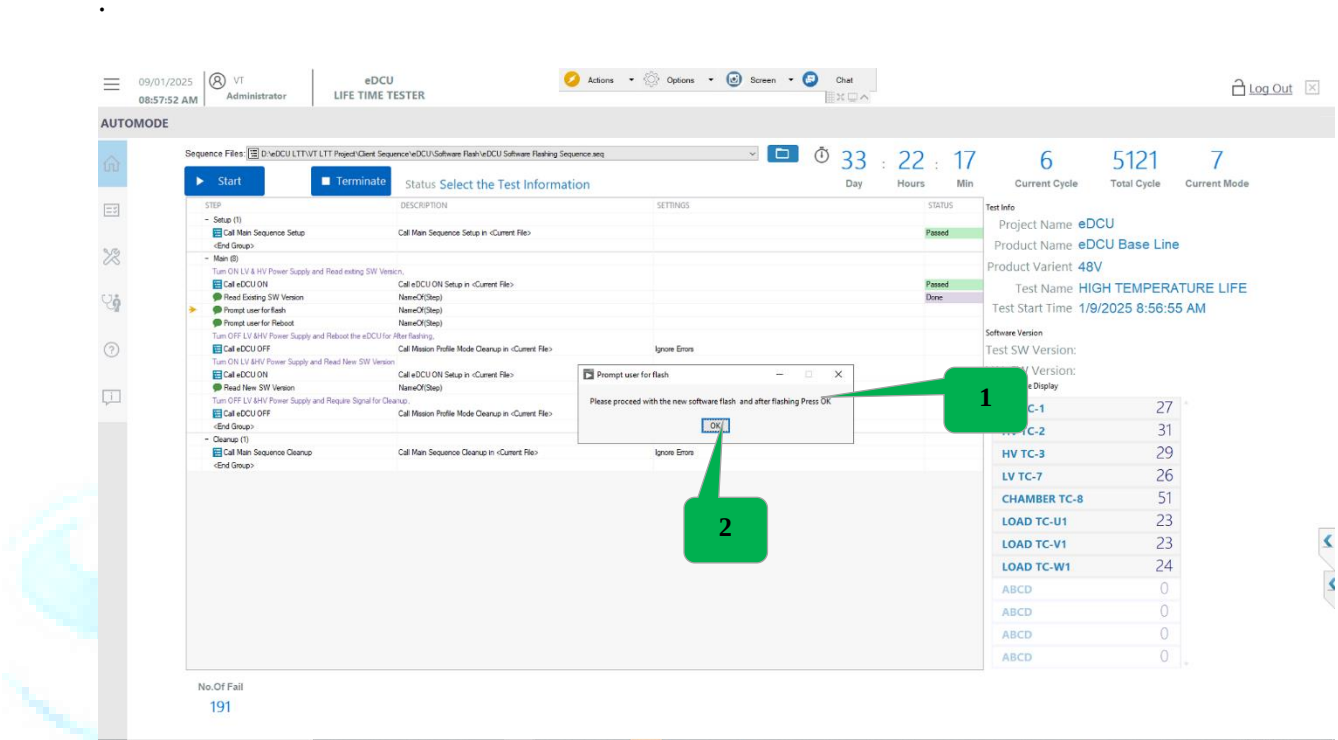


Figure 84: Proceed Flash the Software

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12.3.4 Reboot DUT

- 1) Proceed with the reboot the eDCU and showing Popup
- 2) click the “OK”.

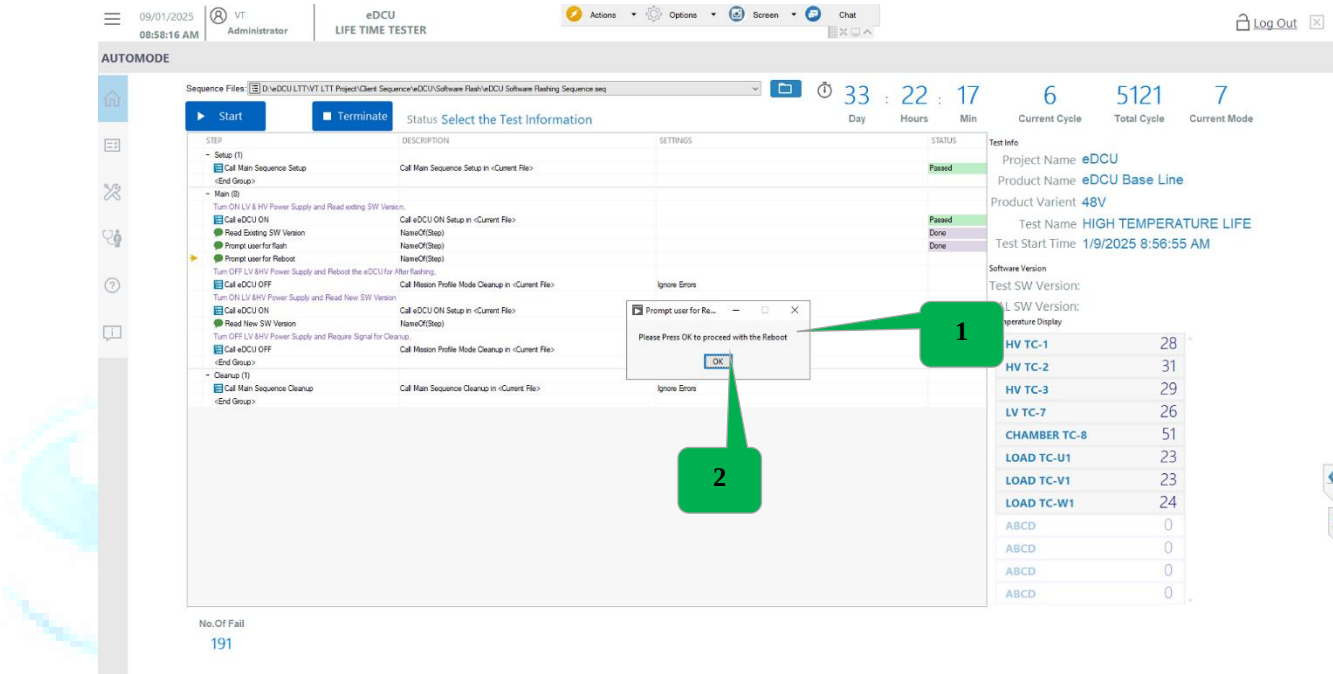


Figure 85: Reboot the Test

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12.3.5 Verify New Software

- 1) New Software read TSW and VAL SW Version showing Popup and
- 2) Click the button “OK”.

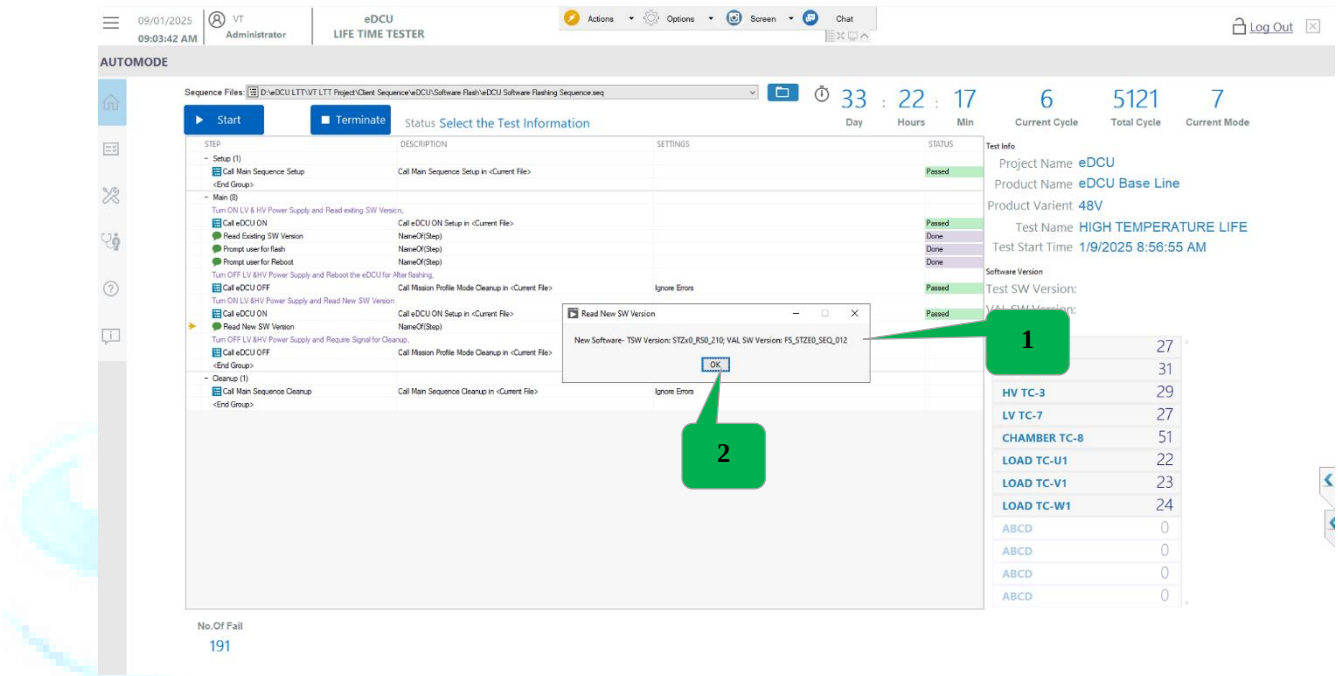


Figure 86: New Software Version Read

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13 Procedure for using custom step types to create a test sequence development

This section outlines the process for using custom step types to create test sequence development. The user follows this procedure to create a client-specific test sequence, utilizing LabVIEW code modules within TestStand. The lifetime tester uses the required instrument drivers, and based on this, custom step types are created. Users can drag and drop steps and edit the configuration details depending on the logic needed for test case development.

13.1 How to Launch TestStand with LTT Environment

- 1) Locate and open the folder on the tester PC at **D:\eDCU LTT\VT LTT Project** and find the **TestStand Launcher.bat** file.
- 2) Right-click the **TestStand Launcher.bat** file and select **Run as Administrator** to launch TestStand. Refer to the figure below for clarification.

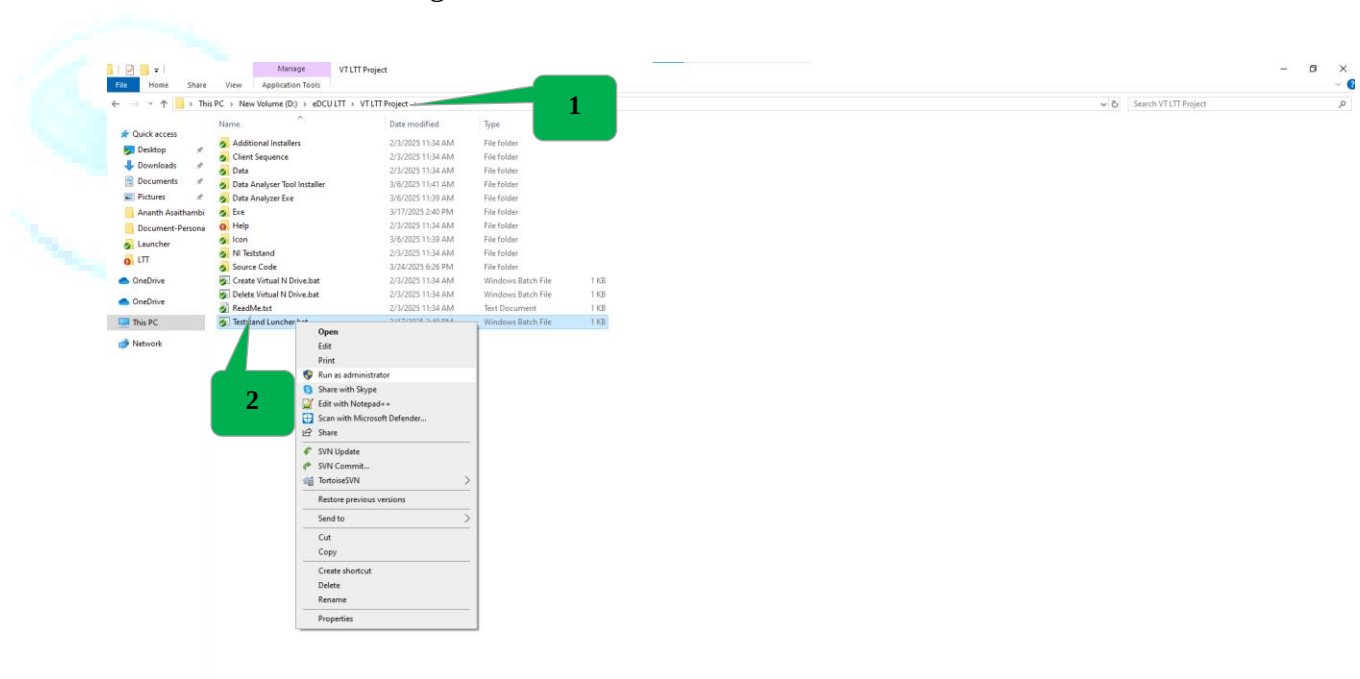


Figure 87: How to Open Test stand

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13.1 Login the TestStand

- 1) The login screen is shown in **Figure 88** below.
- 2) Enter "**Administrator**" as the username. No password is required. After that, click the **OK** button to proceed.

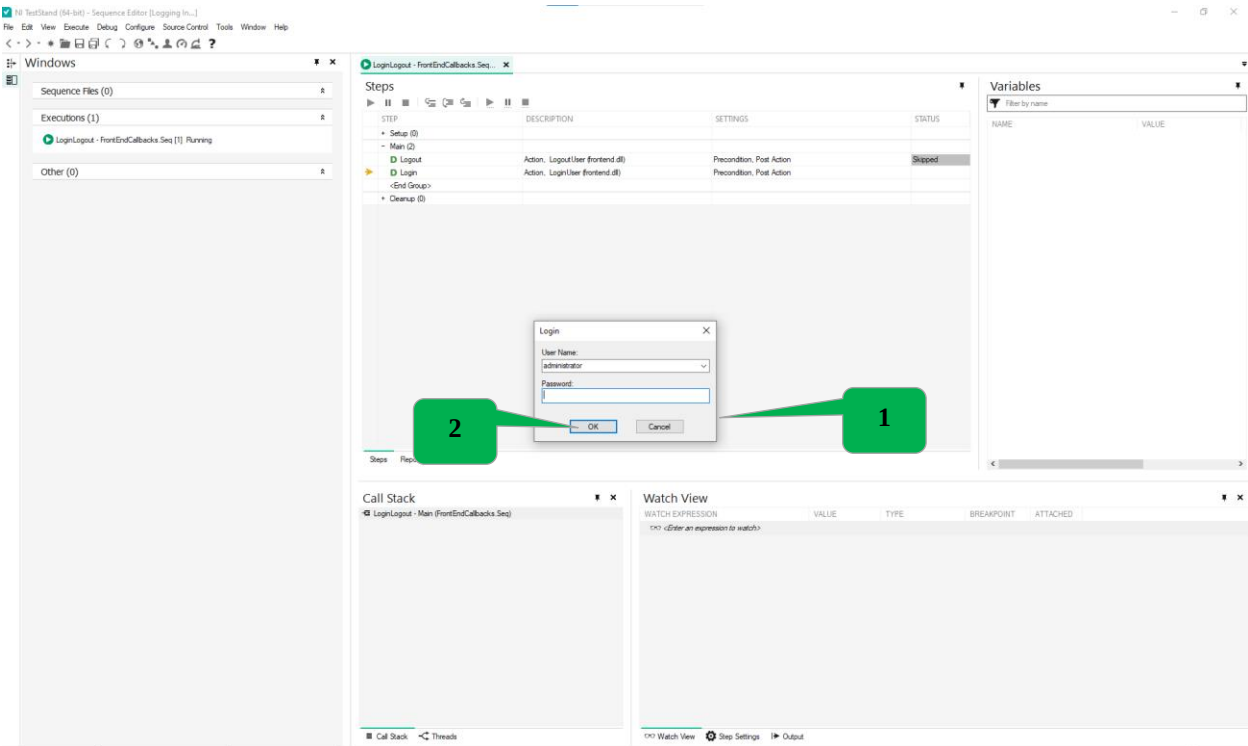


Figure 88: Login the Test stand

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13.2 List of Custom Step types

- 1) The screen displaying the **TestStand Custom Step Types Insertion Palette** is shown in **Figure 89** below.
- 2) List of the Step types below.
 - a) CAN
 - b) Chamber
 - c) eDCU Specific (Wait Monitor)
 - d) Database
 - e) Temperature controller
 - f) Power supply
 - g) DAQ
 - h) Power Monitor
 - i) Mail
- 3) Some step types include additional sub-steps based on the application developed.

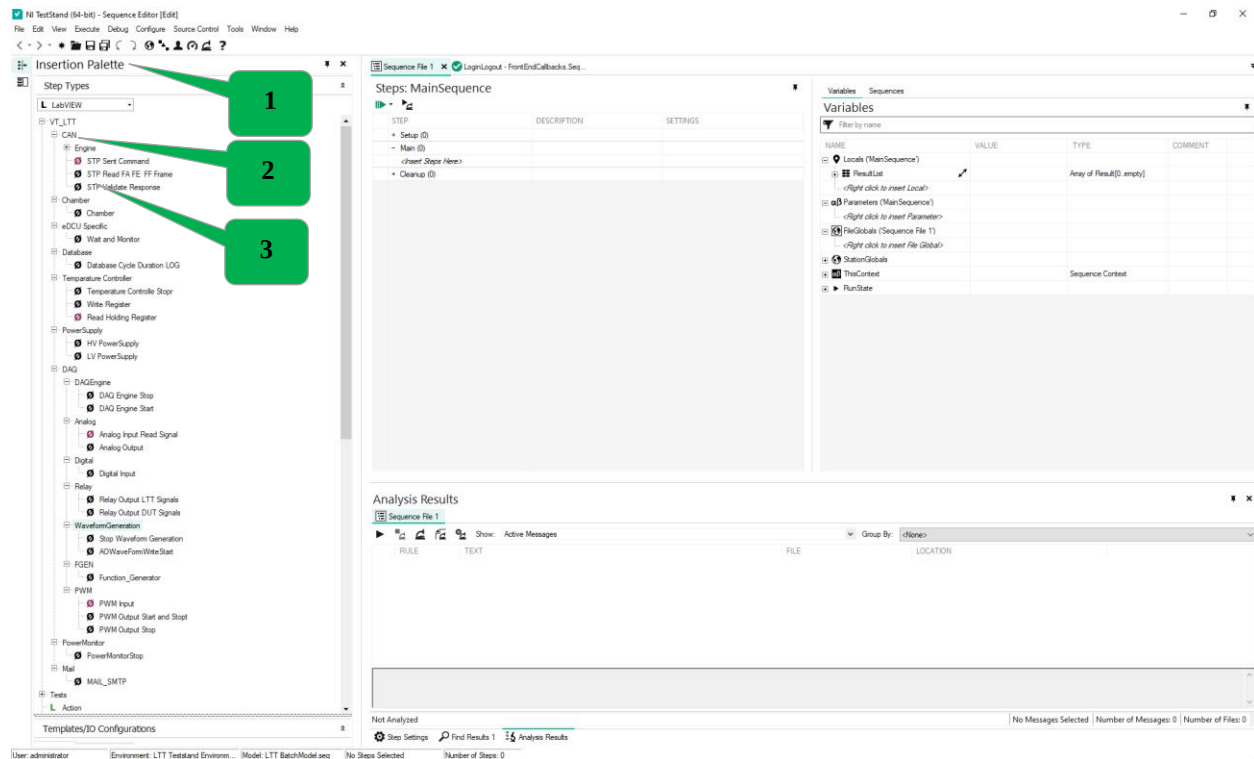


Figure 89: List of Custom step types of view

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13.3 CAN

13.3.1 CAN Engine Start and Stop.

- 1) In the Insertion Palette, click on **CAN** and Click the Engine then drag and drop the **Start CAN Engine and Stop CAN Engine** step.
- 2) The Start CAN Engine should be set up in the sequence specified, without the need for manual editing, as it operates internally.
- 3) The Stop CAN Engine should be Clean up in the specified sequence, without the need for manual editing, as it operates internally.

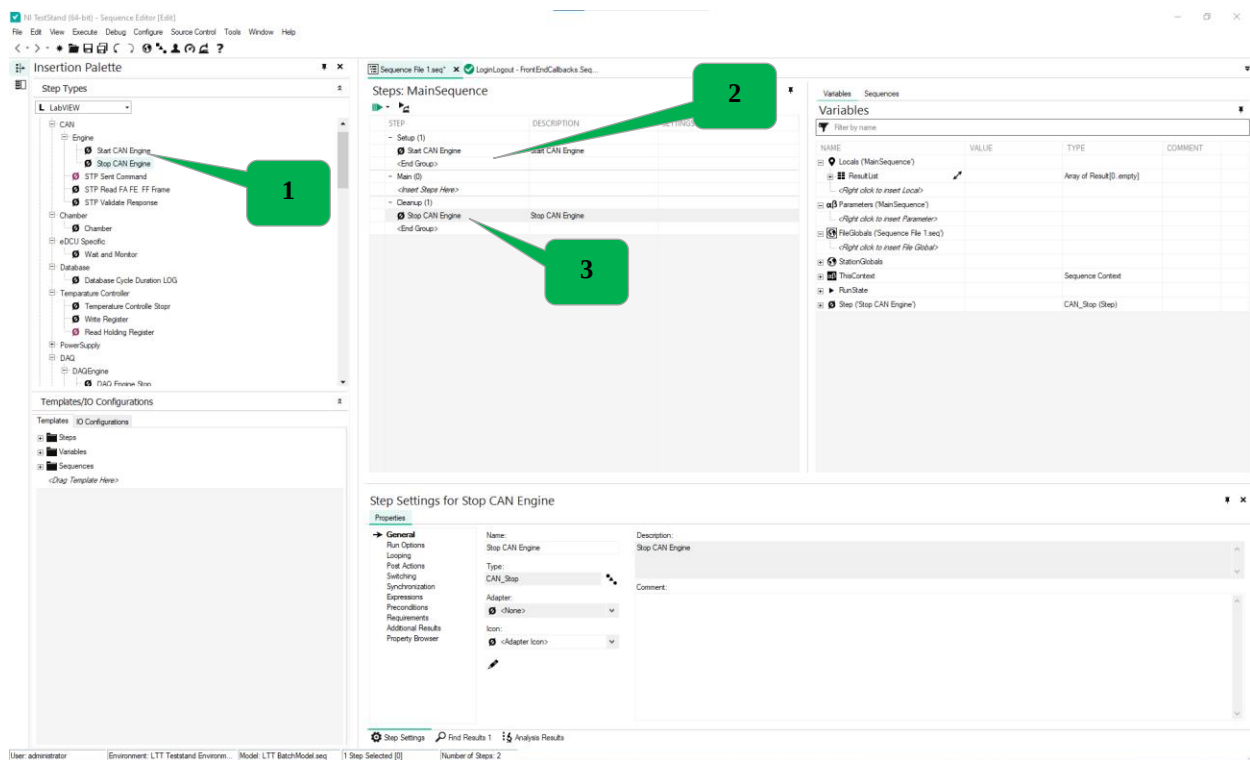


Figure 90: CAN Engine Start and Stop drag and drop.

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13.3.1.1 STP Sent Command.

- 1) Always drag and configure the step in the Main Step sequence.
- 2) Select the ECU CAN Port.
- 3) Choose the Command.
- 4) Write the CAN Write Data.
- 5) Enter the Positive response code and compare it internally during the post-execution step.
- 6) In the Sequence Specific step, click on the STP Sent Command. The Edit Window will appear, displaying the configuration details. Click the tick mark to store the step during execution or click the intended mark to avoid storing the step and retain the old data.

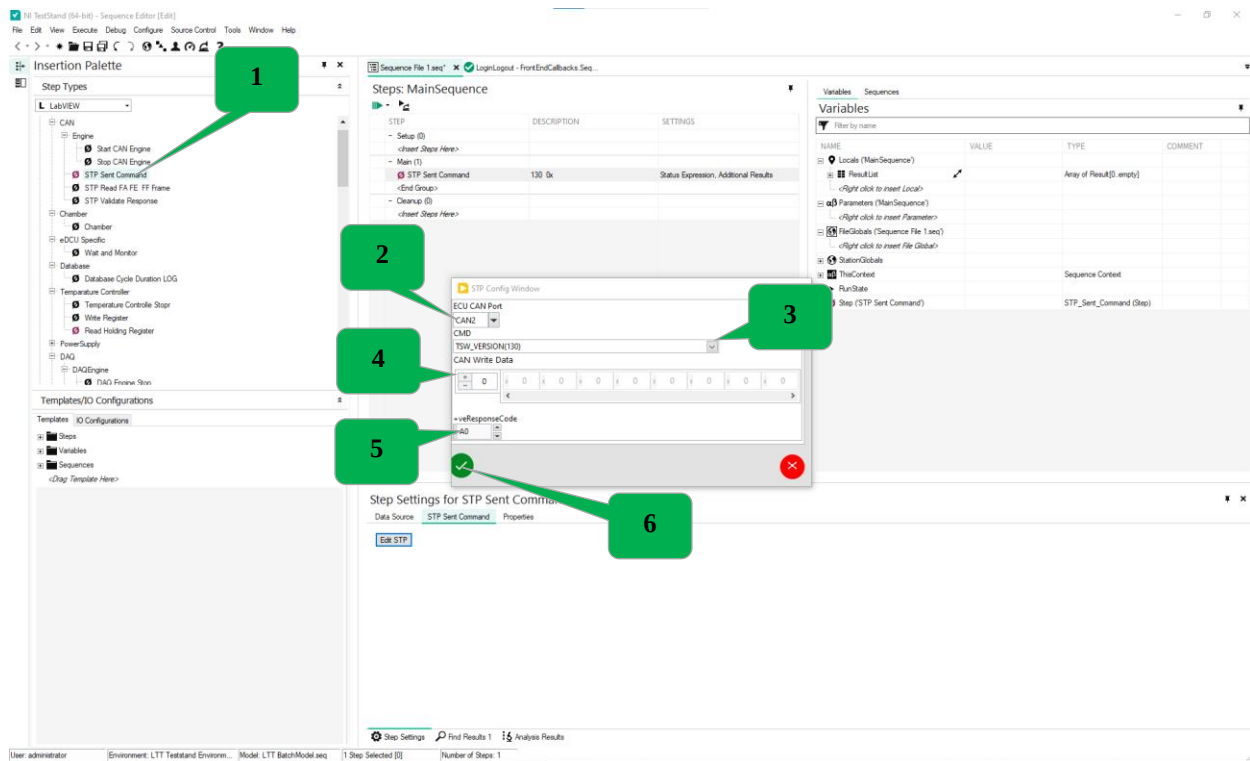


Figure 91: Configure the STP Sent Command step

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13.3.1.2 STP Read FA FE FF Command.

- 1) Always drag and configure the step in the Main Step sequence.
- 2) Select the ECU CAN Port.
- 3) Choose the Command.
- 4) Write the CAN Write Data.
- 5) Choose the Frame Type.
- 6) In the Sequence Specific step, click on the STP Read FA FE FF Command. The Edit Window will appear, displaying the configuration details. Click the tick mark to store the step during execution or click the intended mark to avoid storing the step and retain the old data.

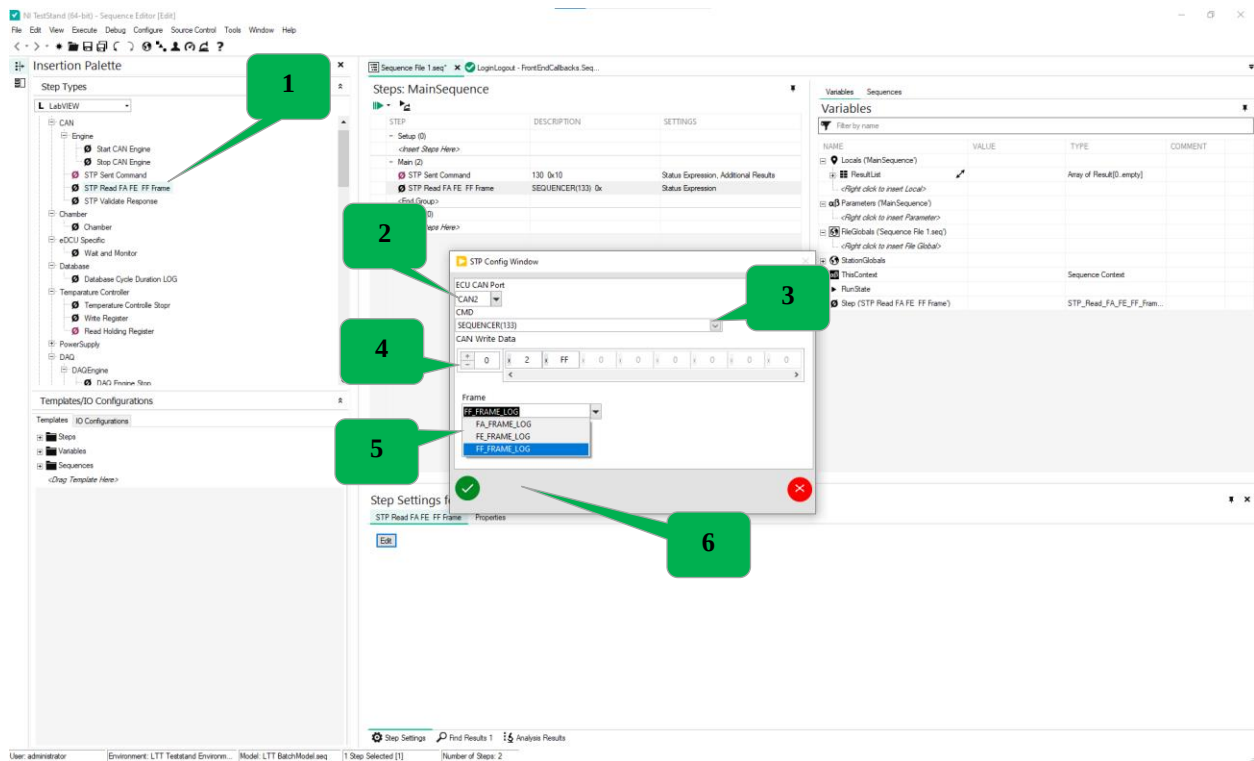


Figure 92: Configure the STP Read FA FE FF Command step

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13.3.1.3 STP Validate Response.

- 1) Always drag and configure the step in the Main Step sequence.
- 2) Select the ECU CAN Port.
- 3) Choose the Command.
- 4) Write the CAN Write Data.
- 5) Choose the Response decoder.
- 6) If the Response Decoder String is selected, fill in the String details.
- 7) If the Response Decoder Number is selected, fill in the Number details.
- 8) If the Response Decoder Boolean is selected, fill in the Boolean details.
- 9) In the Sequence Specific step, click on the STP Validate Response. The Edit Window will appear, displaying the configuration details. Click the tick mark to store the step during execution or click the intended mark to avoid storing the step and retain the old data.

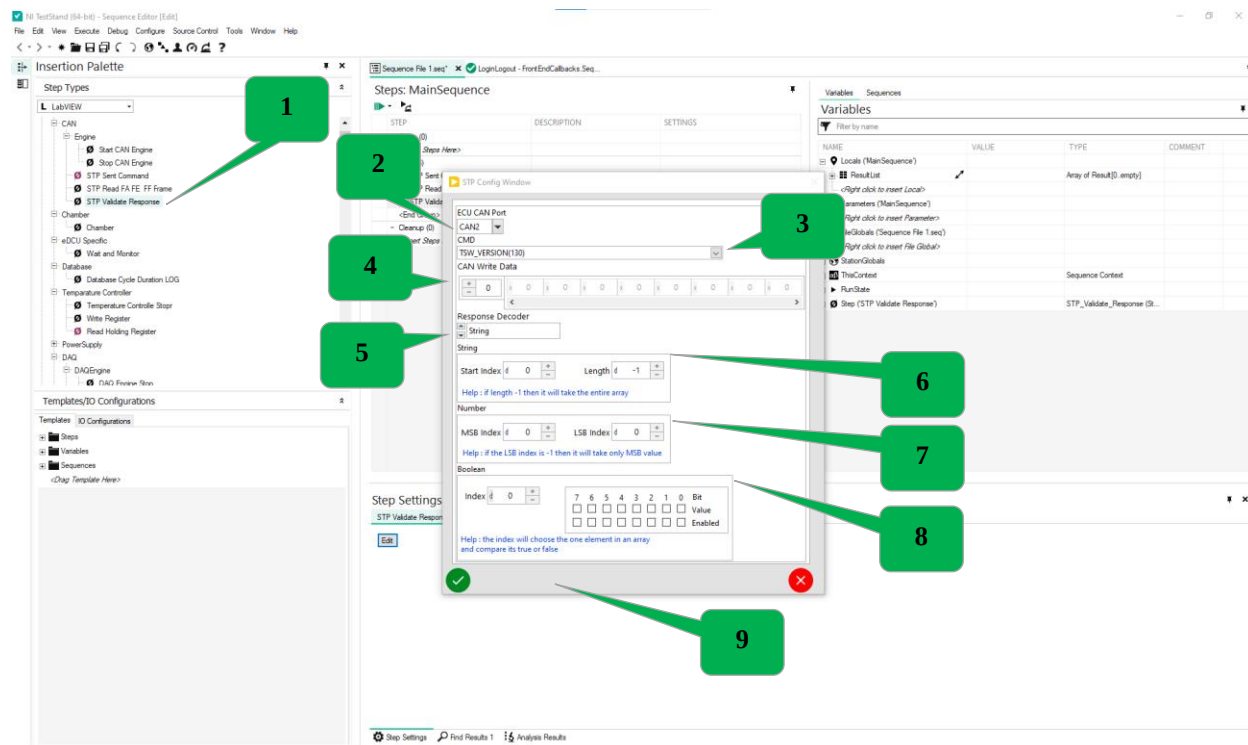


Figure 93: Configure the STP Validate Response step

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13.4 Chamber

13.4.1 Temperature Chamber.

- 1) Always drag and configure the step in the Main Step sequence.
- 2) Click Edit and go to the step properties.
- 3) Choose the Chamber Command.
- 4) In the Sequence Specific step, click on the Chamber. The Edit Window will appear, displaying the configuration details. Click the tick mark to store the step during execution or click the intended mark to avoid storing the step and retain the old data.

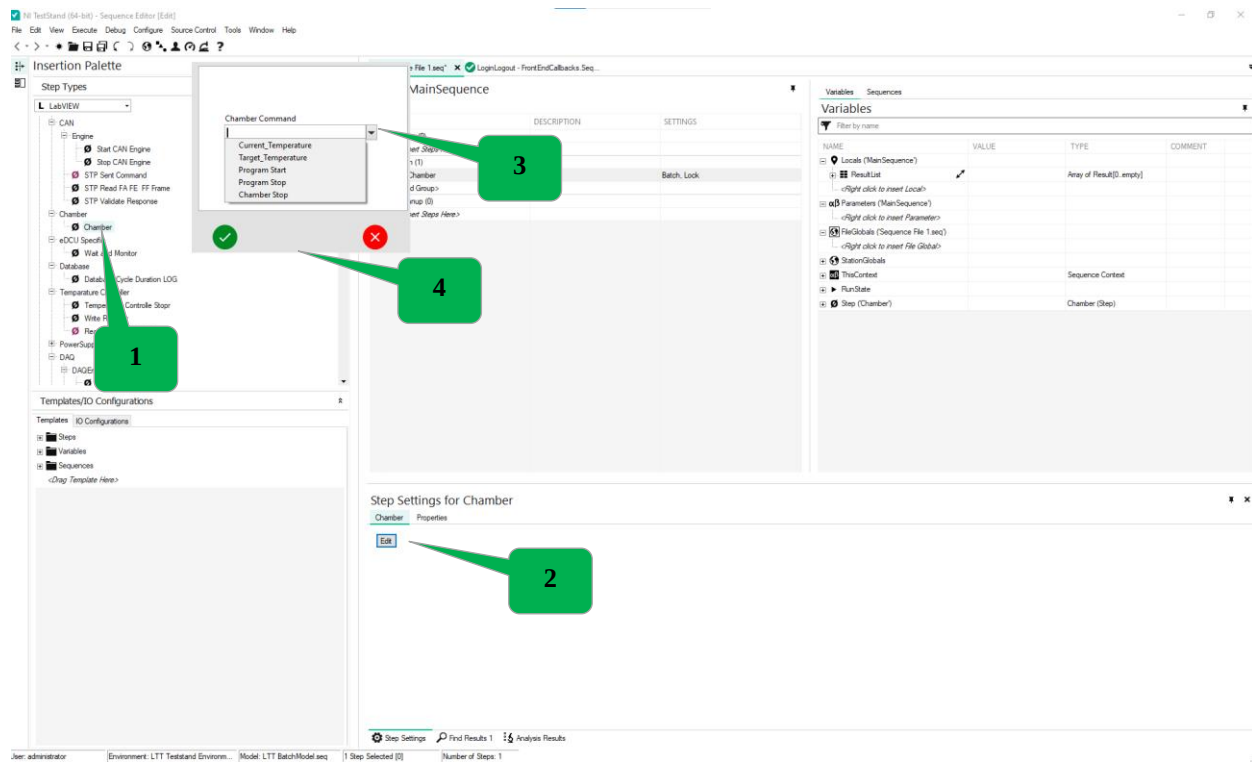


Figure 94: Chamber Step drag and drop.

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13.5 eDCU Specific

13.5.1 Wait and Monitor.

- 1) Always drag and configure the step in the Main Step sequence.
- 2) Click Edit and go to the step properties.
- 3) Enter the option and display the configuration details, which can be customized according to your requirements.
- 4) In the Sequence Specific step, click on the Wait and Monitor. The Edit Window will appear, displaying the configuration details. Click the tick mark to store the step during execution or click the intended mark to avoid storing the step and retain the old data.

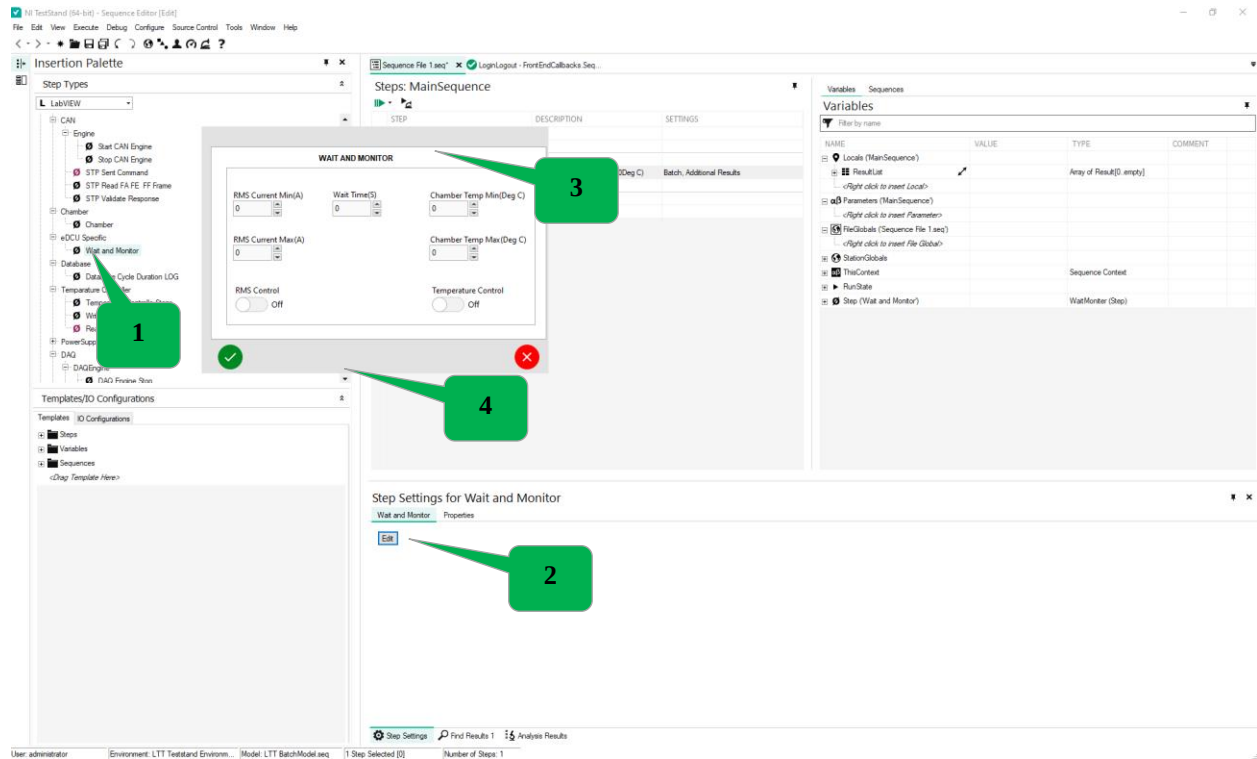


Figure 95: Wait and Monitor drag and drop.

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13.6 Database

13.6.1 Database Cycle Duration Log.

- 1) Always drag and configure the step in the Main Step sequence.
- 2) Click Edit and go to the step properties.
- 3) Choose the Cycle Duration Log.
- 4) In the Sequence Specific step, click on the Wait and Monitor. The Edit Window will appear, displaying the configuration details. Click the tick mark to store the step during execution or click the intended mark to avoid storing the step and retain the old data.

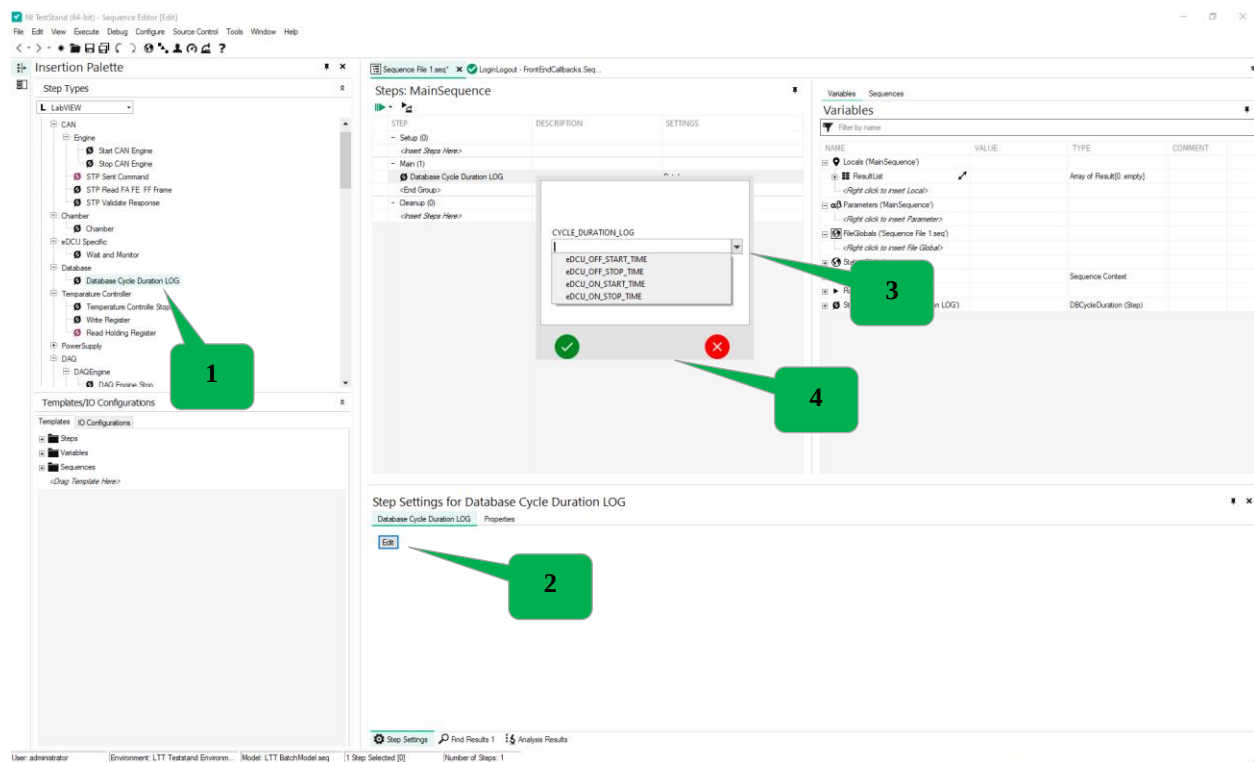


Figure 96: Database Cycle Duration drag and drop.

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13.7 Temperature

13.7.1 Temperature Controller Stop.

- 1) Always drag and configure the step in the Cleanup Step sequence.
- 2) The Temperature Stop should be Clean up in the sequence specified, without the need for manual editing, as it operates internally.

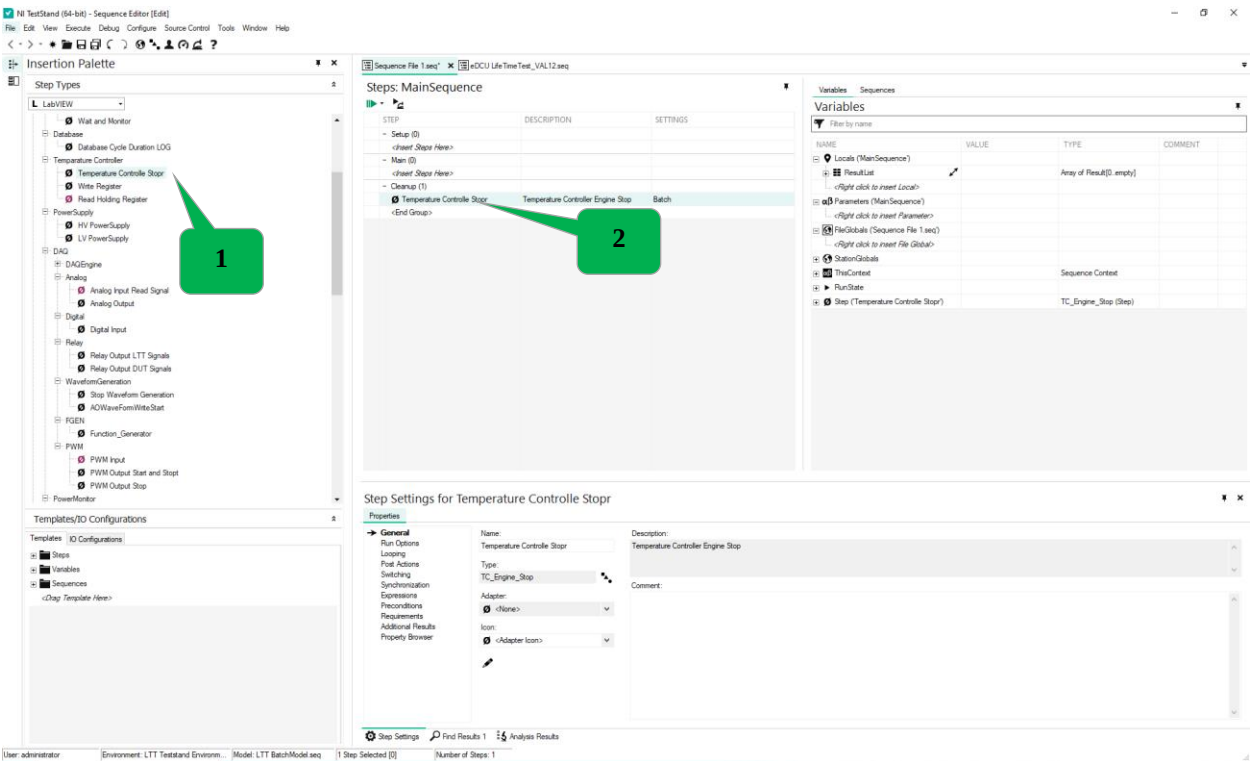


Figure 97: Temperature Controller drag and drop.

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13.8 Power Supply

13.8.1 Low Voltage Power Supplies.

- 1) In the Insertion Palette, click on **PowerSupply**, then drag and drop the **LVPowerSupply** step.
- 2) Select the step and click "Go to Edit Step."
- 3) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details, which can be filled in according to your requirements.

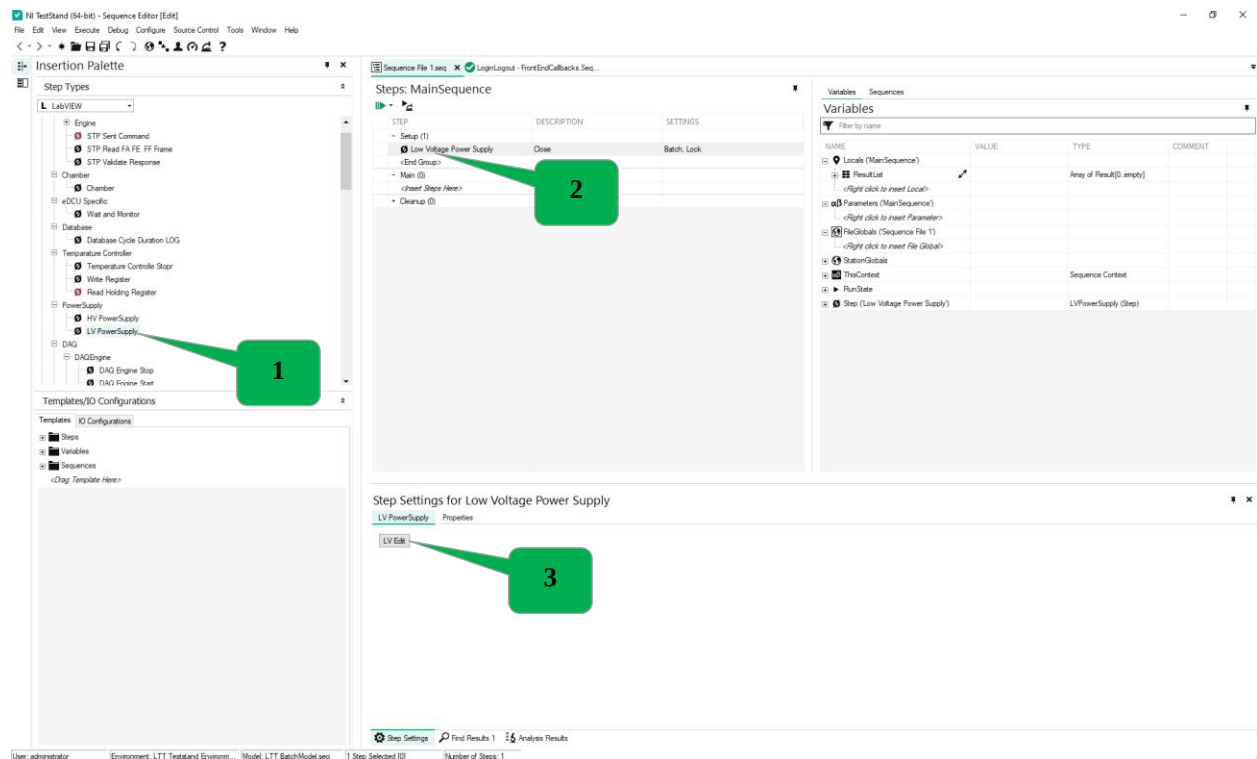


Figure 98: Low voltage power supply drag and drop.

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13.8.1.1 Low Voltage Power Supplies Initialize.

- 1) Click Edit and go to the step properties.
- 2) Always drag and configure the **Init** step in the sequence setup.
- 3) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

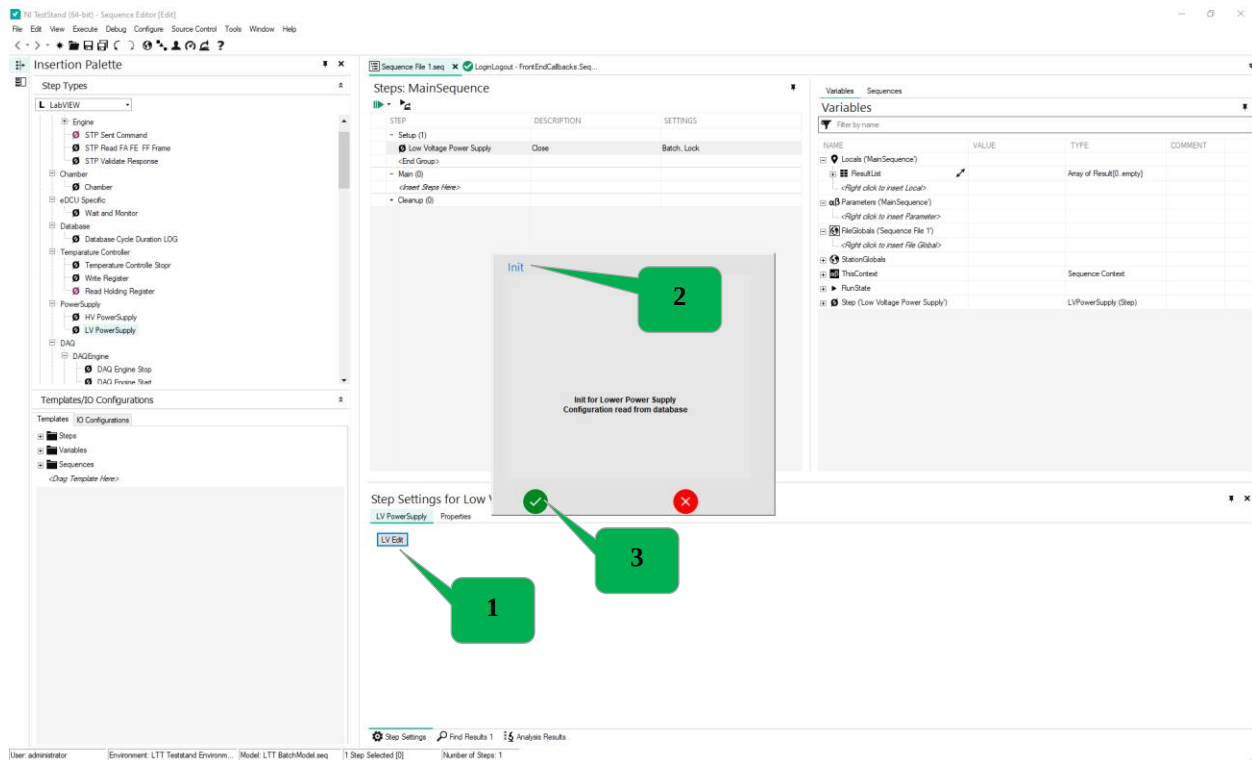


Figure 99: Configure the Init step

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13.8.1.2 Low Voltage Power Supplies Configure.

- 1) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details.
- 2) Select the 'Configure' option and display the configuration details, which can be customized according to your requirements.
- 3) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

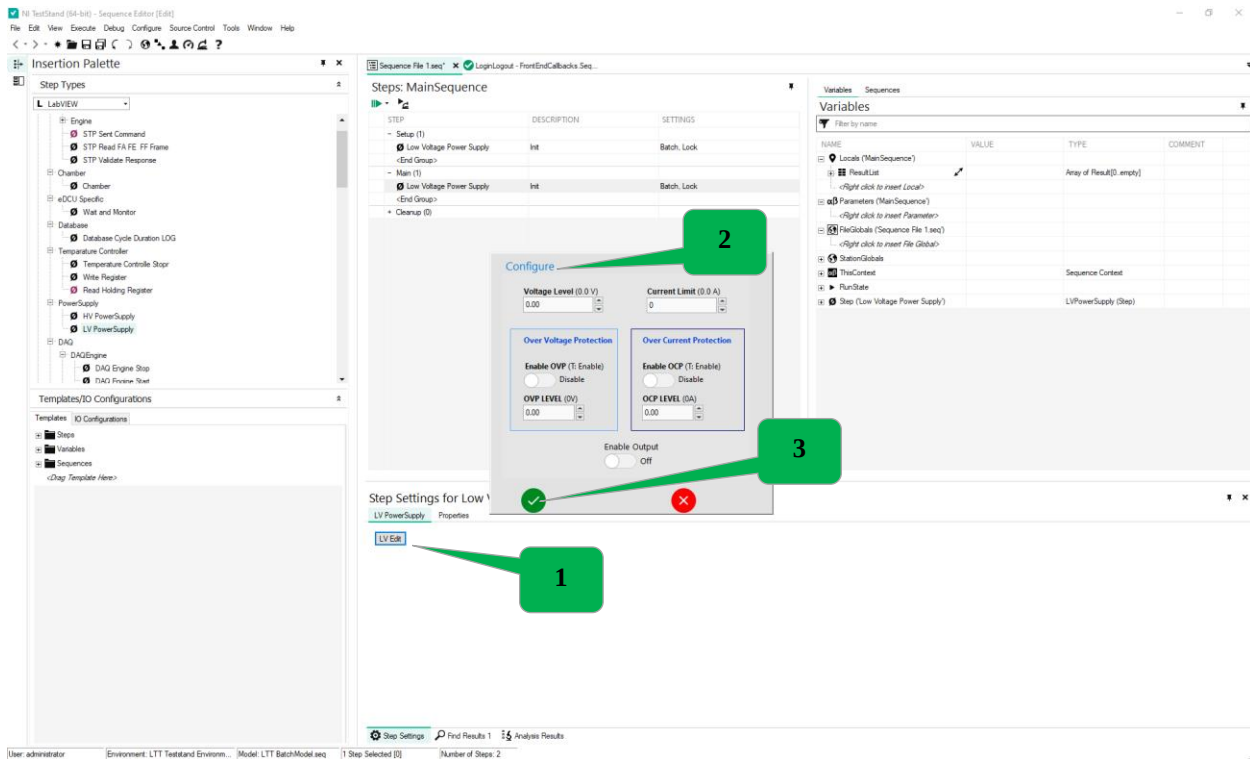


Figure 100: Configure the Configure step

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13.8.1.3 Low Voltage Power Supplies Measure.

- 1) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details.
- 2) Select the 'Measure' option and display the configuration details, which can be customized according to your requirements.
- 3) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

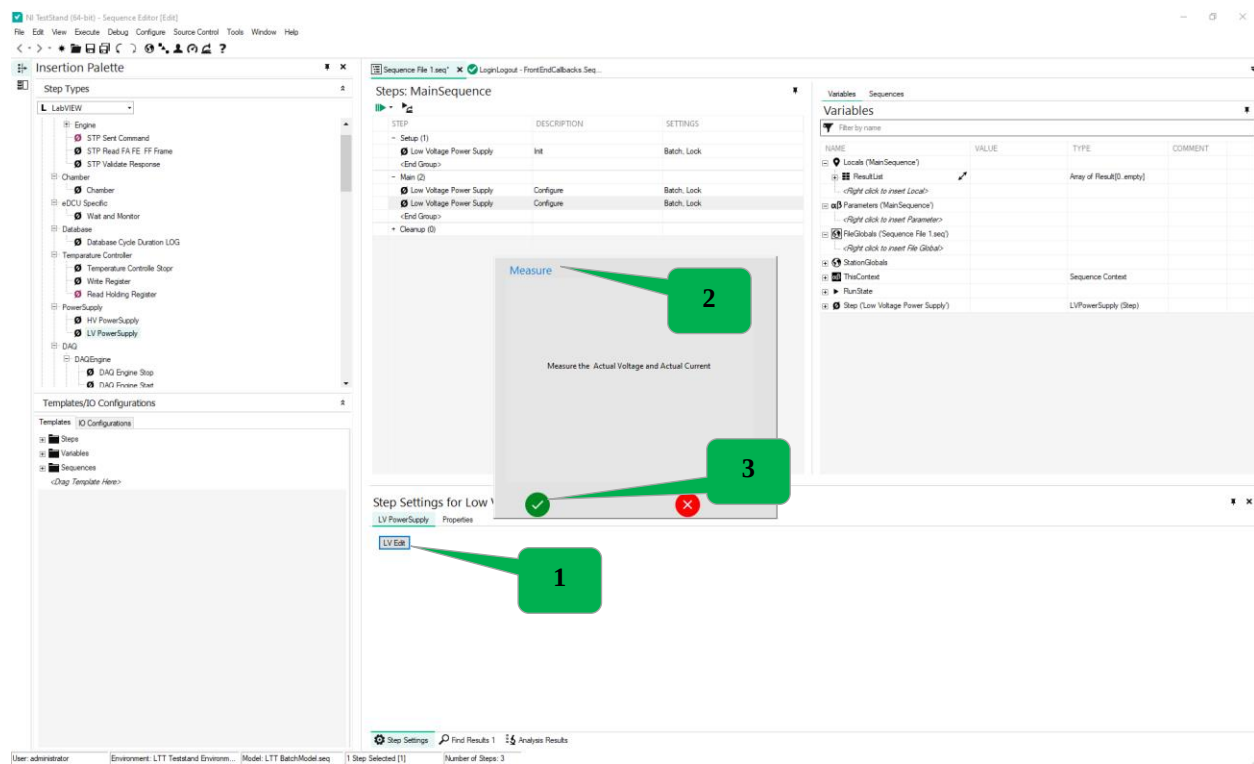


Figure 101: Configure the Measure step

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13.8.1.4 Low Voltage Power Supplies Close.

- 1) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details.
- 2) Select the 'Close' option and display the configuration details, which can be customized according to your requirements.
- 3) In the **Sequence Specific** step, click on **LV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

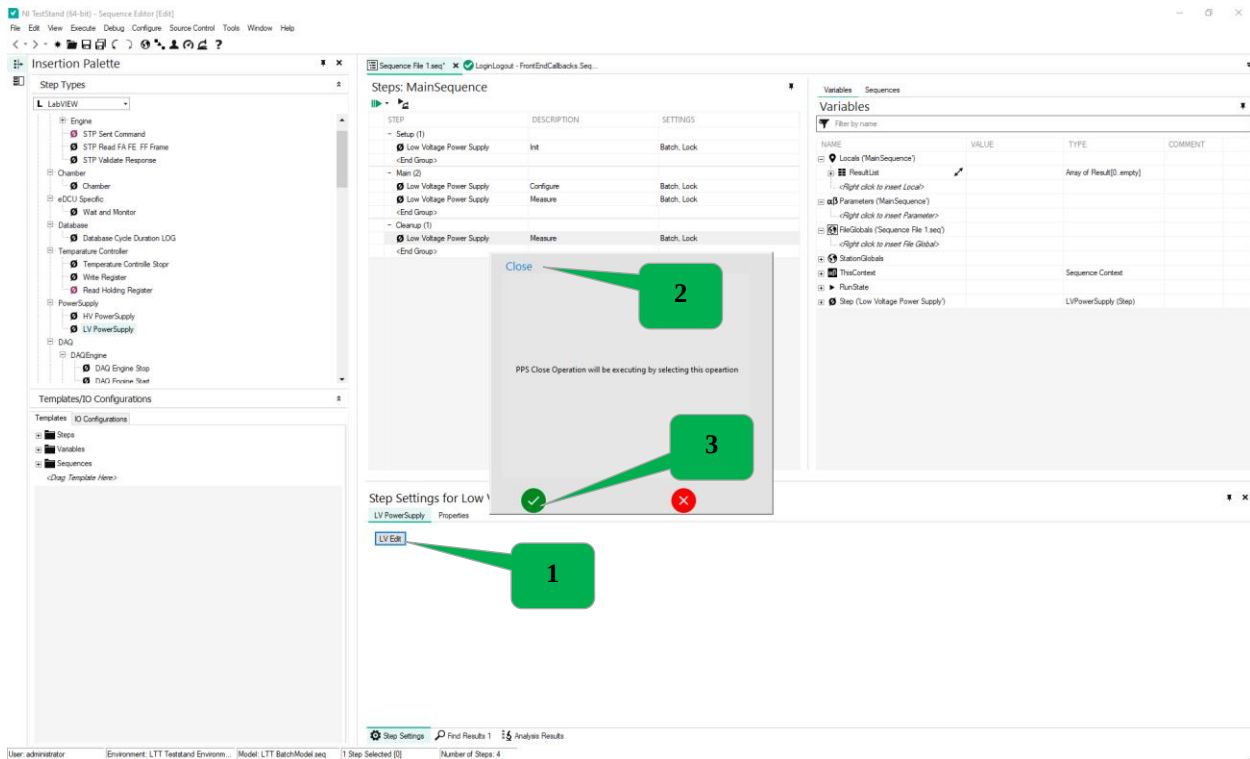


Figure 102: Configure the Close step

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13.8.1.5 Low Voltage Power Supplies Sequence View.

- 1) The initialization of the instrument should follow the sequence setup.
- 2) The sequence configuration should be related to the main section.
- 3) The instrument close should be handled in the sequence cleanup.

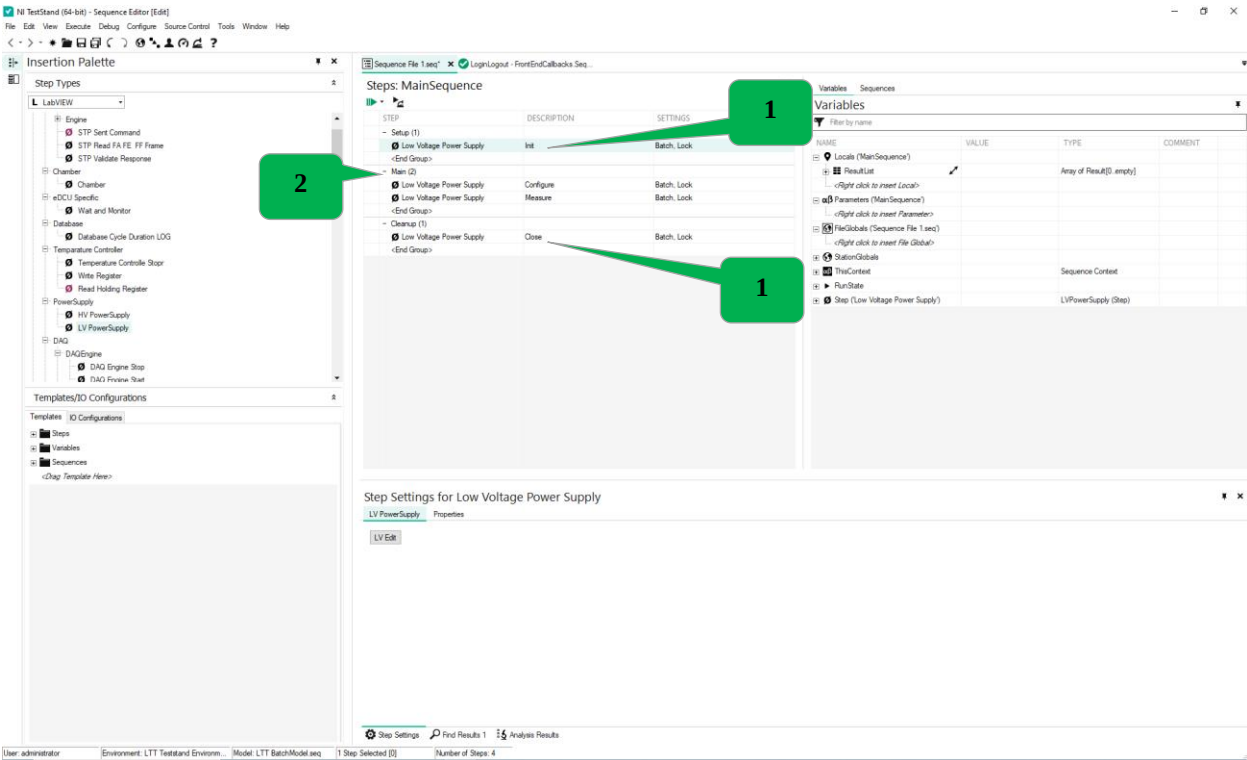


Figure 103:Low Power supply sequence development

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13.8.2 High Voltage Power Supplies.

- 1) In the Insertion Palette, click on **PowerSupply**, then drag and drop the **HVPowerSupply** step.
- 2) Select the step and click "Go to Edit Step."
- 3) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details, which can be filled in according to your requirements.

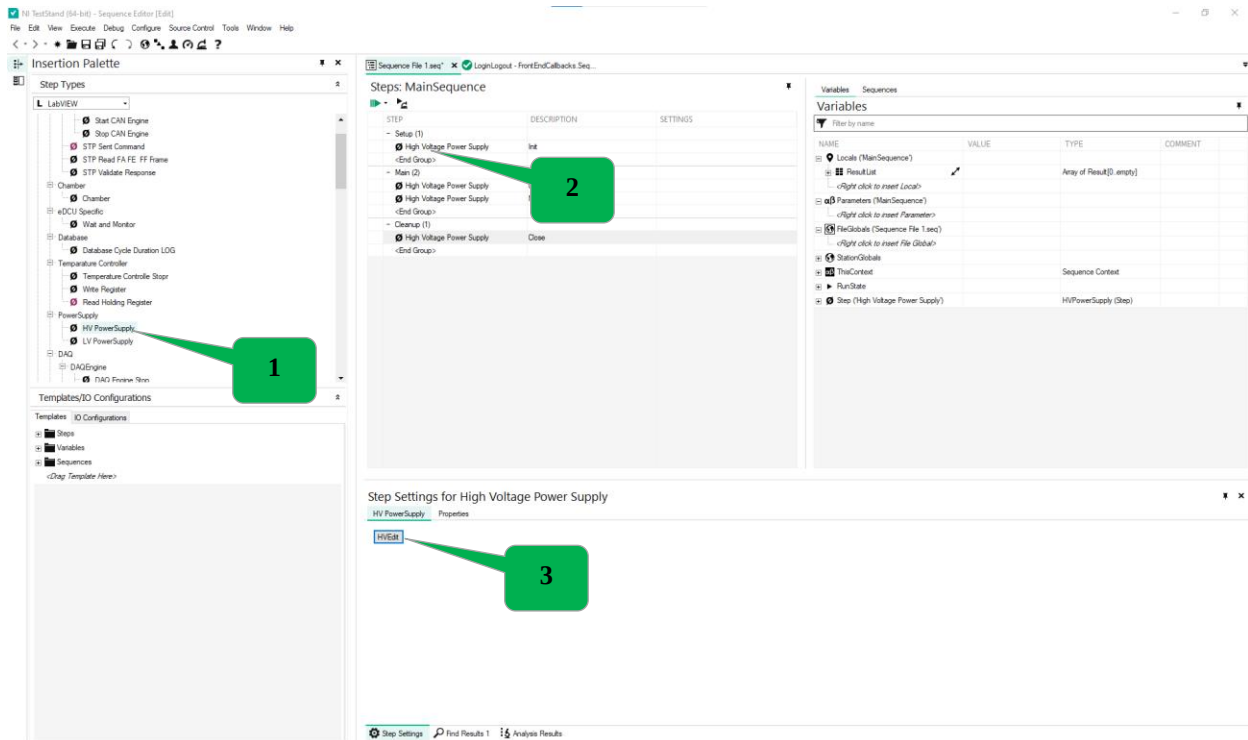


Figure 104: Low voltage power supply drag and drop.

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13.8.2.1 High Voltage Power Supplies Initialize.

- 1) Click Edit and go to the step properties.
- 2) Always drag and configure the **Init** step in the sequence setup.
- 3) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

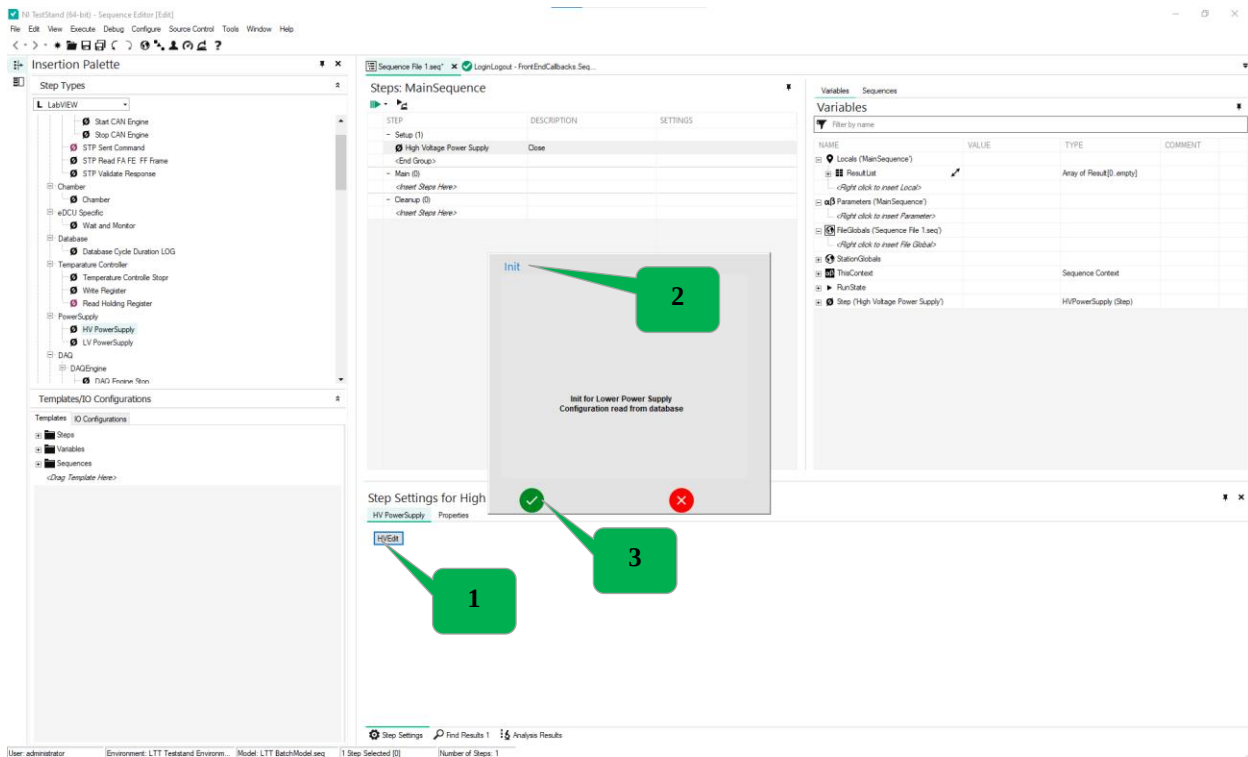


Figure 105: Configure the Init step

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13.8.2.2 High Voltage Power Supplies Configure.

- 1) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details.
- 2) Select the 'Configure' option and display the configuration details, which can be customized according to your requirements.
- 3) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

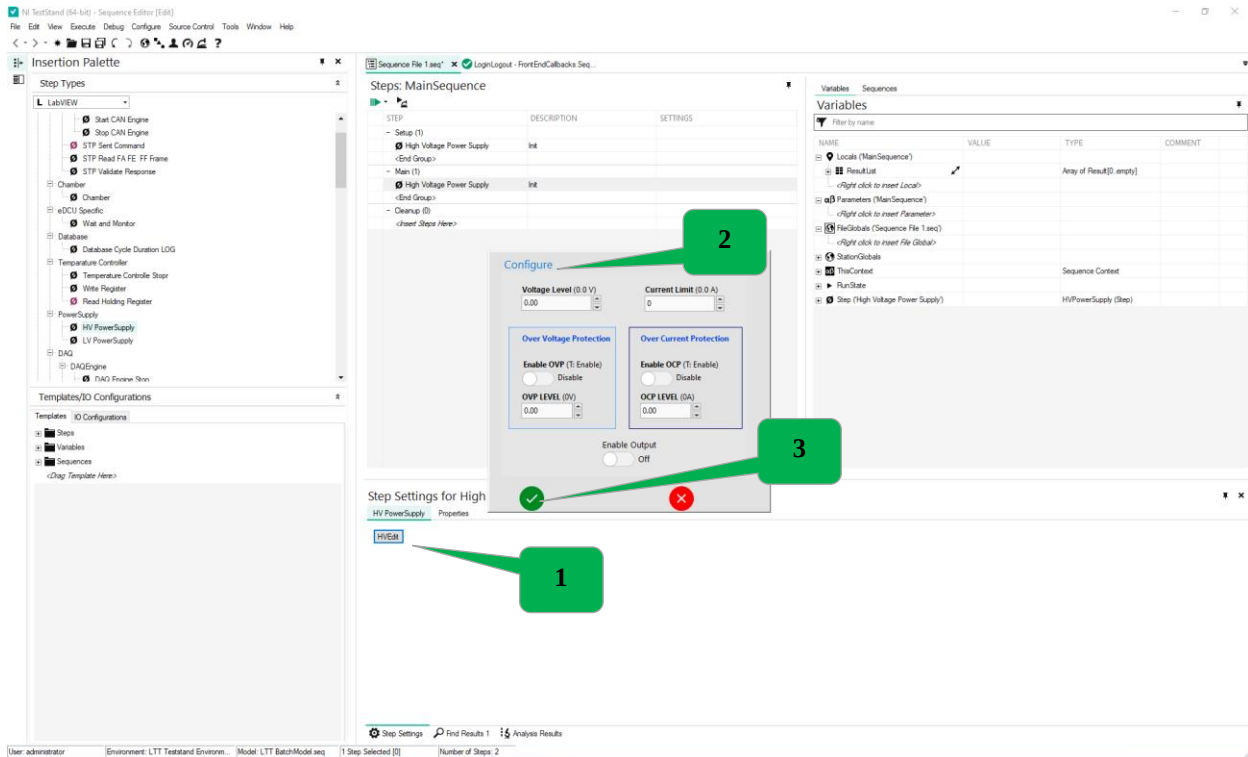


Figure 106: Configure the Configure step

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13.8.2.3 High Voltage Power Supplies Measure.

- 1) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details.
- 2) Select the 'Measure' option and display the configuration details, which can be customized according to your requirements.
- 3) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

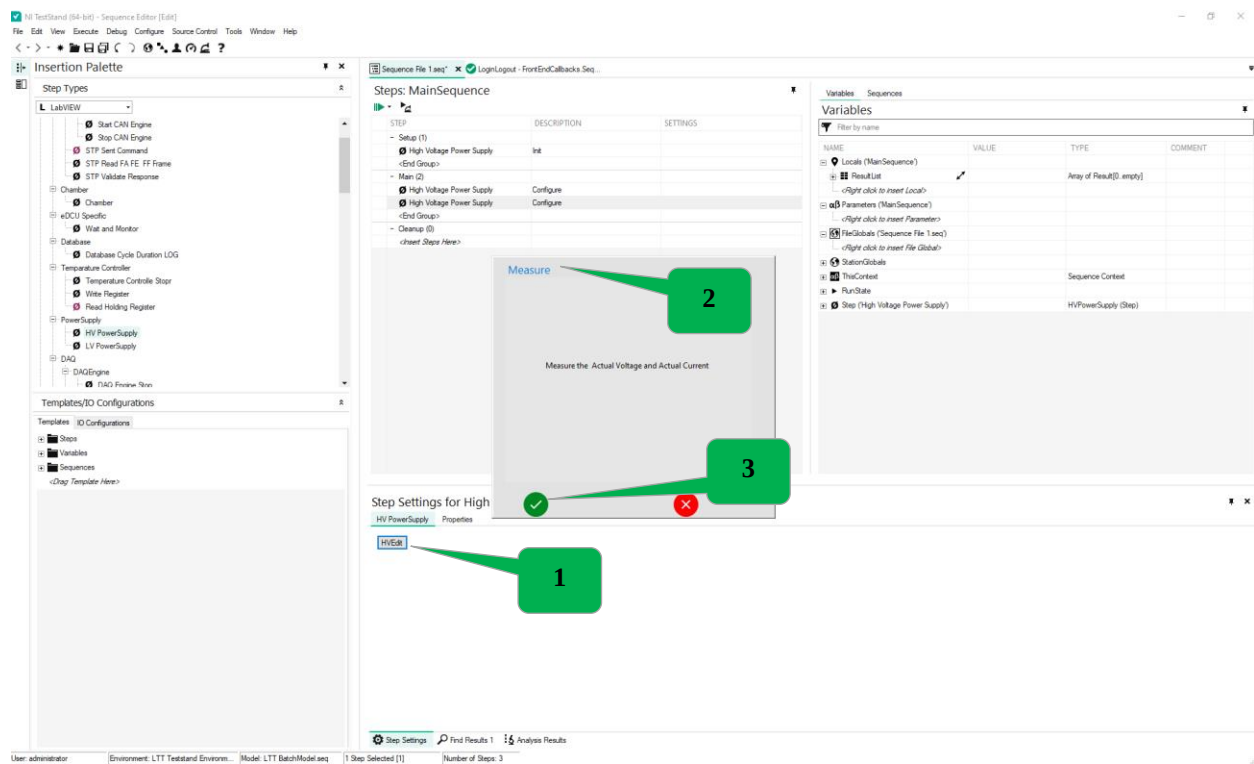


Figure 107: Configure the Measure step

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13.8.2.4 High Voltage Power Supplies Close.

- 1) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details.
- 2) Select the 'Close' option and display the configuration details, which can be customized according to your requirements.
- 3) In the **Sequence Specific** step, click on **HV**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

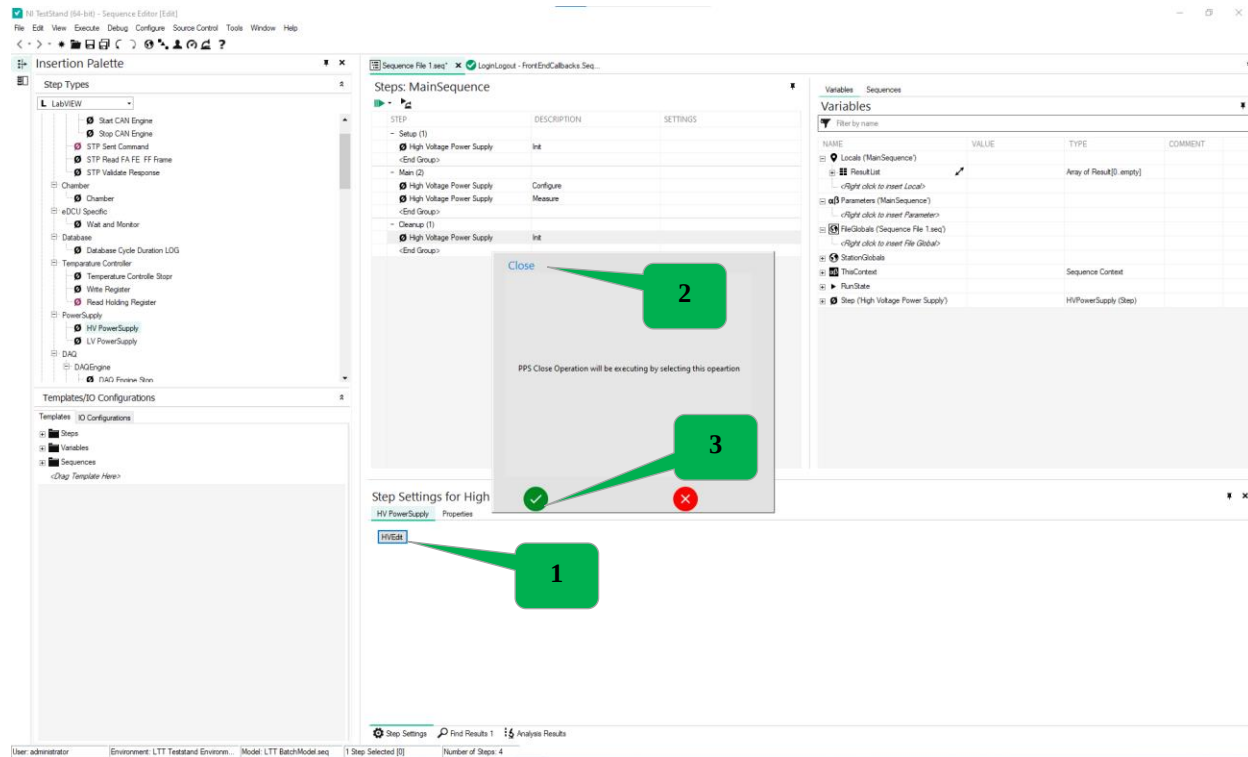


Figure 108: Configure the Close step

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13.8.2.5 High Voltage Power Supplies Sequence View.

- 1) The initialization of the instrument should follow the sequence setup.
- 2) The sequence configuration should be related to the main section.
- 3) The instrument close should be handled in the sequence cleanup.

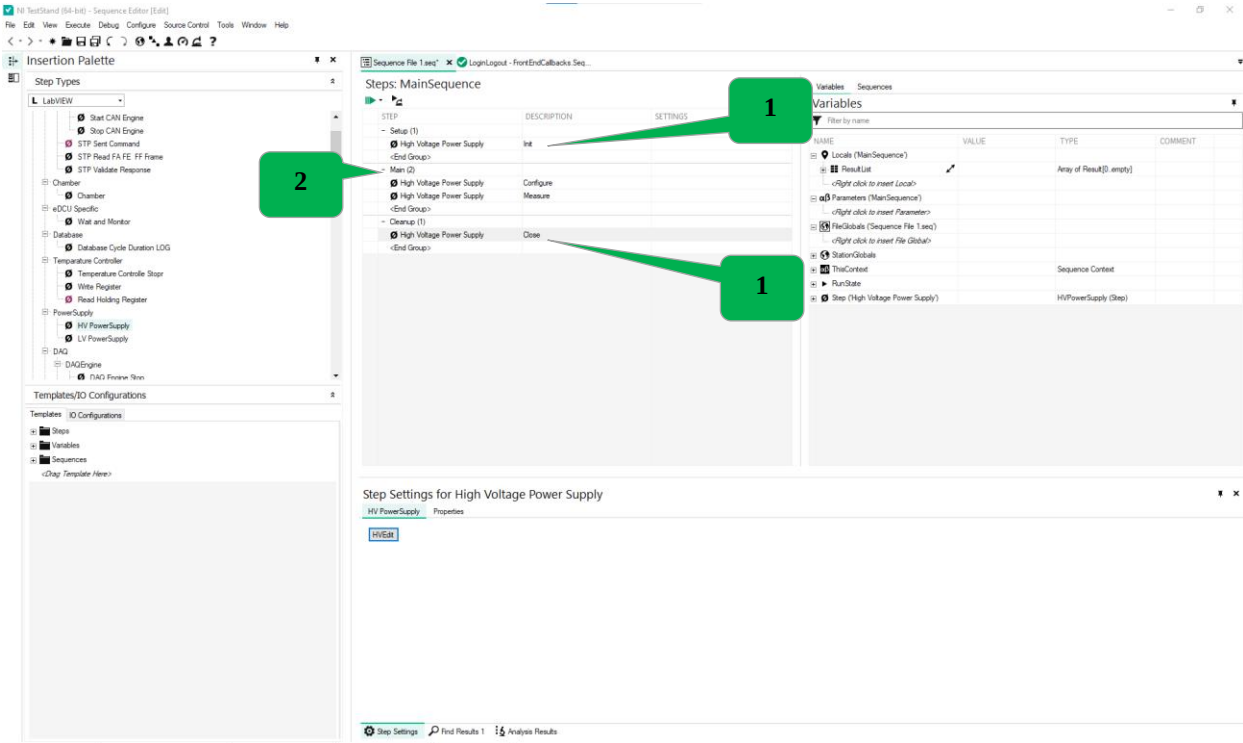


Figure 109:Low Power supply sequence development

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13.9.1.1 Analog Input Read Signal.

- 1) In the Insertion Palette, click on **DAQ** and Click the Analog then drag and drop the **Analog Input Read Signal** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Signal Input list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **AI Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

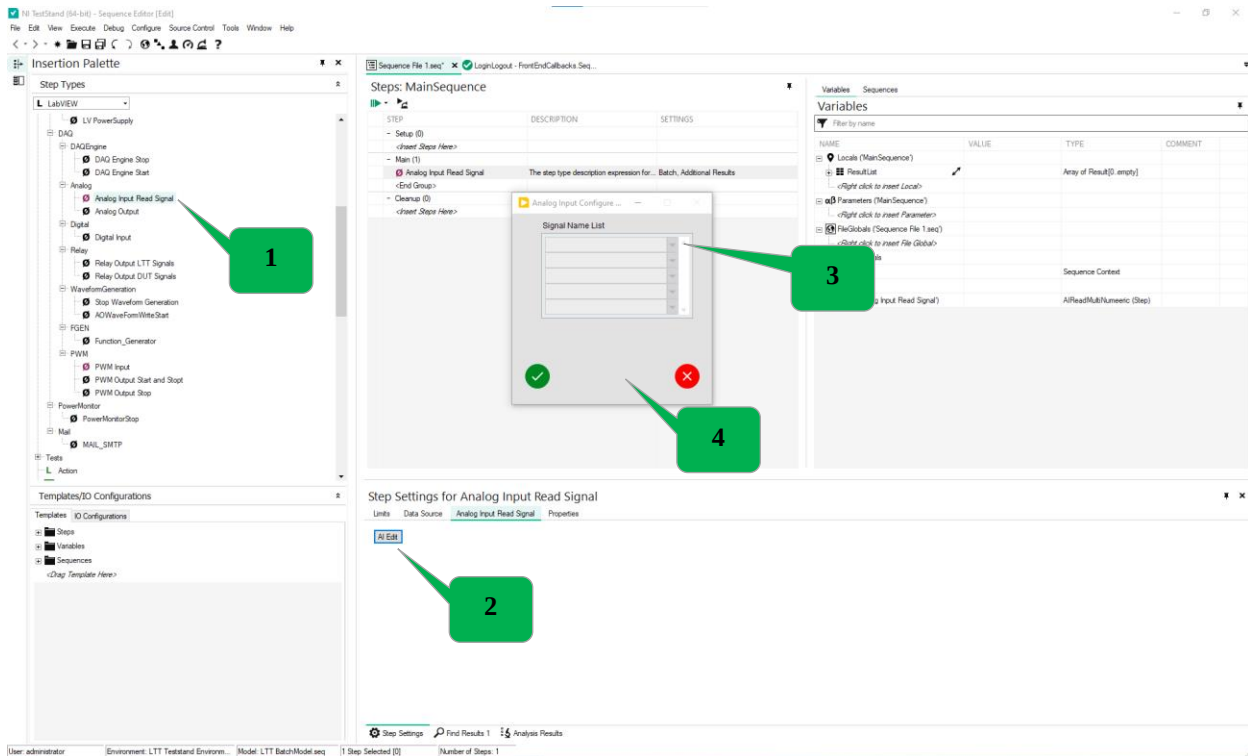


Figure 111: Analog Input Signal step

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13.9.1.2 Analog Output.

- 1) In the Insertion Palette, click on **DAQ** and Click the Analog then drag and drop the **Analog Output** step
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Output Signal list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **AO Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

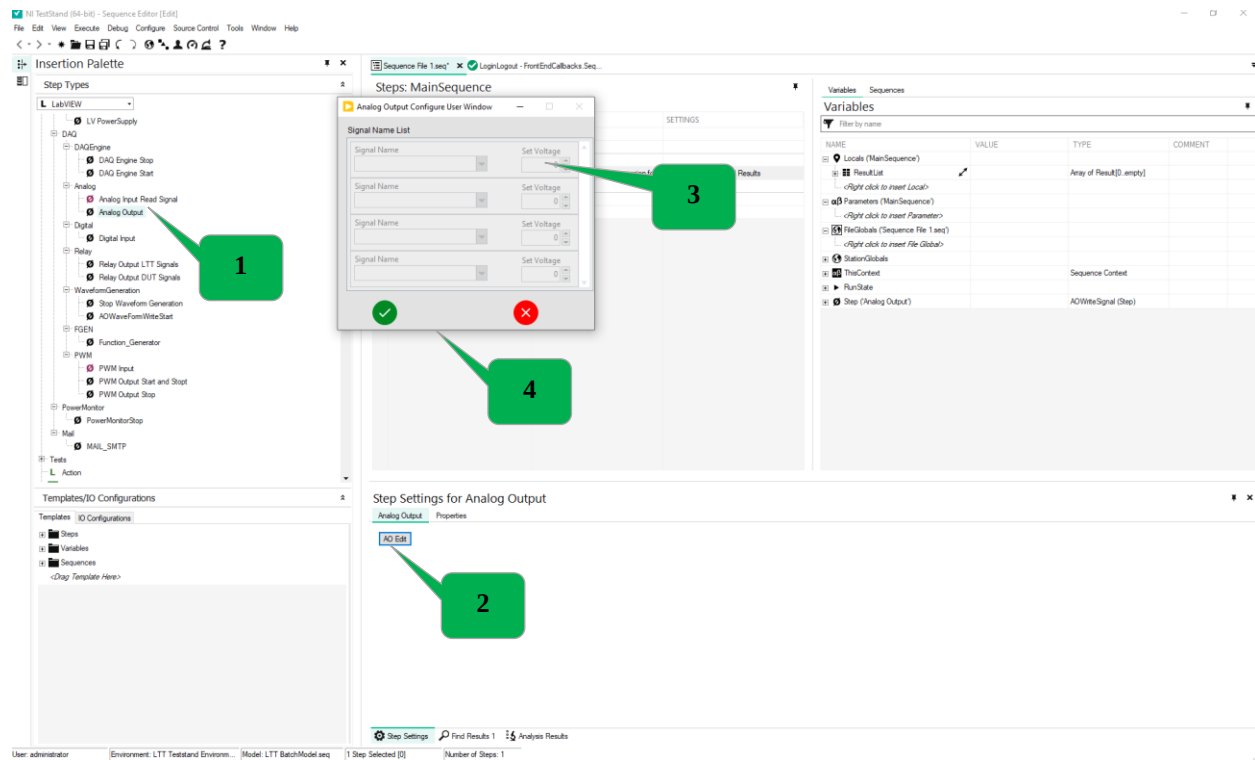


Figure 112: Analog Output Signal step

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13.9.1.3 Digital Input.

- 1) In the Insertion Palette, click on **DAQ** and Click the Digital then drag and drop the **Digital Input** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Input Signal list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **DI Read Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

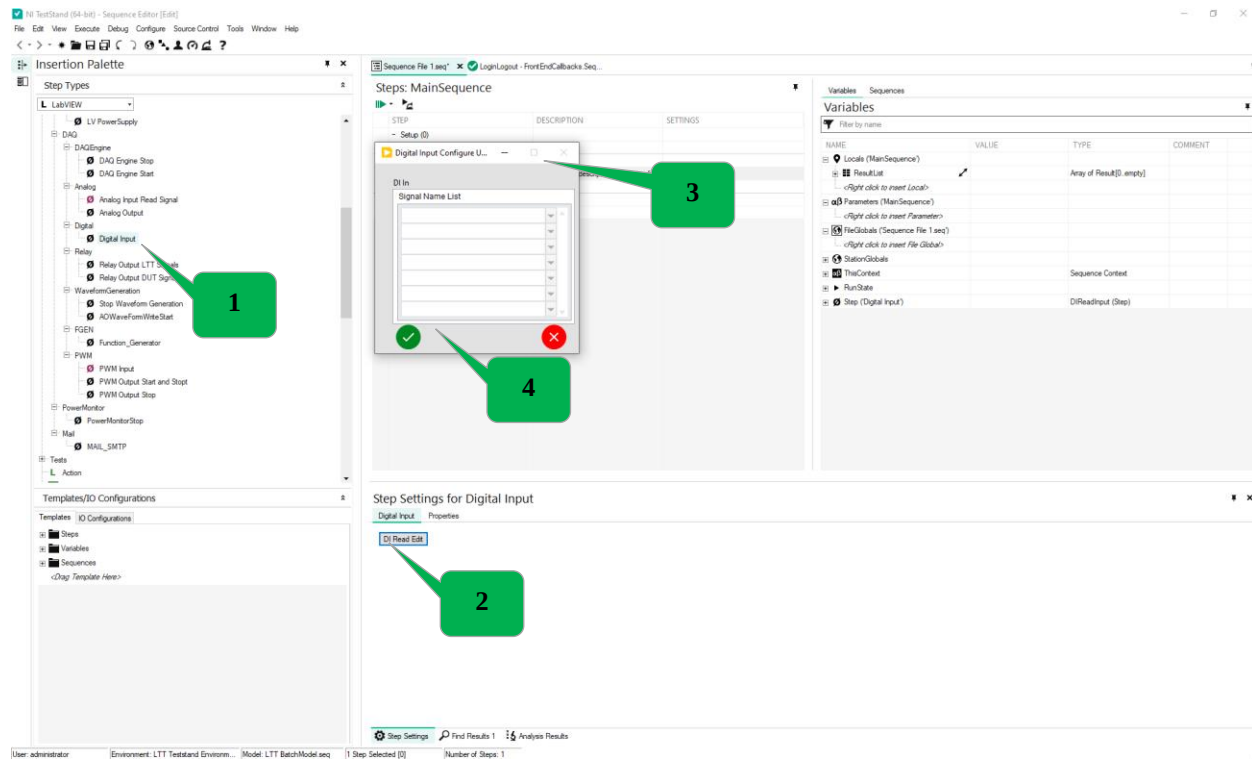


Figure 113: Digital Input Read Signal step

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13.9.1.4 Relay Output LTT Signal.

- 1) In the Insertion Palette, click on **DAQ** and Click the Relay then drag and drop the **Relay Output LTT Signal** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Output Signal list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

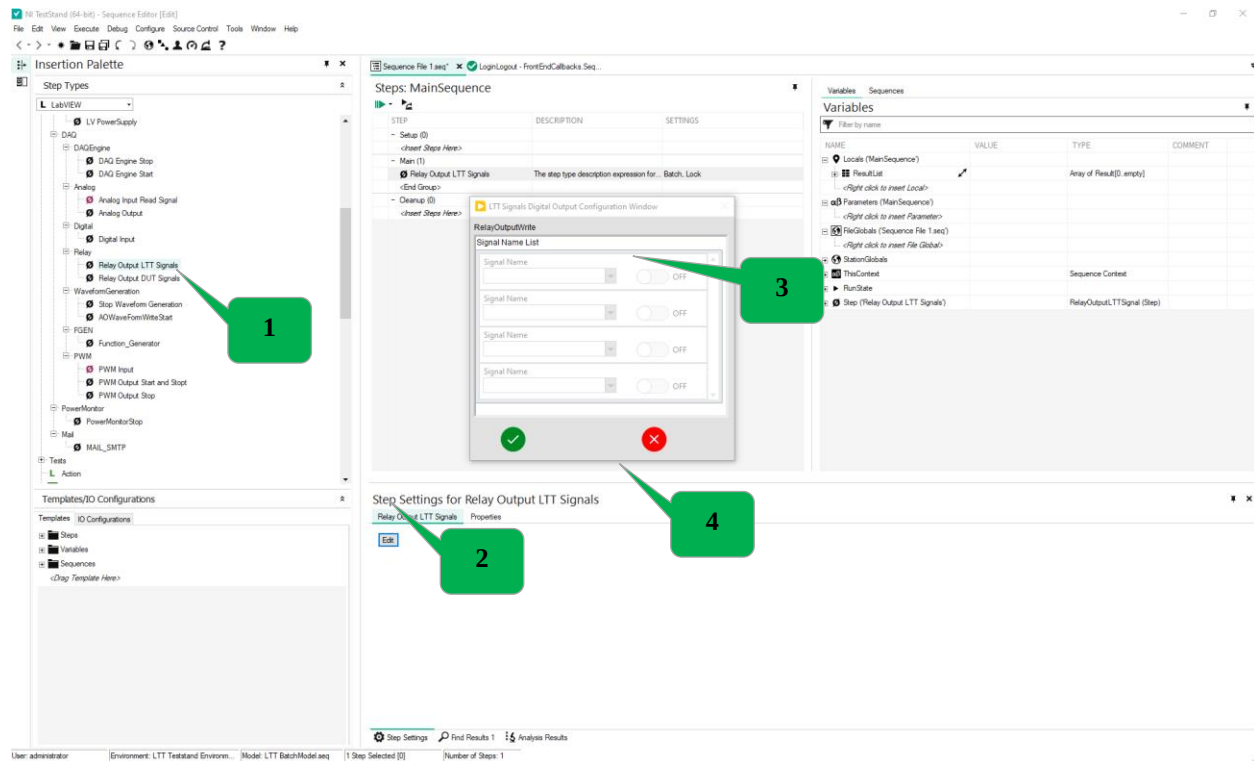


Figure 114: Relay Output LTT Signal step

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13.9.1.5 Relay Output DUT Signals.

- 1) In the Insertion Palette, click on **DAQ** and Click the Relay then drag and drop the **Relay Output DUT Signal** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Output Signal list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

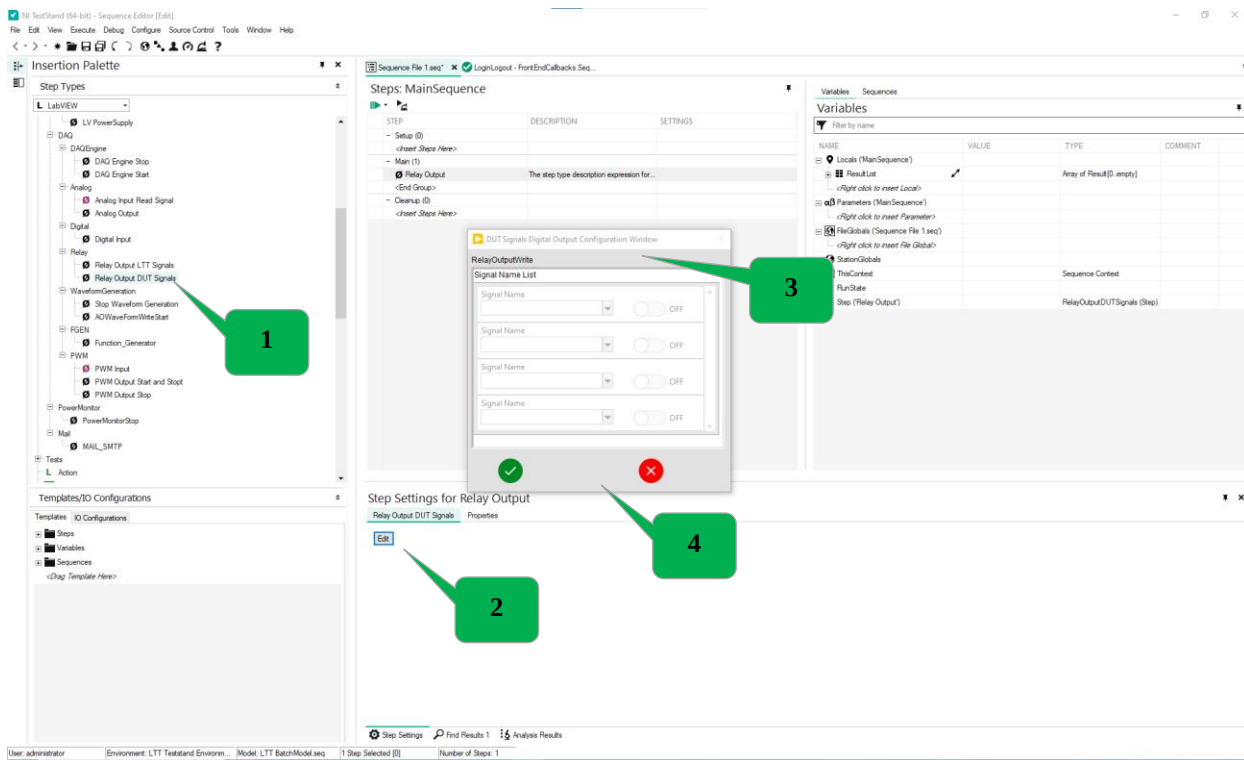


Figure 115: Relay Output DUT Signal step

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13.9.1.6 PWM Input

- 1) In the Insertion Palette, click on **DAQ** and Click the PWM then drag and drop the **PWM Input** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Input Signal list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **PWM Input Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

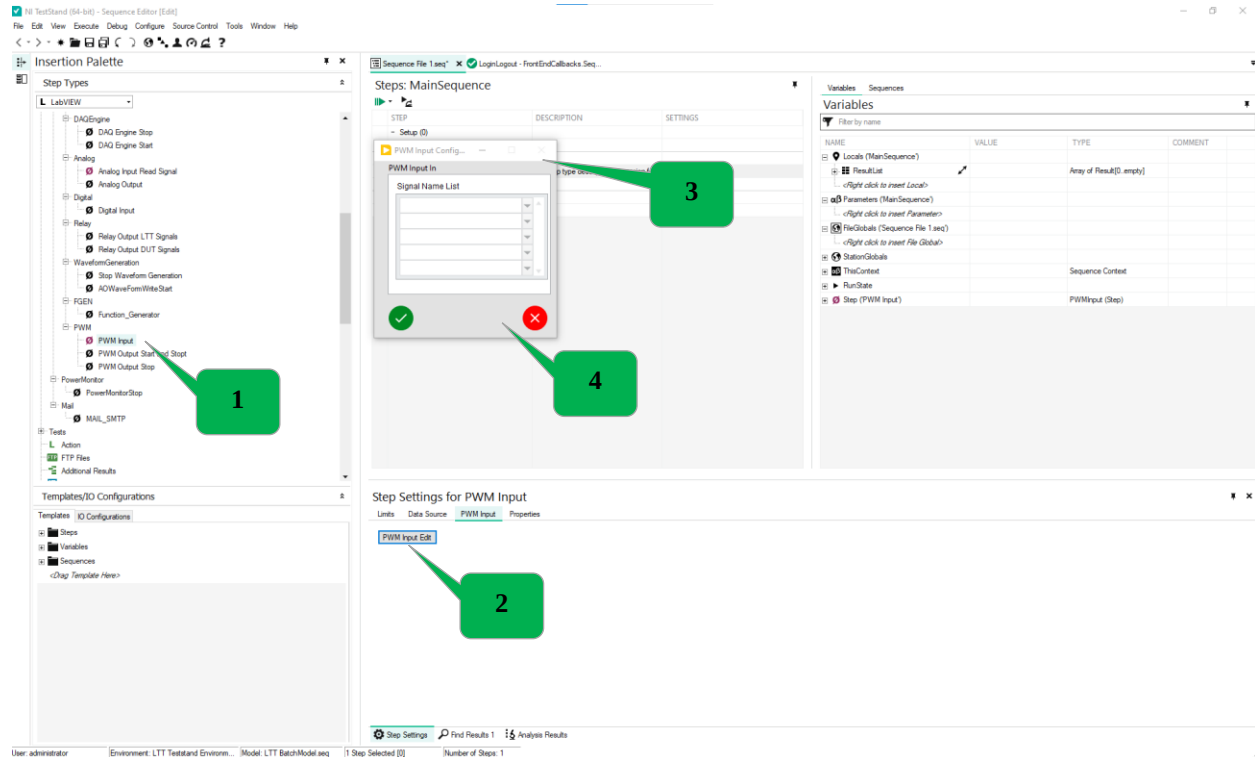


Figure 116: PWM Input Signal step

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13.9.1.7 PWM Output Start and Stop

- 1) In the Insertion Palette, click on **DAQ** and Click the Relay then drag and drop the **PWM Output Start and Stop** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the Output Signal list step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

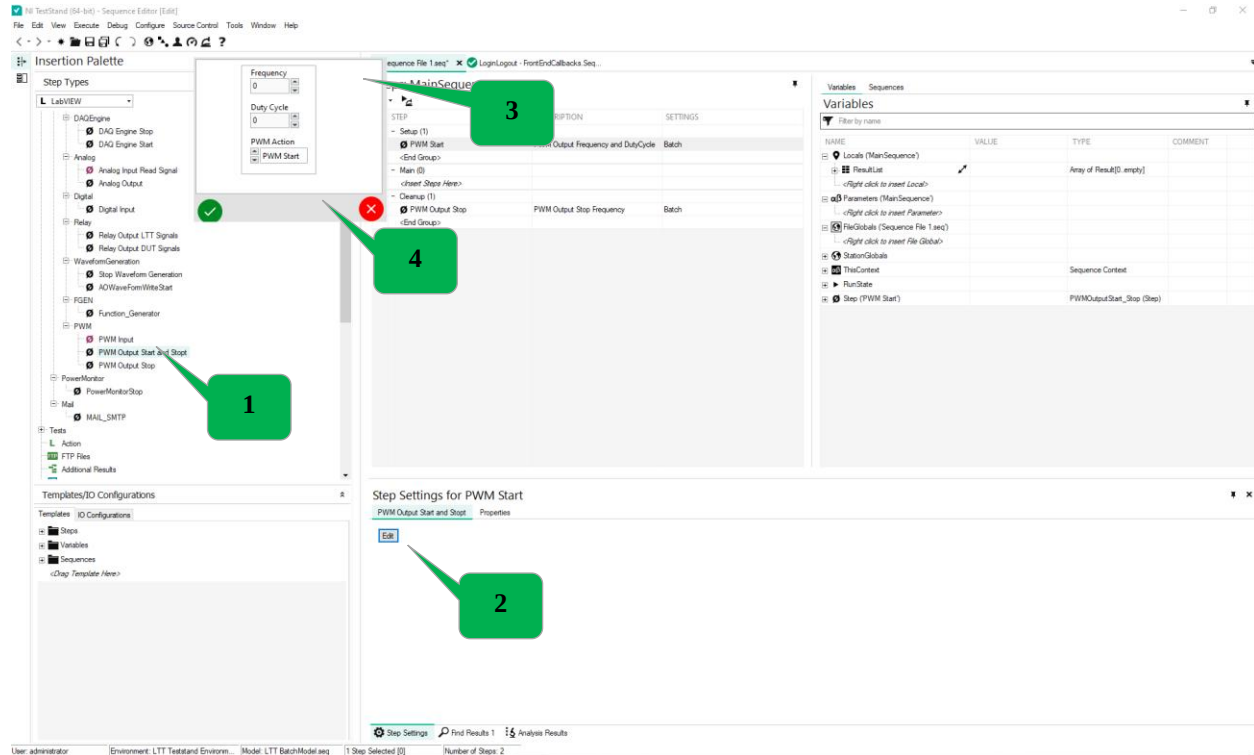


Figure 117: PWM Output Start and Stop Step

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13.10 Power Monitor

13.10.1 Power Monitor Stop.

- 1) In the Insertion Palette, click on **PowerMonitor** then drag and drop the **Power Monitor Stop** step.
- 2) The Stop Power Monitor Engine should be Clean up in the specified sequence, without the need for manual editing, as it operates internally.

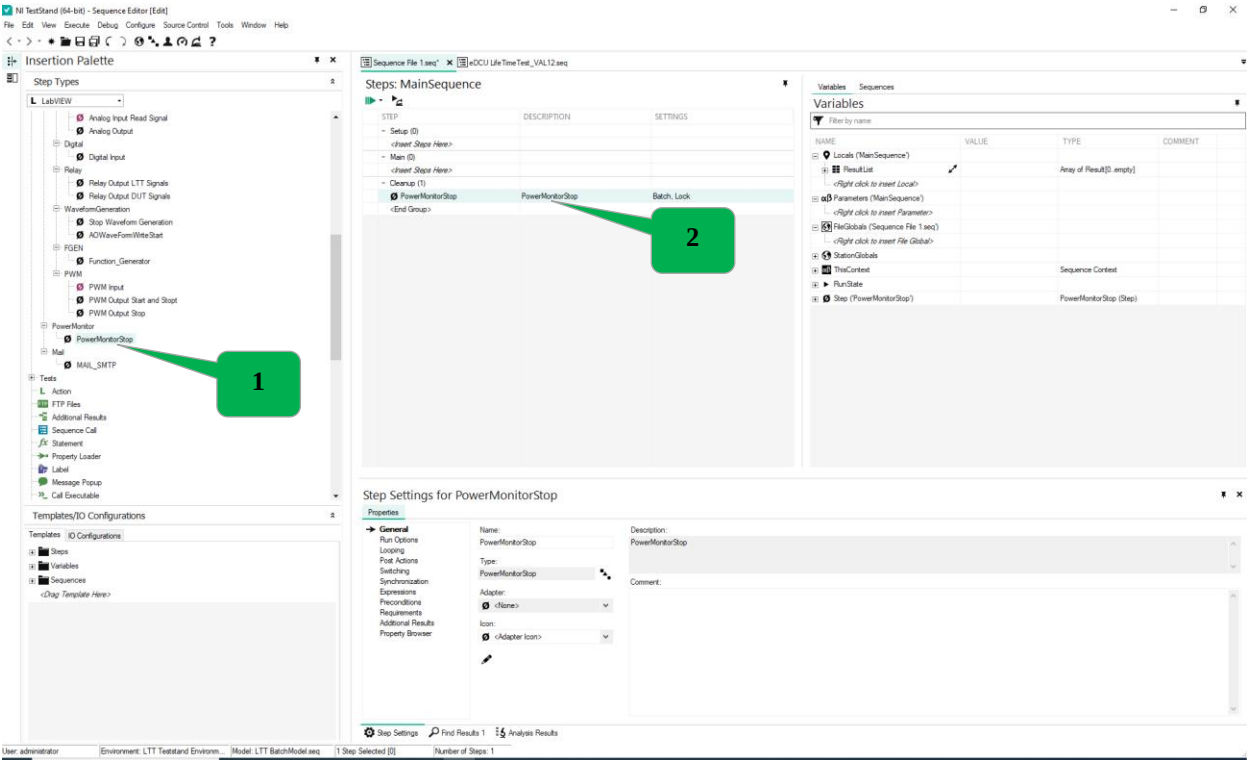


Figure 118 Power Monitor Stop drag and drop.

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13.11 Mail

13.11.1 SMTP Mail.

- 1) In the Insertion Palette, click on **Mail** then drag and drop the **Mail SMTP** step.
- 2) The initialization of the Mail should follow the sequence setup.
- 3) The sequence configuration should be related to the main section.
- 4) The Mail close should be handled in the sequence cleanup.

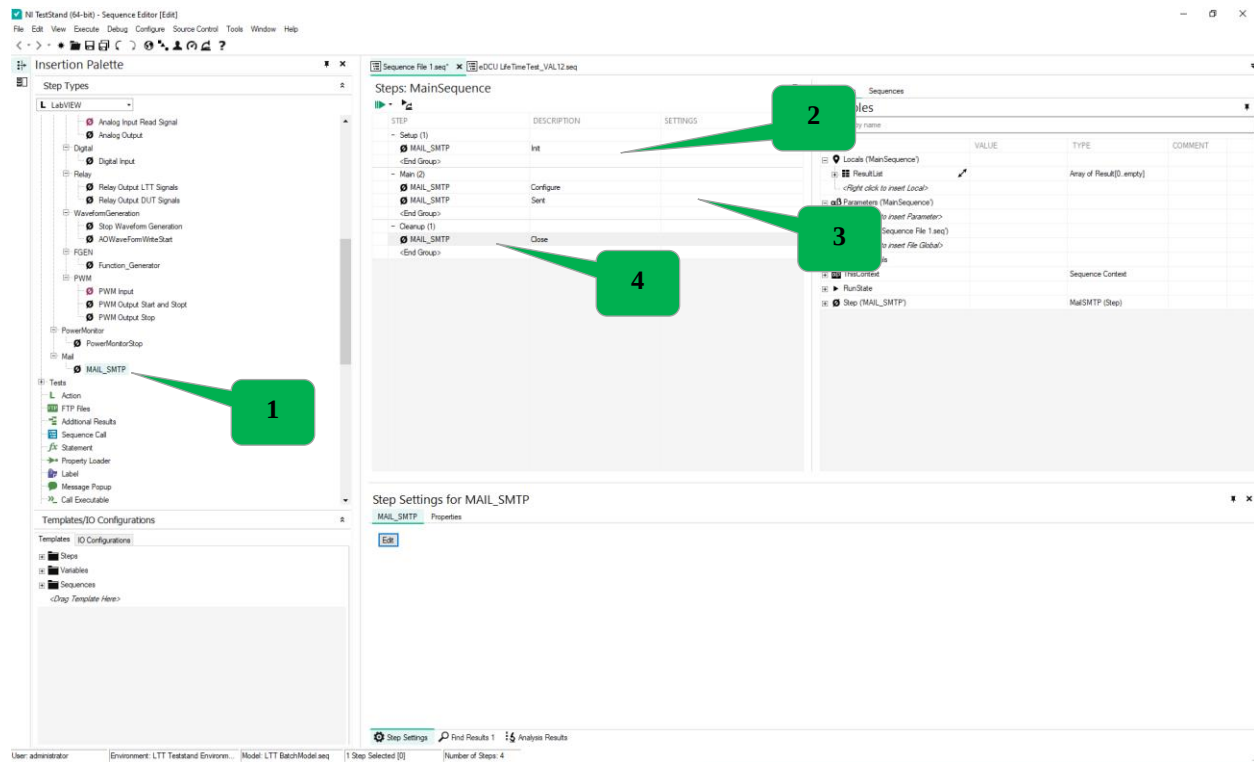


Figure 119 Mail drag and drop.

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13.11.1.1 Mail SMTP Initialize.

- 1) In the Insertion Palette, click on **MAIL** drag and drop the **MAIL SMTP** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the step in the sequence Setup.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

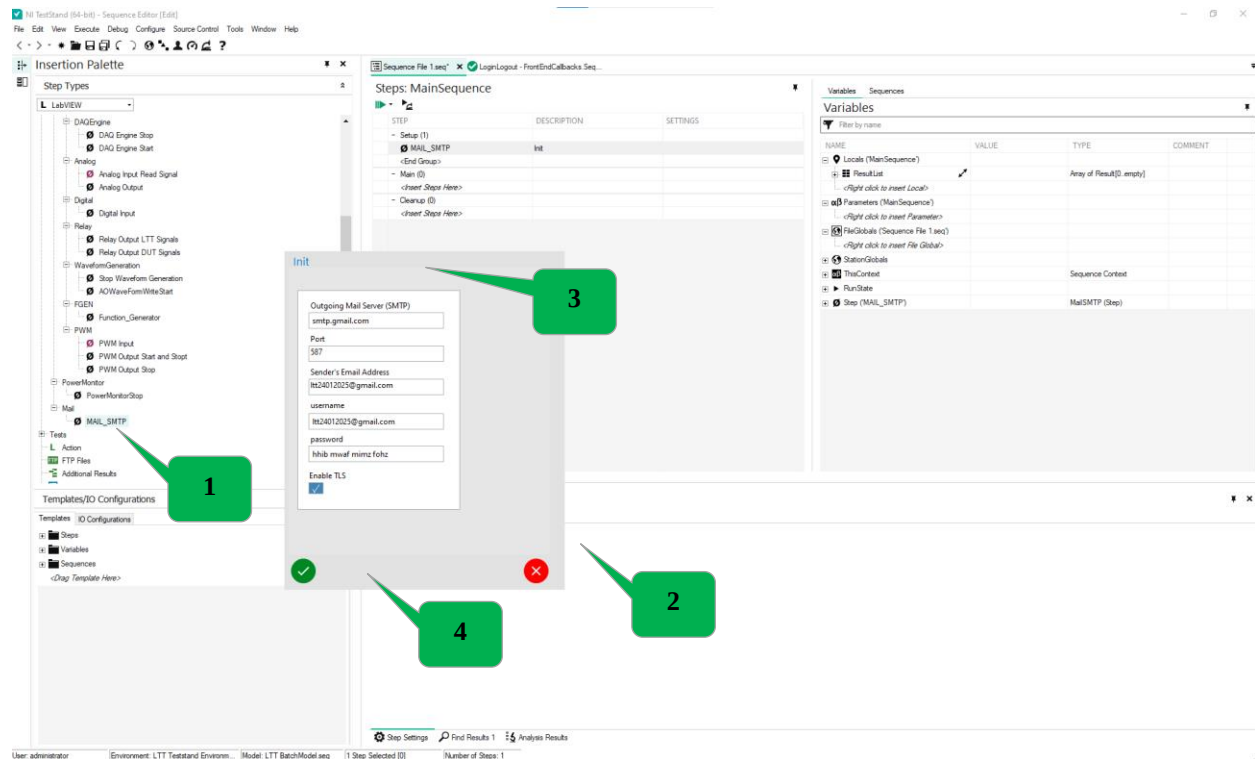


Figure 120: Mail SMTP Init step

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13.11.1.2 Mail SMTP Configure.

- 1) In the Insertion Palette, click on **MAIL** drag and drop the **MAIL SMTP** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

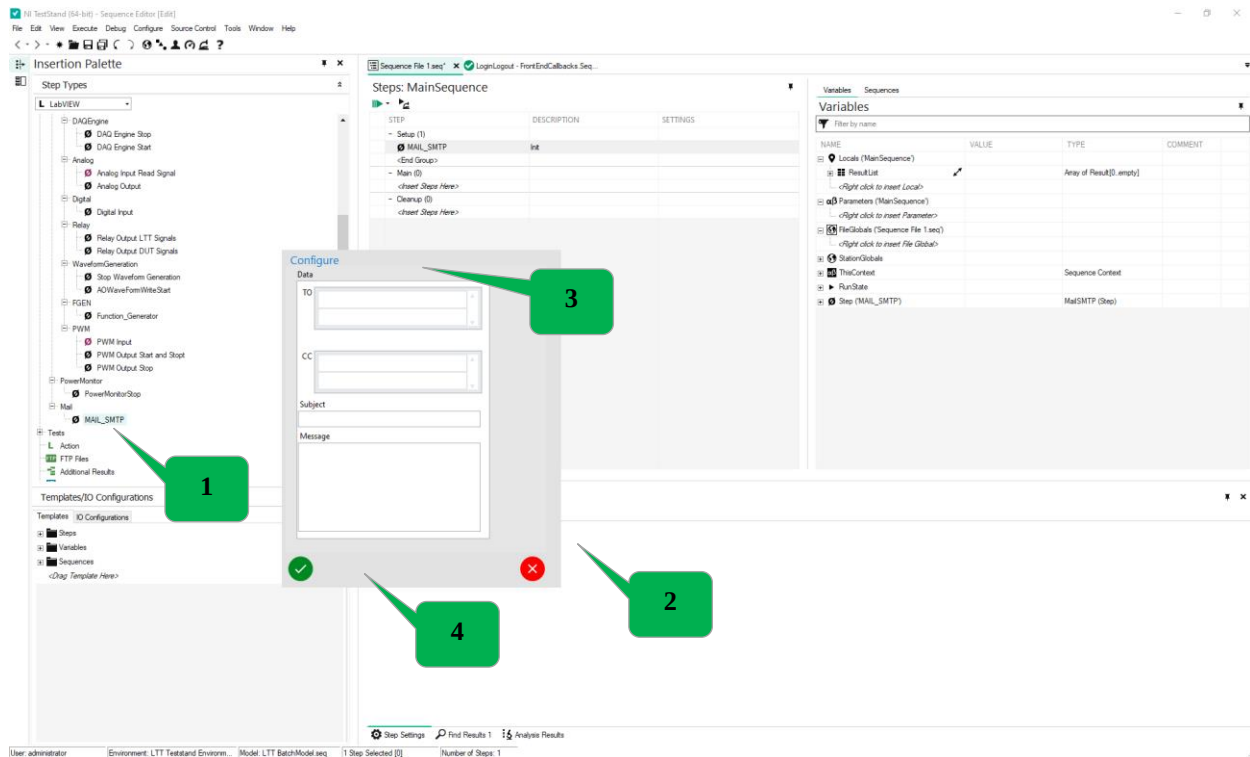


Figure 121: Mail SMTP Configure step

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13.11.1.3 Mail SMTP Sent.

- 1) In the Insertion Palette, click on **MAIL** drag and drop the **MAIL SMTP** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the step in the sequence Main.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

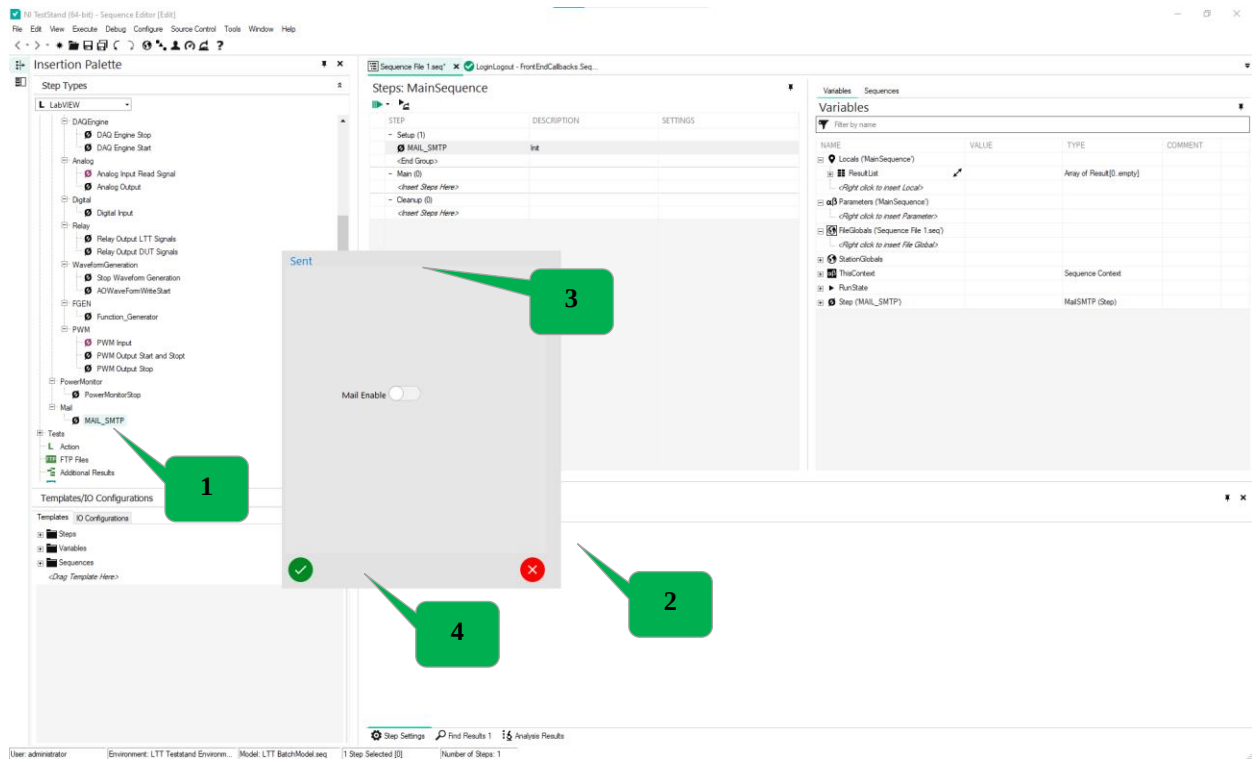


Figure 122: Mail SMTP Sent step

Project Name: Lifetime Tester	Document ID: 23-10-051-VTF-CBAIC-UM	Version:1.0
	PP Form No: R/PP-03/25	Version Date: 03-02-2025
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13.11.1.4 Mail SMTP Close.

- 1) In the Insertion Palette, click on **MAIL** drag and drop the **MAIL SMTP** step.
- 2) Click Edit and go to the step properties option and display the configuration details, which can be customized according to your requirements.
- 3) Always drag and configure the step in the sequence Clean up.
- 4) In the **Sequence Specific** step, click on **Edit**. The Edit Window will appear, showing the configuration details. Click the **tick mark** to store the step during execution or click the **intended mark** to avoid storing the step and retain the old data.

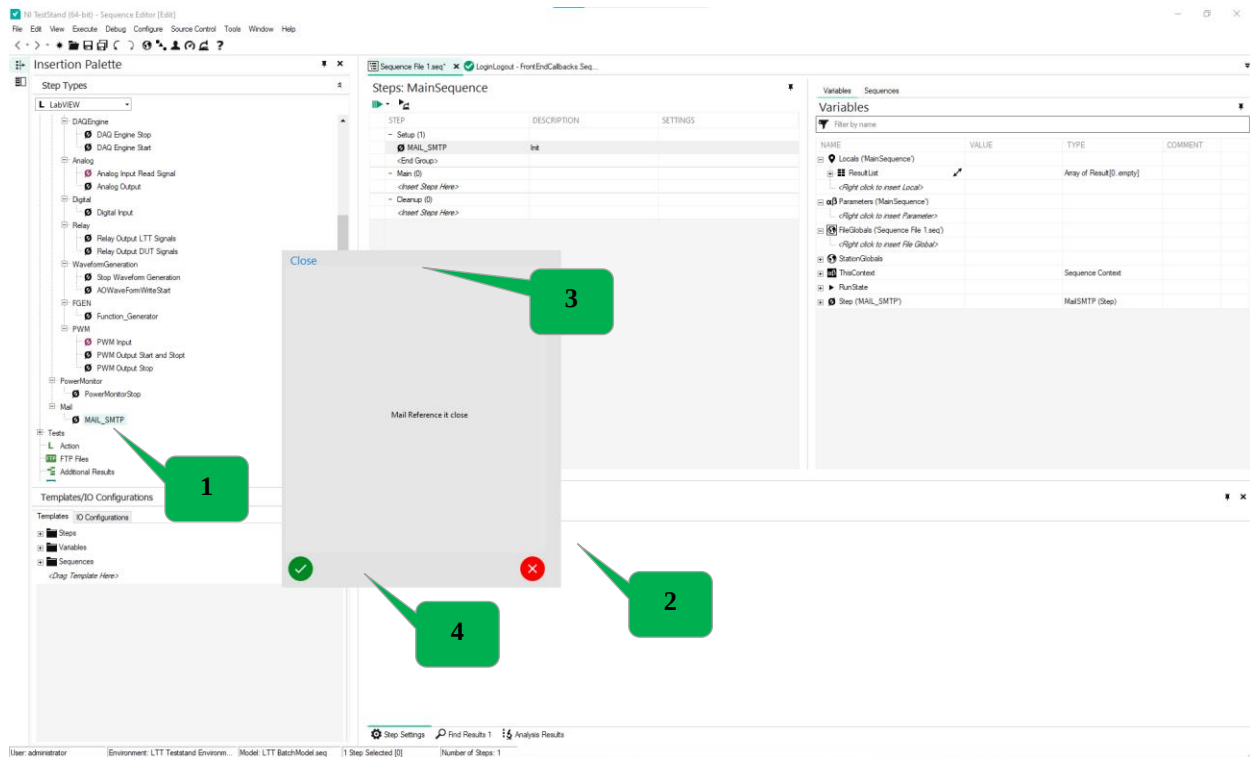


Figure 123: Mail SMTP Close step